

***ADVANCING THE LANDSCAPE RESILIENCE INDEX (LRI)
IN TANZANIA AND MOZAMBIQUE UNDER THE IUCN
SUSTAIN INITIATIVE***

FINAL REPORT
February 2026

LIST OF ACRONYMS

ADEM: Manica Province Economic Development

AFR100: African Forest Landscape Restoration Initiative

AGCOT: Agricultural Growth Corridors of Tanzania

AWF: African Wildlife Foundation

COSTECH: The Tanzania Commission for Science and Technology

EOA: Ecological Organic Agriculture

ILM: Integrated landscape management

IUCN: International Union for Conservation of Nature

LRI: Landscape Resilience Index

MADER: Republic of Mozambique, Ministry of Agriculture and Rural Development

miniSASS: Mini Stream Assessment Scoring System

MITADER: Republic of Mozambique, Ministry of Land, Environment and Rural Development

NbS: Nature-based Solutions

NCCSC: National Climate Change Steering Committee

NCCTC: National Climate Change Technical Committee

NDC: Nationally Determined Contributions

SAGCOT: Southern Agricultural Growth Corridor of Tanzania

SDGs: Sustainable Development Goals

SLM: Sustainable Land Management

SNV: SNV Netherlands Development Organisation

SRI: System of rice intensification

SUSTAIN-Africa: Sustainability and Inclusion Strategy for Growth Corridors in Africa

TARISSfupi: Tanzania River Scoring System

TAWIRI: The Tanzania Wildlife Research Institute

URT: United Republic of Tanzania

VPO: Vice President's Office

WSDP III - Water Sector Development Programme III

WWT: Water and Wetlands team

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EXECUTIVE SUMMARY

This report enhances and advances the Landscape Resilience Index (LRI) developed under IUCN's SUSTAIN-Africa initiative, updating the framework so it can be applied consistently across Tanzania (Kilombero, Ihemi, Sumbawanga) and Mozambique (Barué-Vanduzi), while remaining flexible enough to incorporate new datasets as they become available.

PURPOSE AND APPROACH

The updated LRI integrates indicators under three pillars—Land, Water, and Adaptive Capacity—and applies them across **Site**, **Landscape**, and **National** scales to translate fragmented monitoring data into decision-relevant evidence for integrated landscape management (ILM) and nature-based solutions (NbS) prioritisation.

KEY BASELINE FINDINGS ACROSS LANDSCAPES

Baseline LRI scoring highlights both strengths and priority constraints:

- **Kilombero** (Tanzania): Land indicators were generally good (above- and below-ground biodiversity and carbon stocks), with moderate soil fertility. Water chemistry and soil-water availability were excellent, while landscape water access remained moderate, indicating a social/infrastructure vulnerability despite strong environmental water conditions.
- **Ihemi** (Tanzania): Soil fertility was good, water chemical quality excellent, and water availability good, but landscape water access was poor and below-ground biodiversity, and carbon stocks were weaker (carbon stocks rated poor in the summary).
- **Sumbawanga** (Tanzania): Soil fertility scored good and water chemical properties were good-excellent, but below-ground biodiversity was poor.

Water access was moderate, and adaptive capacity indicators were mostly moderate despite strong scores for extreme weather and perceived shocks/stresses.

- **Barué-Vanduzi** (Mozambique): Soil fertility and water chemistry were comparatively strong (water chemical properties excellent), but carbon stocks were poor, small-stream biological activity was poor, and several adaptive-capacity constraints remained (including poor land cover percentage).

Across all landscapes, results reinforce that resilience is not determined by a single pillar: strong biophysical conditions (e.g., water chemistry) can coexist with major access constraints and/or land degradation pressures.

HOW THE LRI CAN BE USED (UPTAKE PATHWAYS)

The report sets out three practical entry points for LRI uptake:

1. Establishing an evidence base for investment (targeting hotspots and intervention priorities),
2. Monitoring, reporting, and adaptive management (tracking whether interventions deliver intended outcomes and adjusting over time), and
3. Institutional integration and policy alignment (embedding indicators into planning and reporting systems).

CONCLUSIONS AND NEXT STEPS

The report concludes that the LRI is already a useful cross-scale framework for guiding NbS/ILM prioritisation but emphasizes that continuous data updating and stakeholder refinement of scoring thresholds are essential, since some thresholds were necessarily provisional. Two immediate priorities are recommended: (i) a participatory workshop to review and validate thresholds, and (ii) appointing a data coordinator to track new data collection and keep the LRI dataset current and interoperable.

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INTRODUCTION

BACKGROUND

The Water and Wetlands Team (WWT) at the International Union for Conservation of Nature (IUCN) manages a broad portfolio of initiatives aimed at advancing integrated water and land management. One of WWT's flagship initiatives is the Sustainability and Inclusion Strategy for Growth Corridors in Africa (SUSTAIN-Africa). Implemented between 2014 and 2020, the first phase of SUSTAIN-Africa operated in the Southern Agricultural Growth Corridor of Tanzania (SAGCOT) and the Zambezi Valley in Mozambique. It began as a multi-year initiative to catalyse climate-resilient development that balances economic growth, social prosperity, and ecosystem stewardship. The first phase adopted an integrated landscape management approach by establishing multi-stakeholder partnerships with businesses, government agencies, communities, and farmer groups to deliver on this vision.

Building on these achievements, IUCN launched a second phase of SUSTAIN-Africa through two complementary programmes: SUSTAIN Pro and SUSTAIN Eco. Both are underpinned by the principles of equity and inclusion and aim to support a transformative shift away from business-as-usual development models towards approaches that build resilience in economies, people and nature.

SUSTAIN PRO: PRODUCTIVE LANDSCAPES FOR INCLUSIVE GROWTH IN TANZANIA AND MOZAMBIQUE

SUSTAIN Pro focuses on the development of sustainable food systems and healthy, productive landscapes in Tanzania and Mozambique. Designed as a ten-year initiative, the programme is currently implementing a four-year operational plan covering the period 2022-2025.

The delivery strategy is structured around three outcomes:

- (1) upscaling sustainable agricultural production systems,
- (2) promoting landscape resilience through landscape partnerships, and

- (3) investing in inclusive and sustainable value chains to accelerate the transition to sustainable food systems.

In Tanzania, SUSTAIN Pro is being implemented in the Kilombero and Ihemi landscapes within SAGCOT and interventions include promoting the system of rice intensification (SRI) practices, riverine restoration, agrochemical waste management and agroforestry.

In Mozambique, the programme is focused on the Vanduzi and Barué districts within the Beira Agricultural Growth Corridor and interventions include setting up community demonstration plots which promote sustainable agricultural practices such as, application of organic fertilisers, crop rotation and agroforestry. [SUSTAIN Pro \(outcome 1\) is being implemented in collaboration with ADEM in Mozambique.](#)

SUSTAIN ECO: ECOSYSTEM STEWARDSHIP TO BALANCE SUSTAINABILITY AND GROWTH IN TANZANIA

SUSTAIN Eco is a sister programme to SUSTAIN Pro and is being implemented in Tanzania only. It promotes healthy ecosystems and prosperous communities by improving governance, securing rights, advancing sustainable resource management practices, and mobilising investment for the protection and restoration of biodiversity and ecosystem services.

SUSTAIN Eco focuses on three key outcomes:

- (1) strengthening coordination among governance structures for sustainable and inclusive management of natural resources,
- (2) improving ecosystem health and generating livelihood opportunities through integrated landscape management, with particular attention to women, youth, and vulnerable groups
- (3) increasing investment in ecosystem restoration to enhance climate resilience for people and ecosystems.

The programme operates in the Kilombero and Sumbawanga clusters of SAGCOT, implementing environmental and community support interventions. In Sumbawanga, activities include demarcating forest and river buffer zones, promoting alternative

livelihoods such as beekeeping and agroforestry, building local capacity through data collection training and equipment provision, and carrying out restoration actions including tree planting, river monitoring, and invasive species control. In Kilombero, interventions focus on climate-smart agriculture (e.g. terrace farming), citizen science with schoolchildren to strengthen environmental awareness, and nursery establishment for landscape restoration. The programme also supports the “Kizimba” agrochemical waste management system, which provides secure storage for hazardous materials such as used pesticide containers. SUSTAIN Eco is being implemented in collaboration with SNV and AWF.

LANDSCAPE CONTEXTS

IHEMI (SUSTAIN PRO, TANZANIA)

The Ithemi landscape encompasses six districts, including Iringa Municipality and the district councils of Iringa, Kilolo, Mufindi, Wanging’ombe, and Njombe. This diverse region is characterised by competing land uses for agriculture, forestry, urbanisation, and industrialisation. There is growing demand for land, particularly for timber plantations and avocado farming, referred to as the “avocado rush.” Soybean has been identified as a strategic crop supporting large-scale off-takers such as Silverlands, which are developing out-grower schemes to promote sustainable and inclusive agriculture. Sunflower, valued for its drought resistance and local market demand, is also being promoted as a livelihood option for economically disadvantaged farmers.

KILOMBERO (SUSTAIN PRO & ECO, TANZANIA)

The Kilombero Valley lies between key conservation areas, including the Selous Game Reserve and Udzungwa

Mountains, and is noted for both high agricultural productivity and biodiversity value. It is home to a large portion of the world’s Puku antelope and encompasses the Kilombero Valley Floodplain—one of Africa’s largest wetlands and a designated Ramsar Site. However, agricultural expansion and shifting land use are contributing to ecosystem degradation and water management challenges. Programme efforts here are focused on implementing climate-smart agricultural practices (primarily in Kilosa, Mlimba, and Ifakara Town), and on improving governance to sustain ecosystem services across the landscape.

SUMBAWANGA (SUSTAIN ECO, TANZANIA)

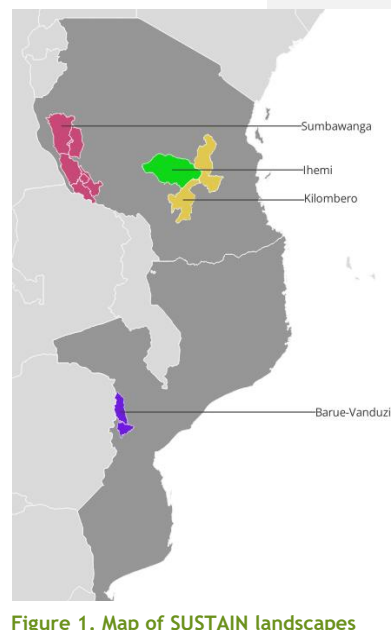


Figure 1. Map of SUSTAIN landscapes

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

Sumbawanga lies between Lakes Rukwa and Tanganyika and contains a rich mosaic of biodiversity, protected areas, and wildlife corridors. Though sparsely populated, recent infrastructure developments have increased access, intensifying pressure on natural resources and raising potential for land-use conflicts. Ongoing efforts seek to strengthen natural resource governance, foster social inclusion, and catalyse green investment across priority value chains such as rice, maize, and sunflower. Activities are focused in the Rukwa and Katavi regions.

VANDUZI AND BARUÉ (SUSTAIN PRO, MOZAMBIQUE)

Situated in the Beira Agricultural Growth Corridor, the Vanduzi and Barué districts of Manica province were selected for SUSTAIN Pro implementation due to their high agricultural potential and strategic importance. Challenges in these areas include limited access to quality seeds, underdeveloped agricultural extension services, and population growth. Major crops include maize (a staple), soy (critical for poultry feed and oil production), and horticultural products (which offer high returns but face market and processing bottlenecks). These dynamics were confirmed through baseline assessments and field visits, including engagement with local cooperatives.

LANDSCAPE RESILIENCE IN SUSTAIN

Landscape resilience is a foundational element across both SUSTAIN programmes. Defined as the capacity of a landscape to sustain ecological function, native biodiversity and critical landscape processes over time, under changing conditions and multiple stressors (1000 Landscapes for 1 Billion People, 2022). It is aligned with the SUSTAIN initiative's Theory of Change, particularly the second pathway: implementing and scaling sustainable agriculture and innovative Nature-based Solutions (NbS) to restore **landscape resilience and health**. This pathway applies integrated landscape management (ILM) principles and aims to scale NbS that support inclusive livelihoods while reversing land degradation. To this end, robust monitoring systems are essential for guiding conservation, governance, and natural resource management decisions, as well as for providing the evidence base needed to attract and scale investment in NbS.

ILM recognises that landscapes are social-ecological systems consisting of interconnected land and water ecosystems with distinct and interacting ecological, historical, economic, and socio-cultural processes and activities (1000 Landscapes for 1 Billion People, 2022). **Building on this framing, SUSTAIN-Africa required a monitoring tool capable of integrating land, water and the socio-economic aspects central to landscapes and their resilience.** However, most existing resilience tools focus heavily on either ecological or socio-economic resilience (Schatt & Meyer 2025; Johnson et al., 2015; Harwood et al., 2022). Few integrate both and even fewer explicitly incorporate hydrological dynamics, despite the centrality of water in shaping ecosystem persistence, agricultural productivity, and community wellbeing (Falkenmark et al., 2019).

For practical measurement, SUSTAIN-Africa required a landscape resilience index with indicators that can be adapted to differing levels of data availability, institutional capacity, and landscape context. Many

existing resilience frameworks rely on fixed indicator sets, standardised survey instruments, or data-intensive modelling approaches, which can limit their applicability in data-scarce settings or across heterogeneous landscapes (Beller et al., 2015; Feldmeyer et al., 2021).

Beyond flexibility in measurement, **SUSTAIN-Africa also required a monitoring approach that operates coherently across multiple spatial scales.** Many existing resilience frameworks are applied either at the site or community level, or at national and global scales, with limited capacity to explicitly link local interventions to landscape-level processes or to national and international commitments. This fragmentation constrains the ability of practitioners, policymakers, and investors to trace how on-the-ground land management actions contribute to progress against broader national and international objectives, such as climate adaptation, biodiversity recovery, or sustainable development.

Despite strong policy commitments, investment in landscape restoration and resilience remains insufficient. Global assessments show that Africa loses an estimated USD 65 billion annually to land degradation, while restoration investments fall far short of the scale required (UNCCD, 2022; IPBES, 2019). While donor funding provides a starting point, **the challenge is that project types and interventions are often dictated by donor priorities rather than the needs of the landscape or local communities; many initiatives are still delivered through short, project-based cycles. Programmes can pause or close, and priorities may shift between funding windows.** Without sustained investment, degradation continues to undermine food security, water availability, and climate adaptation. **Consequently, SUSTAIN-Africa required a tool that enabled investment with multiple points of entry for stakeholder engagement.**

Taken together, existing landscape resilience frameworks lack the flexibility, hydrological integration, and cross-scale coherence needed

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to translate local land management interventions into policy-relevant evidence and investable outcomes for nature-based solutions. This report proposes the Landscape Resilience Index (LRI) as a flexible, integrated tool designed to provide this critical monitoring capacity, as developed within SUSTAIN-Africa.

HOW THE LRI WAS DEVELOPED

Landscape resilience is measured through SUSTAIN's Landscape Resilience Index (LRI). This index combines various indicators to track key components of landscape resilience such as land degradation, recovery potential, fair access to resources, and the resilience of both ecosystems and communities. It acts as a flexible tool that aids in ILM, aligning policies, and supporting adaptive decision-making.

The LRI is encompasses three pillars:

- (1) **Land** - the biodiversity below ground that sustains life above.
- (2) **Water** - Reliable, sufficient water of appropriate quality - ensuring sustainable water use and infrastructure to support agricultural soil moisture.
- (3) **Adaptive capacity**- the ability of social-ecological systems to adapt to climate change while maintaining well-being and productivity.

At the site level which focuses on soil and farms, the index monitors soil biological structures, farm system diversity (like crops and habitats), and soil-water interactions (including infiltration, moisture, and irrigation practices). This is the most detailed scale of

assessment and generates data to guide farm-level decisions and advisory services. It typically involves on-the-ground data collection and lab-based analysis.

At the landscape level, the LRI captures broader ecological and hydrological processes that span across farms and administrative borders. Key indicators include catchment-level water availability and quality, ecological connectivity, use of sustainable land management practices, and exposure to extreme weather events. This level helps link individual actions to broader environmental outcomes which is essential for integrated landscape planning, ecosystem restoration, and climate resilience.

At the national level, the index connects local monitoring efforts to national policies and international goals. While drawing direct links can be complex, qualitative methods and cross-checking data help integrate local knowledge into national reporting systems. Although the national level is not a formal scale for direct monitoring, it plays a critical contextual role in supporting decision-making and understanding how actions in one productive landscape may impact resilience elsewhere. This broader perspective helps ensure that the framework accounts for ecological consequences that extend beyond the immediate target area.

Understanding landscape resilience enables SUSTAIN-Africa to link local interventions to national and global commitments, supporting investment in NbS and catalysing systemic change in the way land is managed. By implementing ILM strategies and NbS, the initiative aims to improve landscape resilience and create inclusive business and livelihood opportunities. Monitoring through the LRI provides the evidence base to inform decisions and accelerate NbS investment.

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@MariaAna

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Commented [GU6]: I would revise the description of adaptive capacity pillar on the landscape scale, as it is not only about the influence of environmental shocks, it looks at the productivity levels, SLM adoption, etc so it does take the overall health of the ecosystem

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

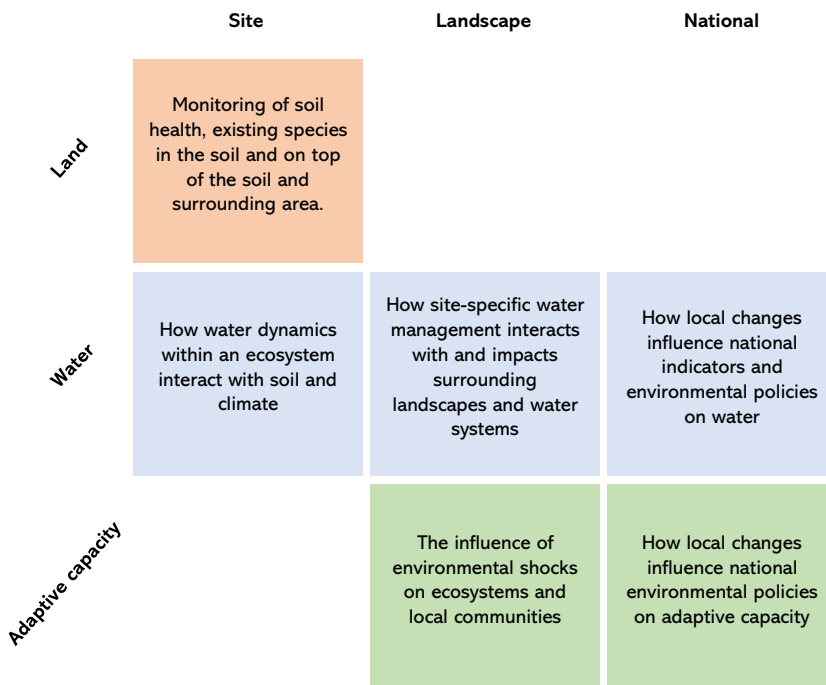


Figure 2. The SUSTAIN LRI, how land, water and adaptive capacity pillars link to the site, landscape and national scales

RATIONALE FOR THE PROJECT

The LRI was designed and tested through a previous consultancy, which led to the selection of 17 key indicators ~~covering soil, water, and ecosystem resilience~~ from an initial list of 110 indicators. Initial baseline values were established for the Ihemi and Kilombero clusters in Tanzania using secondary data sources. Yet data gaps remained for some indicators. Moreover, no baseline assessment has yet been done for Sumbawanga in Tanzania and Vanduzi in Bárue District, Mozambique.

OBJECTIVES AND DELIVERABLE

The objectives of this work are to:

1. Collect and analyse primary and secondary data to complete baseline values for the ~~19 land health~~ indicators previously defined under the Landscape Resilience Index (LRI) methodology in Kilombero and Ihemi and establish new baselines in Sumbawanga, Vanduzi and Bárue;
2. Identify clear pipelines to embed ~~landscape resilience health~~ monitoring indicators and methodologies into national frameworks in Tanzania and Mozambique, particularly considering Tanzania's Soil Health Compact and other policy processes and
3. Generate a clear evidence base to guide future investment in IILM and Nb

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In achieving these objectives, we have delivered the following outputs:

1. **Complete Dataset and Landscape Report:** a full data set and analytical report on the land health status across all four landscapes (Kilombero, Ihemi, Sumbawanga, Bárue- Vanduzi), including gaps filled and new baselines established.
2. **Policy brief:** policy review with key recommendations for how ~~landscape resilience health~~ monitoring can be positioned within national policy and institutional frameworks in Tanzania and Mozambique.
3. **Peer-reviewed scientific article:** development of one scientific article based on the land health monitoring work, submitted to a reputable journal, with the consultant named as lead author or co-author, co-drafted with IUCN focal points for this work.
4. **Presentation of Results:** participation in at least one internal SUSTAIN technical meeting or stakeholder workshop to present key findings and discuss the implications for programme implementation.

POLICY REVIEW

Resilience is a theme across many policy documents for both Tanzania and Mozambique, particularly in the context of climate change and the need for adaptation goals with robust monitoring and evaluation to track progress. Water is a core element running through key policies that highlight water security as essential for health, livelihoods and environmental sustainability. Together, these policies show that effective water management is central to building resilient landscapes that can sustain ecosystems, communities and long-term development.

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TANZANIA

OVERVIEW OF POLICIES AND RELATION TO SUSTAIN LRI

Tanzania's 2023 Voluntary National Review (VNR) Report (URT, 2023)

The Sustainable Development Goals consist of 17 goals coupled with 169 targets and 249 indicators that aim to measure the most pressing issues facing our planet and interactions between goals. We reviewed Tanzania's 2023 Voluntary National Review (VNR) Report (URT, 2023) on the Implementation of the 2030 Agenda for Sustainable Development to understand how currently reported indicators for the SDGs relate to SUSTAIN LRI.

Although the VNR does not explicitly address landscape resilience, the SUSTAIN LRI relates to specific indicators for SDGs 1 No Poverty, 2 Zero Hunger, 4 Quality Education, 5 Gender equality, 6 Clean Water and Sanitation, 8 Decent Work and Economic Growth, 13 Climate Action and 15 Life on Land. The report also has a section on Tanzania's experience with global shocks including climate change where Tanzania will experience rising temperatures, longer dry spells, more intense rainfall and sea level rises. Water resources, agricultural production, livelihoods, food security, ecosystems/biodiversity and human health are all listed as priority areas for building resilience, which reflects the focus of

the SUSTAIN LRI on water, soil/biodiversity and social indicators for resilience. There has been no update on the VNR since Durrant et al., (2024).

Bonn Challenge and AFRI100 (Bonn Challenge 2024)

The United Republic of Tanzania has committed to restoring 5.2 million hectares of land by 2030 under the Bonn Challenge (Bonn Challenge 2024). To implement this, government targets state that 1.5 million trees must be planted in each district each year until 2026. Increasing tree cover within the clusters can be implemented in various ways that will influence Landscape Resilience such as forest restoration, riparian restoration, wildlife corridors or agroforestry. There has been no update on the pledge since Durrant et al., (2024).

Tanzania National Ecological Organic Agriculture Strategy 2023-2030 (Biovision 2024; MuA URT 2023)

This policy was launched in November 2023. It aims to steer food systems away from unhealthy and unsustainable practices and advance a resilient agri-food sector. It contributes to national priorities of agroecological market development to enhance smallholder farmers' livelihoods by increasing access to certification and securing local markets for agroecological products. Specific interventions include plans for

government owned demonstration/training farms and development of a detailed strategy and partnerships to finance, guide and motivate the agroecological sector. Moreover, ecological organic agriculture is a recommended tool in climate adaptation and mitigation strategies. Much of the SUSTAIN LRIHMAI will cover agricultural areas due to it being a predominant land use in the SUSTAIN clusters. Using the SUSTAIN LRIHMAI to monitor the influence of sustainable farming practices on landscape resilience would be beneficial for national targets and offices when they are mobilising funding to implement this policy. There has been no update on this policy since Durrant et al., (2024).

National Biodiversity Strategy and Action Plan 2015-2020 (VPO URT 2015)

The National Biodiversity Strategy and Action Plan does explicitly recognise the intrinsic link between natural and social systems and the relationships between functioning ecosystems, human wellbeing and socio-economic development. This is demonstrated in the vision statement that declares “By 2025, biodiversity and ecosystems are well protected, restored and used sustainably, ecosystem functioning maintained, so that they perpetually deliver sustainable intrinsic benefits for socio-economic development.”

The SUSTAIN LRI specifically links to strategic goal D, targets 14 and 15. Target 14 is focussed on “ecosystem resilience and the contribution of biodiversity to carbon stocks enhanced, through conservation and restoration, thereby contributing to climate change mitigation and adaptation and to combating desertification.” Target 15 states that “by 2020, ecosystems that provide essential services, related to water, and contribute to health, livelihoods and well-being, are restored and safeguarded, taking into account the needs of women, local and vulnerable communities.” Moreover, the policy states that Tanzania is among the top 15 countries for the highest number of threatened species, with at least 900 threatened species under the IUCN Red List, 2013. It also identifies agroecological approaches as among the biodiversity

conservation approaches in agriculture. Both of which are addressed in the SUSTAIN LRIHMAI. There has been no update on this plan since Durrant et al., (2024).

National Environmental Policy, 2021 (URT VPO, 2021)

This policy acknowledges the continued different forms of land degradation including deforestation, loss of vegetation cover, soil erosion, soil pollution, deterioration of aquatic systems and loss of biodiversity - all of which present in the SUSTAIN clusters and are addressed within the SUSTAIN LRI and indicators. There is much discussion on land degradation, which is reported to be increasing at the country level and was measured as almost 50% in 2012 by the Land Degradation Neutrality Report. However, the policy lacks details of what constitutes healthy or non-degraded land or how to monitor this, which is where the SUSTAIN LRIHMAI links and can provide value. There has been no update on this policy since Durrant et al., (2024).

National Climate Change Response Strategy (2021-2026) (URT, 2021)

This policy document devises and implements strategic adaptation and mitigation measures in line with the national circumstances, requirements of climate change related multilateral environmental agreements, Sustainable Development Goals (SDGs), Sendai Framework for Disaster Risk Reduction (SFDRR) and related bilateral and regional agreements. It aims to align climate change interventions with the national development agenda of an industrialised economy. Resilience is a key theme throughout the policy. Therefore, this policy complements the SUSTAIN LRI that takes climate change risks (e.g. drought and flooding) and economics and productivity into account for landscape resilience. It also aims to promote and facilitate the transfer of climate-smart technologies to support climate change adaptation and mitigation. its objectives align with the framing of the SUSTAIN LRI; to ensure sustainable management and resilience of water resources, forests, key ecosystems and the

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agricultural sector for sustainable livelihoods through promoting climate-smart practices. There has been no update on this policy since Durrant et al., (2024).

National Water Policy 2002 version 2025 (MoW URT, 2025)

The 2025 updated policy vision is ‘A water-secure country with equitable, sustainable, accessible, affordable, and reliable water and sanitation services for socio-economic development.’ The 2025 update aims to ensure optimal, reliable, sustainable, and equitable development and use of water resources for the benefit of all in the most cost-effective way whilst ensuring climate resilience. It highlights the challenge of capital investment for rural areas that is beyond capacity of local communities that has slowed progress in service delivery and subsequent poor water quality management. It advocates the need to conserve water sources via Integrated Water Resources Management (IWRM) to ensure water sources are effectively used and conserved.

Water use and demand has risen over the past 20 years due to increased population and economic activities and was 62 billion cubic meters (BCM) in 2024 and is projected to increase to 80.2 BCM in 2035. Tanzania is not internationally recognised as water stressed, but the policy calls to increase productivity and efficiency of water use to ensure water security. In rural areas, access has increased from 50 per cent in 2002 to 83 per cent in June 2024. In urban areas, access to water supply services increased from 73% in 2002 to 91.6% in 2024.

The Policy emphasises on planning and implementation of water resources and other development programmes in an integrated manner and in ways that protect water catchment areas and their vegetation cover, improved management and conservation of wetlands. There is a growing concern of water resource depletion and quality deterioration and water demand for agriculture is rising and a challenge is limited cross-sectoral planning on water issues across sectors such as

agriculture, hydropower, mining and other sectors. The policy lists non-government organisations roles and responsibilities are to (1) sensitise communities of water source protection and management or water schemes and (2) support the provision of water supply and sanitation services.

Specific objectives of the National Water Policy 2022 version 2025 that the LRI feeds into:

1. Enhance sustainable management of water resources.
2. Improve water and wastewater quality management.

These two policy objectives focus on strengthening monitoring and assessment of water quality for the safety and well-being of the people and ensuring water availability, quality and distribution.

Monitoring and assessment of water quality is essential for planning and informed decision-making and is collected from water quality and policy stations in water bodies and water supply/distribution networks across the country. However, there is a lack of regular water quality monitoring and assessment. There is also evidence of deterioration of water quality in many parts of the country due to the presence of harmful chemicals from industrial processing, mining and agriculture. The LRI water indicators at the site and landscape scale relating to water quality and availability could feed directly into the monitoring needed at the national scale.

3. Promote cross-cutting issues including environmental and climate change resilience, gender and good governance.

Environmental and climate change resilience as a cross-cutting issue particularly due to the high dependence of primary livelihood and development strategies on extraction of natural resources. Changing hydrological cycles leading to hazardous flooding and droughts feeds directly into the ~~LRI~~

landscape-LRI indicator assessing extreme weather at the landscape scale.

National Wildlife Policy 2007 (MNRT URT 2007)

The Policy provides for protection of water catchments for wildlife water supply; conservation of biodiversity; protection of fragile ecosystems; and restoration of degraded habitats for flora and fauna. The Policy also promotes sustainable management and use of wetlands, most of which are located in protected areas. It recognises that wildlife resources can be developed and sustainable utilised to raise its contribution to Tanzania's Gross Domestic Product (GDP) and contribute to poverty alleviation. The policy does not explicitly link landscape resilience with wildlife health, but it does underscore the need to conserve habitats, water catchments and soil resources. This links to the LRI through indicators such as threatened species, land cover and soil health metrics. There has been no update on this policy since Durrant et al., (2024).

National Agriculture Policy 2013 (MoA URT, 2013)

The vision of the National Agriculture Policy 2013 is "An agricultural sector that is modernized, commercial, highly productive and profitable; that utilizes natural resources in an overall sustainable manner in Eastern and Central Africa that acts as an effective basis for inter-sectoral linkages by the year 2025." This policy discusses using the agricultural sector to ensure food security and poverty alleviation, and states that agricultural development is dependent on environmental resources. Issues of land degradation and soil erosion are mentioned as among the issues that have affected agrobiodiversity and led to declined land productivity. However, it does not address the interconnectedness of human systems with landscape resilience or how healthy land systems can be sustainably operationalised. Irrigation is deemed essential for increased productivity and production to mitigate increasingly extreme weather but less than 5%

of farming households use irrigation. There has been no update on this policy since Durrant et al., (2024).

AGCOT and SAGCOT Green print (SAGCOT, 2012)

The Agricultural Growth Corridors of Tanzania (AGCOT) is an evolution of the Southern Agricultural Growth Corridor of Tanzania (SAGCOT), launched on April 27, 2025. This nationwide expansion was a direct result of a directive from President Dr. Samia Suluhu Hassan in March 2023, which called for the replication of SAGCOT's successful model across the country. SAGCOT, established in 2010, was a public-private partnership focused on agricultural transformation in the Southern Highlands of Tanzania. Having achieved its 2030 objectives ahead of schedule, it was tasked with a broader scope, leading to the formation of AGCOT (Agricultural Growth Corridors of Tanzania). Comprised of northern, central, southern and Mtwara corridors.

AGCOT's vision is to revolutionize Tanzania's food systems by championing investments in food security, ensuring the inclusion and transformation of smallholder farmers, and promoting sustainable agricultural development. The AGCOT Centre, formerly the SAGCOT Centre, will continue to serve as a facilitator, fostering partnerships among various stakeholders to create inclusive, sustainable, and commercially viable agricultural value chains. AGCOT will expand its corridor approach, tailored to different agricultural zones, and align its strategies with Tanzania's Development Vision 2050 and the New Agriculture Master Plan.

The SAGCOT Green Print identifies climate change, environmental conservation and natural resource management as critical to long-term economic development. Its purpose is to ensure economic development is environmentally sustainable, socially equitable and economically feasible through sustainable intensification of agriculture, natural resources are described as 'investable assets.' Success will see benefits for food

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

production, economic output, local livelihoods and environmental conservation.

The SAGCOT greenprint has some key goals for which the SUSTAIN LHMI may feed into:

1. Agriculture Green Growth (AGG) strategy will generate an additional 2.2 million tons of rice and field crops per year, worth approximately US \$600 million (936 billion TzSh) at recent market prices by 2030.
2. More than 300,000 hectares will be avoided by deforestation by 2030.
3. The carbon contained in these forests, combined with emissions reductions associated with AGG, will generate net emissions reductions of about three million tons of CO₂-equivalent per year by 2030, worth about US \$30 million at current carbon prices.
4. Save nearly one billion cubic meters of water per year due to increased irrigation water use efficiency, while also increasing water use efficiency for rainfed agriculture.

There has been no update on the Green Print since Durrant et al., (2024). However, alongside the Green Print, SAGCOT with key partners such as IUCN has developed the Inclusive Green Growth (IGG) Guiding Tools to promote sustainable agricultural practices. An update to the IGG was published in 2024. The IGG has different guidelines for both small-scale and medium and large-scale producers. There are some aspects of the IGG tool that align with and could be used to provide data for the **LRI HMI** in the future. These include the following questions:

LRI HMI indicator: Adoption of Sustainable Land Management (SLM) Practices

Small-scale producer IGG

- Do you/your members apply conservation farming practices?
- How do you/your members manage soil health?

- Do you/your members farming practices protect water resources / forest reserves/ protected areas? How?

- Are you familiar with climate smart agricultural practices/technologies? Do you implement any?

Medium and large-scale producer IGG

- How do you support regenerative/restorative agriculture practices and Nature-based Solutions in your supply chain?

LRI HMI indicator: Landscape water access

- Does the investment have and implement strategies to enhance efficient water use? Briefly explain. E.g., drip irrigation, crop rotation, choice of varieties

Tanzania wildlife corridors assessment, prioritization, and action plan 2022-2026 (MNRT, 2022)

Tanzania has recently passed Wildlife Conservation Regulations (2018), aka the 'Corridor Regulations' to maintain and restore critical wildlife corridors to sustain biodiversity. The plan identifies and assesses 61 corridors across Tanzania, of which Ruaha Rungwa - Udzungwa scored highest for conservation value. Moreover, Kilombero - Udzungwa Mountains had the fourth highest conservation value score and sixth highest vulnerability score. Udzungwa - Uzungwa Scarp and Udzungwa - Mikumi are also ranked highly for conservation value (top 15). Nyerere Selous - Udzungwa also ranked second highest for vulnerability score, indicating it is a corridor highly threatened by existing or imminent blockage by anthropogenic land use. This plans objectives that relate to the **LRI HMI** such as (1) reduce human-wildlife conflict, (2) maintain sustainable flow of ecosystem services for wildlife and people, (3) provide tangible benefits and livelihoods for local stakeholders. There has been no update on this plan since Durrant et al., (2024).

Draft Tanzania Development Vision 2050 (URT, National Planning Commission 2024)

Commented [GU11]: How does this one related to the wildlife policy 2007 mentioned above

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

Tanzania's Development Vision 2050 is a comprehensive framework for the country's transformation into a prosperous, just, inclusive, and self-reliant nation. The draft vision identifies "Environmental Integrity and Climate Change Resilience" as one of its three core pillars. It explicitly aims to achieve development that safeguards environmental and ecological integrity while enhancing low-carbon development and resilience in the face of climate change. The document acknowledges that climate change poses significant threats to key sectors like agriculture, livestock, fisheries, and tourism, and can trigger internal migration. The vision emphasises building climate resilience, including through optimising land and water use to boost agricultural and rural productivity, increase climate resilience, and reduce greenhouse gas (GHG) emissions. It acknowledges the need to enhance the resilience of ecosystems to climate change challenges.

The draft vision highlights the importance of protecting natural assets, linking conservation and improved natural habitat management to creating economic opportunities. The vision's focuses on environmental integrity and climate change resilience, as well as its mention of the need to protect ecosystems and conserve forests. The vision seeks to modernise the agricultural sector to create more jobs through boosting productivity and resilience, expanding irrigation and promoting climate-smart agriculture.

Tanzania Nationally Determined Contribution (URT 2021)

Tanzania's Nationally Determined Contribution aims to reduce economy-wide greenhouse gas emissions by 30-35% by 2030, relative to a business-as-usual (BAU) scenario, conditional on international support. The updated contribution, submitted in 2021, details both mitigation strategies, including actions in the energy, transport, forestry, and waste management sectors, and adaptation measures focusing on agriculture, water, health, and coastal ecosystems. It aligns with national development goals and international

frameworks, such as the Paris Agreement and the Sustainable Development Goals. Many commitments focused on natural resource management align with the LRI, focussing on land and water resources, climate related shocks, sustainable land management practices, and livestock/agricultural productivity. Commitments that the SUSTAIN LRI could help inform include (1) Up-scaling the level of improvement of agricultural land and water resources management and, (2) Increasing productivity in an environmentally sustainable way through inter alia climate-smart agriculture interventions. However, there is no baseline or quantitative target for the indicators set.

Table 1. Implementation responsibility for each policy analysed for Tanzania

Policy	Responsibility
Tanzania's 2023 Voluntary National Review (VNR) Report (URT, 2023)	SDG National Coordination Taskforce overseas VNR preparation. National Bureau of Statistics is responsible for data collection, research and analysis.
Bonn Challenge and AFRI100	Unclear, but likely Tanzania Forest Service is a key implementing partner.
Tanzania National Ecological Organic Agriculture Strategy 2023-2030 (Biovision 2024; MuA URT 2023)	Ministry for Agriculture, Food Security and Cooperatives
National Biodiversity Strategy and Action Plan 2015-2020 (VPO URT 2015)	Vice President's Office, Division of the environment (VPO-DoE)
National Environmental Policy, 2021 (MoA URT, 2013)	Ministry for Environment
IGG	Existing SAGCOT investors within their operational activities, which are then submitted annually to SAGCOT
National Climate Change Response Strategy (2021-2026) (URT, 2021)	Vice President's Office oversees the strategy which is implemented by the National Climate Change Technical Committee and National Climate Change Steering Committee. The VPO is responsible for monitoring and evaluation.
National Wildlife Policy 2007 (MNRT URT 2007)	Ministry of Natural Resources and Tourism
National Agriculture Policy 2013 (MoA URT, 2013)	Ministry for Agriculture, Food Security and Cooperatives
Tanzania wildlife corridors assessment, prioritization, and action plan 2022-2026 (MNRT, 2022)	Ministry of Natural Resources and Tourism
Draft Tanzania Development Vision 2050	Ministry of Finance
Tanzania Nationally Determined Contribution (2021)	Vice President's Office

RESULTS OF POLICY REVIEW

Resilience is a theme across many policy documents, for instance, the Draft Tanzania Development Vision 2050 because the Development Vision identifies Environmental Integrity and Climate Change Resilience as one of its three core pillars, as it aims to achieve development that safeguards environmental integrity. The vision emphasises optimising land and water use to boost agricultural and rural productivity, increase climate resilience, and reduce greenhouse gas (GHG) emissions. It acknowledges the need to enhance the resilience of ecosystems to climate change challenges. Recently published policy documents, including the Draft Tanzania Development Vision 2050 and the updated 2025 National Water Policy, place strong emphasis on resilience. Both highlight the critical role of landscape resilience in achieving long-term environmental sustainability and socio-economic stability.

Regarding water, there is a large focus within National policy to ensure clean and safe water for all such as in the SDG VNR, The National Climate Change Response Strategy and National Water Policy. Tanzania adopts an approach of integrated water resources management which aims to promote the coordinated development and management of water and land related resources. Tanzania also has a Water Sector Development Programme (WSDP III) aimed at improving sustainable water availability and quality through interventions such as dam construction, strengthening of basin organisations and rehabilitation of water points. Climate resilience to ensure resilient water supply and infrastructure is a key strategy of the WSDP III. Districts and municipal councils are responsible for implementation at the local level, which feeds back down into the site and landscape indicators of the SUSTAIN LRI. The National Water Policy emphasises on planning and implementation of water resources and other development programmes in an integrated manner and in ways that protect water catchment areas and their vegetation cover, improved management and conservation of

wetlands. Monitoring and assessment of water quality is regarded as essential for planning and informed decision-making and is collected from water quality and policy stations in water bodies and water supply/distribution networks across the country. The LRI water indicators at the site and landscape scale relating to water quality and availability could feed directly into the monitoring needed at the national scale.

Despite not specifically mentioning the concept of NbS within any National Policies, NbS can be solutions to many of the environmental and societal challenges throughout and some were specifically mentioned in National Policies such as restoration of land and wetlands and, agroforestry.

Despite no policies defining Land Health or monitoring, the public and private sector has signed a Soil Health Compact, adopting [LRIHMI](#)'s holistic vision of Land Health.

RELATION TO LRI NATIONAL INDICATORS

Table 2 is an updated table from the original policy review that documents some specific targets, indicators, aims or statements that the [LRIHMI](#) could feed into at the national level. Key updates to the table include an updated list of national [LRIHMI](#) indicators, the inclusion of the National Water Policy 2022 updated 2025 and a focus on quantitative national indicators with baseline and/or target values.

Commented [GU12]: I have not read the policy brief yet, so my comments here are only to this section. I would recommend revising it to more or less communicate where these policies are doing well, where they overlap and what gaps exist. From reading the section above, I can see there is great focus on productivity, also on protection of natural habitats, wildlife, integrated management, etc. I can also see similar indicators under these policies being tracked by different entities, which again presents the need for the unification of some of these efforts/indicators under a "data partnership" which we aim to propose to the stakeholders this upcoming trip to Tanzania. So if more details are included in the brief, please include a hyperlink to guide the reader to the full analysis.

Table 2. An overview of some key national indicator suggestions for Tanzania that resulted from the national policy review including a short summary and the specific policy relevance

LRI n	National scale	Indicator	Key quantitative national targets and indicators	Baseline and target values
LRI pillar: Water				
Water quality and quantity			SDG VNR 6.3.2: Number of focused water quality monitoring networks established	Baseline: 430 in 2019 Target: none found
			National Climate Change Response Strategy (2021-2026): Promote water resilience in all basins	Baseline: none found Target: Installation of integrated water resources management tools in all 9 basins by 2026
Access to quality water for different uses: drinking water, sanitation, irrigation			SDG VNR 1.4.1: Population living in households with access to drinking water and sanitation	Baseline: Drinking water 78.8% in 2022 Target: none found
			SDG VNR 2.4.1: Area under irrigation	Baseline: Sanitation 42.7% by 2022 Target: none found
			Tanzania Nationally Determined Contribution 2021: Increase access to clean and safe water in rural areas by 2030	Baseline: 727,281 ha by 2022 Target: none found
				Baseline: 67.7% in 2015 Target: 100% by 2030
LRI pillar: Adaptive Capacity Resilience				
Agricultural productivity			SAGCOT Green Print 2012: Generate an additional 2.2 million tons of rice and field crops per year	Baseline: none found Target: none found
Carbon stored in forest and agricultural land including associated emissions reductions			SAGCOT Green Print 2012: Carbon contained in these forests, combined with emissions reductions associated with AGG, will generate net emissions reductions of about three million tons of CO ² -equivalent per year by 2030	Baseline: none found Target: three million tons of CO ² -equivalent per year by 2030
			Changwon Initiative: Increase soil organic carbon in cropland	Baseline: none found Target: 54.5tons/ha by 2030
Adoption of sustainable land management practices, certification and agroecological markets			Tanzania National Ecological Organic Agriculture Strategy 2023-2030: Number of farmers practicing Ecological Organic Agriculture (EOA)	Baseline: 1,000,000 in 2023 Target: 3,000,000 Farmers received and gained knowledge on EOA Practices by 2030, led by Tanzania Organic Agriculture Movement
			National Climate Change Response Strategy (2021-2026): Improved number of ha of agricultural land	Baseline: None found Target: At least 50% of the community practicing climate smart for agricultural land management by 2026
			Changwon Initiative: Restore forests through sustainable forest management	Baseline: None found

Commented [GU13]: are there further information in Annexe 1?

Commented [GU14]: i assume these are the LRI indicators?

Commented [GU15]: Will, I only see NDCs to be mentioned once in this table. I remember they had a verification session for the third version and for that I did a quick analysis for Anthony. but I recall there was further alignment, please have a look and update the table accordingly: [SUSTAIN_NDCs.docx](#)

Commented [GU16]: This is not included in the analysed policies above? unless it was under a different name

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

Healthy and diverse ecosystems	SDG VNR 15.1.1: Area of natural forest cover (Ha)	Target: 11,011,950 ha by 2030 Baseline: 28,797,926 in 2017 Target: none found
	Bonn Challenge: Restore 5.2 million hectares of land by 2030	Baseline: none found Target: 5.2 mn ha in restoration by 2030 Baseline: none found Target: 1.5 million trees must be planted in each district each year until 2026
	National Biodiversity Strategy and Action Plan 2015-2020 Strategic goal D, target 14: "by 2020, ecosystems that provide essential services, related to water, and contribute to health, livelihoods and well-being, are restored and safeguarded, taking into account the needs of women, local and vulnerable communities."	Baseline: none found Target: none found
	National Climate Change Response Strategy (2021-2026): Percent of forest communities implementing nature-based solution initiative	Baseline: None found Target: At least 3 million ha of degraded forest land restored by 2026
	Resilient communities to shocks, disasters and climate change	Baseline: none found Target: none found
	SDG VNR 1:2:1 Proportion of population living below the national poverty line	Baseline: 31.3% rural in 2017 Target: 0% poverty by 2030
Poverty rate	AGCOT/SAGCOT: Lift 2 million out of poverty	Baseline: none found Target: 2 million by 2030

Commented [GU17]: is this a sub-target to the 5.2 million one?

MOZAMBIQUE

OVERVIEW OF POLICIES AND RELATION TO SUSTAIN LRI

Mozambique Voluntary National Review 2020 (República de Moçambique, Ministério das Finanças 2020)

The 2020 report is the most recent available Voluntary National Review of the Sustainable Development Goals for Mozambique. The SUSTAIN LRI relates to specific indicators for SDGs 1 No Poverty, 2 Zero Hunger, 4 Quality Education, 5 Gender equality, 6 Clean Water and Sanitation, 8 Decent Work and Economic Growth, 13 Climate Action and 15 Life on Land.

Indicators were linked to National goals through a National Framework of SDG indicators, however, the only indicator relating to 13 Climate Action is the ‘*Number of deaths, missing persons and directly affected persons attributed to disasters per 100,000 population*’, and there are no indicators relating to 15 Life on Land. However, Mozambique ranks third among the African countries most exposed to climate-related risks and suffers from cyclones, droughts, floods and epidemics periodically that are becoming more frequent and intense.

Despite having no indicators for 15 Life on Land, the VNR reports that biodiversity represented the livelihoods of over 90% of the human population for food, health, housing and energy. Deforestation has been a recent issue, with Mozambique losing 7.3% of its forest cover in just 3 years between 2015 and 2018. Agriculture is listed as the main driver of deforestation. Moreover, 46.1% of the population in Mozambique live below the poverty line, with that increasing to 53.1% for those living in rural areas.

Update of the First Nationally Determined Contribution to the United Nations Framework Convention on Climate Change, Mozambique 2020-2025 (República de Moçambique, Ministério da Terra e Ambiente 2021)

Mozambique’s updated NDC (2021/2022) outlines both mitigation and adaptation pathways, recognising its high vulnerability to climate impacts and limited historical emissions. It commits to reducing about 40 million tCO₂-eq between 2020 and 2025, conditional on international financial, technological, and capacity-building support. On the adaptation side, the plan emphasises strengthening resilience across key sectors, including agriculture, forests (especially mangroves), coastal zones, water resources, early warning systems, and infrastructure. Critically, Mozambique links mitigation and adaptation through NbS (e.g., sustainable land use, reforestation, and mangrove restoration), which fosters landscape resilience by restoring ecosystem function, enhancing carbon sinks, and reducing vulnerability to climate risks.

The NDC also establishes a strengthened MRV (measurement, reporting, and verification) system to track progress transparently. However, the total finance need is very large (approx. US\$ 7.6 billion for 2020-2025) and there is a significant funding gap, highlighting the challenge of operationalising landscape-level resilience in a resource-constrained context.

The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 (NBSAP: 2015-2035) (MITADER 2015)

The vision of the strategy is that ‘*In 2035, the ecological, socio- economic and cultural value of biodiversity in Mozambique will contribute directly to improve the quality of life of Mozambicans, derived from its integrated management, conservation and fair and equitable use*’. This strategy defined strategic objectives, established targets in line with Agenda 2030 and other international commitments and a detailed action plan and

Commented [GU18]: did we look at the NBSAPs for Tanzania? if not, can we make sure it is included in the analysis.

its implementation involves various development actors. Mozambique has focussed on in-situ conservation measures and 26% of the country is covered by Conservation Areas, 13 inland and 2 marine. States how biodiversity is fundamental to poverty alleviation and economic development within the country, particularly for those in rural areas. The strategy also recognises the importance of agrarian biodiversity for both ecological and socio-economic importance.

Mozambique's National Adaptation Plan (República de Moçambique 2023)

Highlights that climate change may lead to changes that can be monitored in the SUSTAIN LRI such as (1) decrease quantity of quality water available for various uses, (2) greater risk of loss of crops, forests, and soil erosion and (3) decreased soil fertility. The plan creates a framework for integrated climate change adaptation action for its implementation. The plan highlights the need for a robust monitoring and evaluation plan to track progress towards adaptation goals that is operated by the National System for Monitoring and Evaluation of Climate Change. The SUSTAIN LRI could be aligned and fed into this. The plan states that development partners could align technical and financial interventions with the adaptation measures and actions defined in the National Adaptation plan so that stakeholders work to a common objective, the LRI could help facilitate this.

National Water Policy (República de Moçambique, Conselho de Ministros 1995)

The National Water policy 1995 has a decentralised approach to water management. It assigns water resource management to basin and provincial level organisational structures and emphasises integrated water resource management priorities. Emerging issues in the sector including climate change adaptation and ecosystems approaches to water management which aligns with the LRI. A key objective is to meet the basic needs of the people in terms of water and sanitation, in particular for rural people and low-income groups.

National Development Strategy (ENDe) 2015 -2035 (República de Moçambique 2014)

This strategy aims to guide the country's development over the next two decades. The main objective of the ENDe is *"to improve the living conditions of the population through structural transformation of the economy, expansion and diversification of the productive base."* The ENDe underlines the need for resilience by acknowledging Mozambique's vulnerabilities especially to demographic pressures, climate shocks, and regional inequality, and by promoting inclusive growth that reduces inequalities across regions. It therefore aims to prioritise creating resilience in vulnerable sectors such as agriculture. Agriculture and mining are both priority areas for national development within the strategy.

Bonn challenge (Republic of Mozambique / IUCN 2016)

Mozambique has pledged 1,000,000 ha to restoration by 2030 under the Bonn Challenge, 1.27% of its land area.

National Environmental Policy (República de Moçambique, Conselho de Ministros 1995)

This policy provides the overarching framework for national environment plans and legislation, with the goal of reconciling development with environmental protection. It led to the creation of a Ministry for Co-ordination of Environmental Action and is the basis of the Environment Law (No 20/97, 1 October). It has no specific objectives and indicators that can be related to the SUSTAIN LRI.

Strategic Plan for the Development of the Agrarian Sector 2030 (PEDSA II) (MADER 2022)

This plan aims to accelerate the transformation of Mozambique's agricultural sector by modernising production, enhancing value chains, promoting sustainable resource management and improving competitiveness. The plan incorporates climate adaption strategies such as agroforestry, conservation

Commented [GU20]: no recent updates? was it also decentralised perhaps to provincial levels?

Commented [GU19]: it has not been updated since 1995? or are there more recent policies on the basin/provincial level of Barué and Vanduzi or the for the Manica province, if yes then we should account for it.

agriculture, sustainable water management as key strategic actions to achieve its goals. Priority value chains include oilseeds, beans, cereals, cassava, potato, sugar, cashew and macadamia nuts, vegetables, tobacco.

Strategic objective 1 is to promote the accelerated transformation of the agricultural sector through rapid, competitive, inclusive, and sustainable growth, ensuring greater inclusive engagement of the private sector, and contributing to improving food and nutrition security, strengthening disaster resilience, and reducing poverty. Specific goals of this objective which relate to the SUSTAIN ~~LRI~~^{HMI} include:

- Guaranteed annual growth of at least 10.0% in livestock production of strategic animal species, including their by-products (e.g. eggs and milk).⁺
- Doubling the average agricultural productivity of strategic food and cash crops.⁺
- Increased reforested area (native forest and forest plantations) up to one million ha.
- Irrigated area increased by 22 thousand (medium) and 9,900 (low) thousand hectares.

Strategic objective 2 is to double agricultural productivity levels by focusing on the active inclusion of family farmers in functional value chains using research and technological innovation in a sustainable and climate-resilient manner.

Specific goals of this objective which relate to the SUSTAIN ~~LHMI~~^{LRI} include:

- Ensure that at least 30.0% of arable land benefits from sustainable and climate-resilient management practices.
- Ensure 0.05% of forests are restored.
- Ensure that at least 30.0% of agricultural holdings are resistant to shocks and disasters.

Table 3. Implementation responsibility for each policy analysed for Mozambique

Policy	Responsibility
Strategic Plan for the Development of the Agrarian Sector 2030 (PEDSA II)	Agricultural Sector Coordination Committee composed to public and private organisations, civil society organisations, academia and research organisations, cooperation and development partners. Coordinated by the Ministry of Agriculture. Monitoring progress in implementing the PEDSA II and its achievements is designated at the national, sectoral, provincial, district, local authority, and lower levels of government administration.
National Environmental Policy (1995)	Ministry of Land, Environment and Rural Development
Bonn Challenge	Unclear
National Development Strategy (ENDe) 2015 -2035	Ministry of Economy and Finance (MEF)
National Water Policy (1995)	Ministry of Public Works, Housing, and Water Resources
Mozambique's National Adaptation plan (2023)	National Directorate of Climate Change within the Ministry of Land and Environment
The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 (NBSAP: 2015-2035)	Ministry of Land, Environment and Rural Development
Update of the First Nationally Determined Contribution to the United Nations Framework Convention on Climate Change, Mozambique 2020-2025	National Directorate of Climate Change within the Ministry of Land and Environment
Mozambique Voluntary National Review 2020	DG National Reference Group (GRN-SDG)

Commented [GU21]: Are there specific policies/strategies that look at disaster risk reduction? if yes, then we should include it for both countries.

Commented [GU22]: Where does the work of the MRV team we connected with fall? - if not under the NDCs and rather under another strategy/policy we should include an analysis/reference to it.

RESULTS OF POLICY REVIEW

Some key themes emerged from the Mozambique policy analysis that can be used to help identify alignment of the SUSTAIN [LHAMU LRI](#) with national frameworks and policies and provide insight into implementation mechanisms at the national scale. There is a large focus on climate adaptation and [resilience](#) across many policies, perhaps given Mozambique's susceptibility to extreme weather. Mozambique's National adaption plan highlights the need for a robust monitoring and evaluation plan to track progress towards adaptation goals that monitor impacts such as (1) decrease quantity or quality water available for various uses, (2) greater risk of loss of crops, forests, and soil erosion and (3) decreased soil fertility. This aligns well with many of the SUSTAIN [LHAMU LRI](#) indicators.

The Strategic Plan for the Development of the Agrarian Sector 2030 (MADER 2022) and the Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 (MITADER 2015) in particular, have many indicators and targets that are relevant for the LRI national indicators (Table 5). Environmental and biodiversity policies are all framed within the context of using natural resources for economic growth and poverty alleviation through agriculture.

Water is a core element running through all of Mozambique's key policies. The National Water Policy provides the foundation for integrated water resources management, while the NDC and National Environmental

Policy position water as both a climate risk and a resilience opportunity through watershed restoration, drought management and coastal protection. The National Development Strategy (ENDe) links sustainable water use to economic transformation, agricultural productivity and inclusive regional development. Even the SDG Voluntary Review highlights water security as essential for health, livelihoods and environmental sustainability. Together, these policies show that effective water management is central to building resilient landscapes that can sustain ecosystems, communities and long-term development.

Agriculture is a priority area within the National Development strategy (República de Moçambique 2014), which also recognises climate change may hinder development due to an increase in extreme weather events. The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 (MITADER 2015) also recognises the importance of agrarian biodiversity for both ecological and socio-economic importance.

RELATION TO LRI NATIONAL INDICATORS

Building on the policy analysis, we have made suggestions as to how the [LHAMU LRI](#) relates to national indicators in Table 4. This demonstrates how site and landscape monitoring can be contextualised within broader national and international policies and commitments. The table shows specific national targets and indicators that relate to the proposed national level indicators for the SUSTAIN [LHAMU LRI](#) in Mozambique.

Commented [GU23]: the same comments i left about for the Tanz. section apply here

Commented [GU24]: I am not sure when you say resilience here if you refer to overall landscape resilience as we define it or if you mean the third pillar "adaptive capacity" which was previously called "resilience". just to be clear and avoid confusion. since you speak about the water pillar afterwards

Table 4. An overview of some key national indicator suggestions for Mozambique that resulted from the national policy review including a short summary and the specific policy relevance. See Annex 1 for the detailed national policy analysis.

LRI pillar	National scale indicator	Key quantitative national targets and indicators	Baseline and target values
LRI pillar: Water			
Water quality and quantity		SDG indicator 6.3: Goal 6.3 By 2030, improve water quality by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally	Baseline: 0.65 m3/s of wastewater safely treated in 2017 Target: 6 m3/s of wastewater safely treated by 2030
		SDG indicator 6.4: Goal 6.4 By 2030, substantially increase water-use efficiency across all sectors and ensure sustainable collection and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity	Baseline: 6.45 usd/m3 change in water use efficiency over time in 2017 Target: 9.4 usd/m3 change in water use efficiency over time by 2030
		The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035: Reduce the pollution levels at critical locations and ecosystems (MITADER 2015)	Baseline: None found Target: Reduce the pollution levels at critical locations and ecosystems by 20% by 2025
Access to quality water for different uses: drinking water, sanitation, irrigation.		SDG indicator 6.1.1: Proportion of population using safely managed drinking water services	Baseline: 45% in rural areas in 2018 Target: 100% by 2030
LRI pillar: Adaptive Capacity			
Agricultural productivity		Strategic Plan for the Development of the Agrarian Sector 2030: Double agricultural productivity tons/ha (MADER 2022)	Maize Baseline: 1 tons/ha in 2022 Target: 2 tons/ha in 2030 Soy Baseline: 1.5 tons/ha in 2022 Target: 3 tons/ha in 2030
Carbon stored in forest and agricultural land including associated emissions reductions		Strategic Plan for the Development of the Agrarian Sector 2030: Contribution of the agricultural sector to reducing greenhouse gas emissions: Agricultural and Forest (MADER 2022)	Agriculture Baseline: 17.3 MtCO2 in 2022 Target: 9.8 MtCO2 by 2030 Forest Baseline: 12 MtCO2 in 2022 Target: 6.8 MtCO2 by 2030
Adoption of sustainable land management practices, certification and agroecological markets		The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035: % of agricultural area land benefiting sustainable management practices (MITADER 2015).	Baseline: none found Target: 30% by 2030

Healthy and diverse ecosystems.	The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035: % of forest area land benefiting sustainable management practices (MITADER 2015).	Baseline: none found Target: 0.05% by 2030
	Strategic Plan for the Development of the Agrarian Sector 2030: Percentage area under climate-smart agriculture (MADER 2022)	Baseline: 1% in 2022 Target: 30% by 2030
	The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035: By 2035, rehabilitate at least 15% of the degraded ecosystems /habitats, restoring their biodiversity and ensuring its sustainability, with a view to mitigating the effects of climate change and combating desertification (MITADER 2015).	Baseline: None found Target: 15% of degraded ecosystems by 2035
	Bonn Challenge: 1,000,000 ha to restoration by 2030 under the Bonn Challenge, 1.27% of its land area	Baseline: None found Target: 1,000,000 ha by 2030
Resilient communities to shocks, disasters, and climate change	Strategic Plan for the Development of the Agrarian Sector 2030: % of producers resilience to shocks and disasters (MADER 2022)	Baseline: 15% of agricultural producers in 2022 Target: 30% of agricultural producers by 2030
	Strategic Plan for the Development of the Agrarian Sector 2030: % of producers resilient to climate change (MADER 2022)	Baseline: 15% of agricultural producers in 2022 Target: 30% of agricultural producers by 2030
Poverty rate	Strategic Plan for the Development of the Agrarian Sector 2030; National Development Policy: Poverty rate % (MADER 2022; República de Moçambique 2014)	Baseline: 46% in 2022 Target: 40% by 2030
	SDG VNR Goal 1.1: Proportion of population living below the national poverty line	Baseline: 53.5% in 2015 Target: 0% by 2030

METHODOLOGY

METHODS SUMMARY

The SUSTAIN LRI indicators, initially proposed by Durrant et al. (2024), were first reviewed and modified to ensure their relevance to the landscapes and to strengthen their link to water and climate resilience. Subsequently, secondary data collection began to conduct baseline assessments for all landscapes. To supplement data gaps, primary data collection was conducted in Mozambique and Tanzania. Primary data collection was focused in areas of the landscapes where IUCN SUSTAIN is active.

A participatory workshop involving IUCN staff members and local stakeholders was held in Mozambique on Tuesday, August 26th 2025. The workshop's purpose was to determine which relevant data had been previously collected, was pending collection, or could be collected by local stakeholders active in Vanduzi and Barue districts.

To ensure the SUSTAIN LRI aligns with national policy and commitments, a review of key national policies, targets, and indicators was conducted. For Tanzania, the policy review from Durrant et al. (2024) was updated to reflect changes to the Tanzania policy landscape, including the expansion of the SAGCOT to other regions of Tanzania with the establishment of Agricultural Growth Corridors of Tanzania (AGCOT). A qualitative summary is provided to show how national policies and commitments relate to the SUSTAIN LRI. The SUSTAIN LRI's national-level indicators were also linked to specific national targets to which they align and to which they can contribute. Where possible, national-level baseline data and targets have been included.

REVIEW AND ADAPTATION OF ORIGINAL LRI INDICATORS FROM PHASE I

The original Landscape Resilience Index comprised 17 indicators across three pillars: land, water, and adaptive capacity for direct monitoring at the site and landscape level. The selection of these specific indicators was

based on their scalability, feasibility, cost-effectiveness, and required skill level for data collection, aiming to ensure that data collection for the LRI was both financially and practically viable. In preparation for this study, we comprehensively reviewed and revised the proposed LRI indicators. This revision incorporated valuable feedback from IUCN staff and Tanzanian stakeholders, who emphasized a greater focus on water and climate resilience. We also ensured that all indicators are relevant and feasible for both Tanzania and Mozambique, aligning them with ongoing activities to facilitate practical LRI implementation. The national level indicators were specific to Tanzania so have been broadened to be relevant for both countries' policy landscape. Whilst we do not collect data at the national scale, these indicators are included to provide insight on how LRI aligns with and can be implemented within existing national policies and strategies within each country.

ADAPTING LRI TO EXISTING DATA AND CONTEXT

Discussions with IUCN local offices revealed that substantial relevant data had already been collected for each of the SUSTAIN LRI pillars, though not always in the exact format of our previously proposed indicators. Recognising that primary data collection is extremely costly and that this existing data is highly relevant to the broader concept of landscape resilience, we revised the LRI (table 1) to include existing data where possible.

We propose an updated framework using indicators and sub-indicators that are no longer limited to specific units of measurement (e.g. bird species richness, soil pH, soil C:N etc), making it adaptable to diverse contexts. Consequently, our first key revision was to broaden the original set of LRI indicators to represent overarching themes. This allows for a variety of different sub-indicators to be included depending on data availability and the specific landscape context and also ensures the LRI can be updated in the future to incorporate newly collected data.

For example, the indicators *soil C:N ratio* and *soil pH* have been replaced by the indicator *soil fertility*. In Kilombero, the soil fertility indicator consists of secondary data on soil pH; C:N ratio, total nitrogen and available phosphorus, whereas in Mozambique soil pH and electrical conductivity are used.

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Where secondary data was not available and field sampling was required, the sub-indicators were chosen from the original indicator list. For example, the above ground biodiversity indicator is comprised of bird

species richness surveys in Ihemi, Sumbawanga and Barue-Vanduzi landscapes.

THE UPDATED LRI

Table 5. The update Landscape Resilience Index

Pillar	Indicator
Site	
Land	Above-ground biodiversity
	Below-ground biodiversity
	Carbon stocks
	Soil fertility
Water	Soil-water availability
	Water quality: biological activity
	Water quality: chemical properties
Landscape	
Water	Landscape water access
	Landscape water availability
	Water quality: chemical properties
	Adoption of Sustainable Land Management (SLM) practices
	Agricultural, agroforestry and tree plantation productivity
Adaptive capacity	Extreme weather
	Household poverty
	Landcover percentage
	Landscape biodiversity
	Perception of shocks and stresses
National	
Water	Water quality and quantity
	Access to quality water for different uses: drinking water, sanitation, irrigation.
	Agricultural productivity
Adaptive capacity	Carbon stored in forest and agricultural land including associated emissions reductions
	Adoption of sustainable land management practices, certification and agroecological markets
	Healthy and diverse ecosystems.
	Resilient communities to shocks, disasters and climate change
	Poverty rate

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SITE: LAND INDICATORS

The original LRI from phase 1 included 7 indicators for the land pillar measured at the site scale (Table S. 1). This pillar has been restructured to 4 indicators with the original indicators included as sub-indicators:

Above-ground biodiversity

(bird, mammals and invertebrate species richness)

Bird species richness is an ideal indicator for ecosystem health because they are sensitive to land use change (Davis et al., 2024) and are relatively easy to detect and identify, making them comparatively cheaper than other indicator species. Bird species population dynamics, behaviour, life history and ecosystem functions are well studied and given they are sensitive to disturbance at lower trophic levels the composition of a landscape's bird species can indicate functional diversity and ecosystem service/disservice delivery. Ecosystem services/disservices related to bird species include as predation, pollinators, scavengers, seed dispersers, seed predation, and ecosystem engineers (Whelan et al., 2008).

Invertebrates are integral to many ecosystem services including pollination, pest control, nutrient cycling and regulation of water (Quiroga, 2023). Macroinvertebrates that live within the soil are useful indicator species of soil health (Stork and Eggleton, 1992). If a soil has poor biological, chemical and physical properties, this will be reflected in the diversity of macroinvertebrates present. Pollination is a key ecosystem service that benefits livelihoods, in one study in northern Tanzania close to 65% of crop species are at least partially dependent on insect pollinators (Sawe et al. 2020).

Mammals occupy diverse ecological niches (Mazel et al., 2017) and support critical ecosystem services (Lacher et al., 2019), including seed dispersal (Matias et al., 2010), nutrient cycling (Abraham, 2021), and population control of prey species (Prevedello et al., 2013), ensuring functional redundancy that enhances resilience.

They often occupy higher trophic levels which require large, connected habitats and so are sensitive to environmental changes such as habitat degradation or human disturbance (Gorczyński et al., 2021), making shifts in their richness an early warning of ecosystem stress.

Together, these factors make mammal species richness a reliable proxy for an ecosystem's capacity to withstand and recover from disturbances.

Below-ground biodiversity

(earth worm count)

Earthworms contribute to decomposition of organic matter (Schon et al., 2020), which increases nutrient availability (Bertrand et al., 2015; Blouin et al., 2013), improves soil structure (Joschko et al., 1989) all of which benefits above ground primary productivity. They are also an important food source for many species (Scheu, 2001) and can be sampled relatively easily making them a suitable indicator of soil quality (Fründ et al., 2010).

Carbon stocks

(above-ground and below-ground carbon)

Carbon stocks which encompass both above-ground carbon and soil organic carbon, is an estimate of the total carbon stored within an area, reflecting multiple ecosystem services and serving as a critical component of climate change mitigation.

Soils alone store two to three times more carbon than is present in the atmosphere as CO₂ (Minasny et al., 2017). Soil organic carbon influences virtually all chemical, physical, and biological soil properties including microbial diversity, pH buffering, aggregate stability, nutrient cycling, and water retention (Libohova et al., 2018; Murphy, 2015). By enhancing the retention of water and nutrients, soil organic carbon improves soil quality and boosts plant productivity, while reductions in soil organic matter can diminish rainfall infiltration and soil moisture storage, both of which are important for flood mitigation (Ontl & Schulte, 2012).

Plants also sequester atmospheric CO₂, and above-ground carbon stocks have been positively linked with tree species diversity (McNicol et al., 2020; Sintayehu et al., 2020).

Soil fertility

(electrical conductivity, pH, total nitrogen, soil phosphorus and soil C:N ratio)

The soil fertility indicator is applied primarily to agricultural soils, as fertility is a subjective term whose interpretation depends on an intended objective which primarily is, crop production. Intensive agriculture can disrupt critical soil processes and deplete essential nutrient stocks. For example, total soil nitrogen is an important indicator of the soil's nitrogen-supplying capacity (Al-Kaisi et al., 2005) and is often significantly lower in recently developed intensive farmland compared with natural and semi-natural soils (Leul et al., 2023). Soil phosphorus loss, driven largely by erosion (Carpenter et al., 2010) following extreme rainfall events (Quinton et al., 2011), is expected to constrain future agricultural production, particularly in regions such as Africa, South America, and Eastern Europe where reserves of critical P-bearing minerals are already limited (Alewell et al., 2020). Intensive land use has also been associated with soil acidification (Zamanian et al., 2024), a major concern given the key regulatory role of pH in microbial activity (Malik et al., 2018), root functioning, nutrient cycling, and soil organic carbon (SOC) accumulation (Neina, 2019).

The soil carbon-to-nitrogen ratio provides insight into the balance of carbon and nitrogen, influencing microbial activity and the decomposition of organic matter (Brust, 2019). Similarly, soil pH governs a wide range of biogeochemical processes, including mineralisation, ammonia volatilisation, nitrification, denitrification, and the solubility of nutrient and toxic ions (Neina, 2019). Soil electrical conductivity (EC) further reflects salinity levels that affect plant growth and microbial communities and is widely used to guide soil nutrient and water management (Corwin & Leech, 2003).

SITE: WATER INDICATORS

The water pillar indicators assessed at the site level have been restructured to reflect different country contexts and data availability.

Soil-water ~~content~~ availability

(soil moisture, plant-available water content; volumetric water content soil)

Soil moisture, plant-available water content, and volumetric water content are all measures related to the amount of water held in soil. Soil moisture, typically measured *in situ* using sensors, reflects water content at a specific point in time and is highly sensitive to irrigation and precipitation patterns. In contrast, volumetric water content is measured in the laboratory and indicates the soil's water-holding capacity at a given pressure (typically field capacity). Plant-available water content is also determined in the lab and is commonly estimated as the difference in volumetric water content between field capacity (around 30 kPa) and the permanent wilting point (around 100 kPa).

Water quality: Small stream biological activity

(TARISSfupi, MiniSASS)

The macroinvertebrate assemblages are assessed using two similar protocols: the Mini Stream Assessment Scoring System (miniSASS) and the Tanzania River Scoring System (TARISSfupi). TARISSfupi and MiniSASS are simple rapid bioassessment methods that can be used to monitor the health of a river and measure the general quality of the water in that river. They use the composition of macroinvertebrates living in rivers (E.g. Crabs, Snails, Dragonflies and Stoneflies etc). The main differences between TARISSfupi and miniSASS lie in the sensitivity scores assigned to stoneflies (Table S. 2) and the inclusion of Brush legged Mayflies and Oval Mayflies in TARISSfupi, which are not included in miniSASS.

These methods do not measure the contamination of the water by bacteria and viruses and thus does not determine if the river water is fit to drink. However, the macroinvertebrate assemblage will be a product of key water properties and so water quality is reflected in the diversity of macroinvertebrates (Alavaisha, Lyon & Lindborg 2019).

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Although TARISSfupi was an original indicator from phase 1 and remains the sub-indicator for Sumbawanga, for Barue-Vanduzi, the sub-indicator will be the miniSASS. For Ihemi and Kilombero, we will also use miniSASS assessments, since this data was already available at <https://minisass.org/>.

Site level water chemistry

(pH, electrical conductivity and total dissolved solids)

This indicator has been added to reflect a deeper focus on water in the LRI.

Water pH, electrical conductivity (EC) and total dissolved solids (ppm) are critical indicators of water quality and ecosystem resilience.

Water pH affects nutrient availability and organism survival (Pinheiro *et al.*, 2021), while EC reflects dissolved salts and can rise due to fertilizer leaching or runoff, limiting irrigation use (Yan *et al.*, 2024). Dissolved solids, influenced by soil erosion and mineral runoff, indicate total solute load, and turbidity signals suspended particles that reduce light and harm aquatic life (Weber-Scannell *et al.*, 2007). Together, these measures reveal water suitability for ecosystems and human use, with changes indicating stress that can reduce resilience.

In Tanzania, the site-level water chemical properties indicator were collected in streams of Strahler order 2 or 3. In Mozambique, there was a noticeable lack of rivers in close proximity to the sampling sites, so the site level water chemical properties was obtained from man-made wells, typically dug in farmland.

LANDSCAPE: WATER INDICATORS

Landscape water availability

(River discharge)

A new indicator on water availability has been introduced at the landscape level.

River discharge is a key indicator of landscape water availability, representing water that is not bound to soils or vegetation and is thus available for ecosystem processes and human use. Historical and real-time discharge measurements are essential for managing water resources, supporting applications such as flood protection, irrigation, and environmental flows.

This will be assessed using river discharge (i.e. flow rate, m³/S) estimates from the GEOGLOWS (Group on Earth Observations Global Water Sustainability) river forecast system. This indicator replaced the *depth to water table* as access to wells/data is not certain without visiting the landscape.

Landscape water access

(household consumption, irrigation)

Interventions to increase the access to water require infrastructure development that occur at the landscape scale; hence this is a landscape-level indicator.

Access to water will influence the system's ability to cope with stress and shocks and can have outcomes for human health and wellbeing. This is explicitly discussed in SDG 6 Clean Water and Sanitation, which states that 'access to safe water, sanitation and hygiene is the most basic human need for health and well-being' (United Nations, 2024).

Landscape level water chemistry

(pH, electrical conductivity (EC) and total dissolved solids (ppm))

This was previously proposed as TARISSfupi for large rivers. However, given its impracticality for deep rivers we have changed this indicator to focus on water chemistry, similar to that at the site-level. The sub-indicators will consist of measurements of pH, electrical conductivity (EC) and total dissolved solids (TDS). In Mozambique, water chemistry in surface waters (irrigation channels and rivers) were used as the landscape-level indicators, whereas in Tanzania, the landscape-level water chemistry was determined using rivers of Strahler order 4 and above.

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LANDSCAPE: ADAPTIVE CAPACITY INDICATORS

Several changes were made to improve feasibility and regional relevance of the indicators.

Agricultural, agroforestry and tree plantation productivity

(maize, rice, sunflower, beans, sugarcane, soybean, pigeon pea, green bean, cabbage)

Agricultural, agroforestry and tree plantation productivity is used as a measure of economic productivity, upon recommendation of stakeholders. Crop yield is a standard measurement of the amount of agricultural production harvested per unit of land area. It is an important indicator of food security, and it can be used to compare the relative performance of different crops under different management and the effectiveness of sustainable management practices (Shah and Wu, 2019).

Adoption of Sustainable Land Management (SLM) practices

(soil fertility, soil erosion, apply manure, crop rotation, add compost, agroforestry)

Sustainable land management (SLM) practices give insights into the sustainability of current management activities practised and highlight specific areas for improvement. For instance, irrigation of pasture or rangelands may increase carbon levels beyond historic soil organic carbon (SOC) stocks (Ontl & Schulte 2012) while reducing tillage, controlling erosion, applying organic amendments and using cover crops can also influence soil organic carbon by either increasing it or reducing loss (Ontl & Schulte 2012).

Household poverty

(poverty probability index; insufficient income to buy food)

Including household poverty in the LRI acknowledges and allows exploration of the explicit link between landscape resilience and poverty. Reducing poverty is a major goal of environmental and sustainable development policy, links to national targets and is monitored within SUSTAIN's LRI framework.

Land cover percentage

(percentage tree cover loss)

A variety of species, along with the ecosystem services they offer, depend on the ability to move between ecosystems (Mitchell et al., 2013) to access key resources such as water food, cover, breeding sites and security (Ministry of Natural Resources and Tourism, 2022). Monitoring land cover percentage and tree cover loss in particular, can identify threats to the processes that support biodiversity and can be used to prioritise conservation and assess sustainability of land use. This indicator can be assessed using open-source satellite data. It will replace Natural ecosystem connectivity, previously derived from secondary data on wildlife corridors in Tanzania, as it was not a feasible indicator for Mozambique.

Landscape Biodiversity

(Biodiversity Habitat Index)

Biodiversity is important for resilience and security, social relations, health and freedom of choices and actions (Millennium Ecosystem Assessment, 2005), in addition to its contribution to goods and other material provisions that have economic value for livelihoods (Hanley and Perrings, 2019). The Biodiversity Habitat Index from the UN Biodiversity Lab is the sub-indicator. The originally proposed indicator—percentage of critically endangered species—was questioned by stakeholders, due to concerns over the accuracy of species lists in the Ihemi and Kilombero clusters and the limited likelihood of detecting measurable change. The BHI, on the other hand, employs biologically scaled environmental modelling to estimate the impacts of habitat loss, degradation, and fragmentation on the retention of terrestrial biodiversity worldwide. It draws on remotely sensed forest and land-cover change data. Importantly, the BHI is updated every five years (from 2000 to 2020), making it a practical tool for assessing current conditions relative to historical baselines.

Perceptions of shocks and stresses

(drought, flooding)

The perception of shocks and stresses is the percentage of those surveyed who are experiencing different shocks and stresses. Monitoring perception over time can indicate the effectiveness of different NbS interventions and is far more feasible than collecting data on individual shocks and stresses. This will continue to be assessed via household surveys, but now a weighted average of significant shocks or stressors from drought and flooding will form the baseline metric, rather than focusing solely on flooding.

Extreme weather

(drought, flooding)

The impact of extreme weather events is an important consideration when determining landscape resilience. In extreme weather events, farmers report crop losses that can reduce household food security (Ghambi et al., 2024, Durrant et al., 2024b). The extreme weather indicator, previously based on data from the Tanzania Meteorological Authority, was found to be unscalable to Mozambique. It has been replaced with a new sub-indicator: proportion of land area affected by flooding, using fortnightly data from the Humanitarian Data Exchange (HDX).

To account for extreme weather events in drier landscapes where periods of drought are more typical, the proportion of land area affected by drought will be used as a sub-indicator.

NATIONAL: WATER INDICATORS

The indicators represent where the LRIHMI can feed into national and international targets. These are:

Water quality and quantity at the national level

Critical for informing national policies which aim to ensure optimal, reliable, and sustainable water development in the face of rising demand and quality deterioration. At the national level, this indicator expands to include water and wastewater. This is a specific target of SDG 6 Clean Water and Sanitation. For Tanzania, the National Water

policy has a specific goal to improve water and wastewater quality and management. For Mozambique, pollution levels are a focus.

Access to quality water for different uses at the national level

Essential for measuring progress on Sustainable Development Goals (SDGs) and national development targets (Mozambique and Tanzania). For example, monitoring irrigation access is key to mitigating increasingly extreme weather and boosting agricultural productivity in both nations. At the national level, this indicator is expanded to also include sanitation and irrigation. Improving water access for different uses is a specific target of SDG 6 Clean Water and Sanitation. For Tanzania, access to water is a target of the SDG indicators and their Nationally Determined Contribution 2021.

NATIONAL: RESILIENCE — ADAPTIVE CAPACITY INDICATORS

Agricultural, agroforestry and tree plantation productivity

A major priority for both nations' economies and for poverty reduction, as agricultural development contributes significantly to their respective Gross Domestic Products and provides livelihoods. At the national level, this is to be expanded for any nationally relevant crops. For Tanzania, increasing agricultural productivity in an environmentally sustainable way is a focus of their Nationally Determined Contribution 2021 and SAGCOT/AGCOT. Mozambique has ambitions to double the productivity of their agricultural sector in tons/ha.

Carbon stored in forest and agricultural land including associated emissions reductions

Critical for meeting Nationally Determined Contributions (NDCs), supporting REDD+ national readiness strategies, and attracting climate finance by valuing forests and agriculture as investable assets. At the national level, this indicator reflects country-level emission reductions for the overall

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for me this communicates more on the level of readiness or preparedness of inhabitants and possibly the sectors they belong to i.e., if they work in farming at the landscape level.

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annual carbon budget and reducing emissions of the agricultural and forest sector.

Adoption of sustainable land management practices, certification and agroecological markets

Key to implementing national strategies, such as Tanzania's National Ecological Organic Agriculture Strategy and Mozambique's agrarian sector plans, to secure livelihoods and mobilize funding for the sector. This national indicator will contribute to how forests and cropland can be restored through sustainable management practices. The practices would be defined at the national level, rather than ones that are locally relevant and indicators are usually measured as percentage of land area under these practices.

Healthy and diverse ecosystems

Fundamental for poverty alleviation and socio-economic development, as biodiversity and ecosystems supports the provision of food, health, and energy. This indicator relates to national policy targets from the SDGs, Bonn Challenge, National Biodiversity and Climate Change policies, bringing together natural and restored land areas across the country.

Resilient communities to shocks, disasters and climate change

Crucial for adapting to the increasing frequency of extreme weather events (e.g., floods, droughts) that severely impact vulnerable sectors like agriculture. This indicator specifically focuses on shocks, disasters and climate change and will reflect the building of a climate-resilient society that is able to successfully adapt to the impacts of climate change.

Poverty rate

Directly linked to national development goals and core to the success of initiatives like SAGCOT, which aims to lift millions out of poverty and improve the living conditions of the population. At the national level this

would be reflective of how each country measures poverty and not necessarily use the PPI like the landscape indicators do.

INDICATOR SCORING AND CLASSIFICATION

Scoring thresholds were established for each sub-indicators. The specific thresholds for each classification were determined based on a combination of scientific literature and, where applicable, locally relevant policy standards, and are presented in the attached table.

SUB-INDICATOR SCORING

Each sub indicator was scored as poor (1), moderate (2), good (3) or excellent (4) based on the established thresholds. This systematic approach ensures a consistent evaluation of individual data points.

INDICATOR AGGREGATION

For indicators composed of multiple sub-indicators (e.g., Water Quality), the scores of the assigned sub-indicators are averaged. This average score represents the overall health of the broader indicator, providing a single, representative value.

FINAL CLASSIFICATION

The final average score for each indicator is then classified into one of four distinct categories based on the following ranges:

- **Poor:** <1.50
- **Moderate:** 1.50 - 2.49
- **Good:** 2.50 - 3.50
- **Excellent:** >3.50

DATA COLLECTION OVERVIEW

Data collection was carried out for both site and landscape-level indicators. Site-level indicators represent the most detailed scale of assessment and provide data to inform farm-level decisions and advisory services. Landscape-level indicators, on the other hand,

capture broader social, ecological, and hydrological processes that extend across farms and administrative boundaries. The indicator scales in the LRI therefore reflect the scope and type of processes they represent, rather than the precise geographic scale of data collection. Wherever possible, data were collected in areas where ~~IUCN~~-SUSTAIN is active; consequently, results should be generalised to other geographic areas with caution.

Data used in this report includes primary (field sampling) and secondary data. In Kilombero, only secondary data was used (table 2) since sufficient data was available.

For **all indicators** we searched for secondary data from publicly available, spatially explicit studies, IUCN collected data or global databases using multiple sub-indicators for each indicator, where possible.

The following indicators rely solely on secondary data across all landscapes because it has either already been collected within the landscape or global databases.

Landscape scale- water pillar

- Landscape water availability
- Landscape water access

Landscape scale- adaptive capacity pillar

- Agricultural, agroforestry and tree plantation productivity
- Sustainable land management
- Household poverty
- Land cover percentage
- Landscape biodiversity
- Perception of shocks and stresses
- Extreme weather - flooding and drought

For all other indicators, a combination of primary and secondary data was relied on, depending on data availability.

Primary data was collected through field sampling in targeted areas where secondary data was **not available, to fill data gaps** and provide specific baseline assessment of the respective indicator in a specific site in the landscape. Table 2 summaries the type of data used for each indicator and methods are detailed in the annex.

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FIELD DATA COLLECTION

MOZAMBIQUE

A field sampling campaign was conducted from 11th to 22nd August, 2025 in the Vanduzi and Barue districts of Manica, Mozambique. The campaign aimed to collect data for indicators related to below-ground biodiversity (earthworm counts), above-ground biodiversity (bird species richness), and water quality (miniSASS and chemical properties). Data for the remaining indicators was obtained from SUSTAIN Pro and/or global online datasets. The study focused on specific farmer associations and clubs previously engaged by SUSTAIN Pro. Associations are co-operatives of neighbouring farms, clubs are considered their pre-cursor, being slightly smaller and existing in dispersed patches.

In Vanduzi, sampling was concentrated in the village of Belas across five associations (7 Abril 1, 7 Abril 2, Campo 4, and Nhamadembe). In Barue, sampling took place around clubs belonging to the Samora Machel association (Chibuto, Maputo, Macossa, Nhatsanze, Matundo).

TANZANIA

A field sampling campaign was conducted from 13th - 28th October 2025 in the Ihemi landscape and between 10th - 27th November 2025 in the Sumbawanga landscape.

The campaign aimed to collect data for indicators related to below-ground biodiversity (earthworm counts), above-ground biodiversity (bird species richness), carbon stocks (above and below ground carbon stocks), soil fertility (pH, total N) and chemical (pH, TDS, EC) water quality.

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Biological water quality (TARISSfupi) was also collected in Sumbawanga.

The study focused on villages previously engaged by SUSTAIN Pro and SUSTAIN Eco.

In Ihemi, sampling was carried out across 7 villages in the Kilolo district: Ruaha Mbuyuni, Mtandika, Mahenge, Wotalisoli, Lulindi, Idete and Masisiwe.

In Sumbawanga, sampling was carried out across 4 villages in Sumbawanga district (Mshani, Kalambanzite, Kilembo, Mleche); 1 village in Nkasi district (Chala B); 2 villages in Kalambo district (Chisenga and Kafukoka); 1 village in Nsimbo district (Sitalike) and 2 villages in Mpanda district (Kakese and Kabwaga).

Table 6. Updated indicator list with example sub-indicators that may be used per landscape, and the source of data. List of example sub-indicators is not exhaustive, relevant data will be added during the project. Indicators for which field sampling is required highlighted in yellow. 1 = primary data; 2 = secondary data from literature; 3 = secondary data from on-going datasets

Indicator name	Landscape specific sub-indicator			
	Kilombero	Ihemi	Sumbawanga	Barue-Vanduzi
Site				
Land				
Above-ground biodiversity	Bird species richness ² ; invertebrate spp richness ² ; mammal spp richness ²	Bird species richness ¹	Bird species richness ¹ ; Mammals species richness ² ; Reptiles species richness ²	Bird species richness ¹
Below-ground biodiversity	Earthworm count ²	Earthworm count ¹	Earthworm count ¹	Earthworm count ¹
Carbon stocks	Soil organic carbon ² ; above ground carbon ²	Soil organic carbon ¹ ; above ground carbon ¹	Soil organic carbon ¹ ; above ground carbon ¹	Soil organic carbon ²
Soil fertility	Soil C:N ² ; soil pH ² ; soil total N ² ; soil available P ²	soil total N ¹ ; Soil pH ¹	soil total N ¹ ; Soil pH ¹	Soil pH ² ; soil EC ²
Water				
Soil-water content	Plant available water content ³	Water holding capacity ¹	Water holding capacity (g/g)	Soil moisture ²
Site-level water biological activity	Macroinvertebrate assessment (miniSASS) ²	Macroinvertebrate assessment (miniSASS) ²	Macroinvertebrate assessment (TARISSfupi) ¹	Macroinvertebrate assessment (miniSASS) ¹
Site-level water chemistry	pH ² ; EC ² ; PPM ²	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹
Landscape				
Water				
Landscape water availability	River discharge (m3/s) ³	River discharge (m3/s) ³	River discharge (m3/s) ³	River discharge (m3/s) ³
Landscape water access	Access to water for home consumption ² ;	Access to water for home consumption ² ;	Percentage of people reporting access problems ²	Access to water for home consumption ²
Landscape-level water chemistry	pH ² ; EC ² ; PPM ²	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹ ; temp ¹	pH ¹ ; EC ¹ ; ppm ¹
Adaptive capacity				
Agricultural, agroforestry and tree plantation productivity	Rice ² ; sunflower ² ; soybean ² ; sugarcane smallhold ² ; sugarcane plantation: irrigated ² ; sugarcane plantation: rainfed ²	Rice ² ; sunflower ² ; soybean ²	Rice ² ; sunflower ² ; beans ² ; maize ²	Maize ² ; soybean ² ; pigeon pea ² ; green bean ² ; cabbage ² ; potato ²
Adoption of Sustainable Land Management (SLM) practices	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost; agroforestry	Percentage of SLM practices ² : soil fertility; manure; crop rotation; soil erosion; compost

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Household poverty	PPI score ²	PPI score ²	PPI score ²	PPI score ²
Land cover percentage	Percentage tree cover loss ³	Percentage tree cover loss ³	Percentage tree cover loss ³	Percentage tree cover loss ³
Landscape biodiversity	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³	Biodiversity habitat index: proportion of species (2020) ³
Perception of shocks and stresses	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²	Percentage respondents impacted by flooding ² or drought ²
Extreme weather: flooding + drought occurrence	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought	Percent land area ² affected by flooding; percentage land area affected by drought
National				
Water				
Water quality and quantity	SDG VNR 6.3.2 National Climate Change Response Strategy (2021-2026)			SDG indicator 6.3: Goal 6.3 SDG indicator 6.4: Goal 6.4 The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035
Access to quality water for different uses.	SDG VNR 1.4.1 SDG VNR 2.4.1 Tanzania Nationally Determined Contribution 2021			SDG indicator 6.1.1
Adaptive capacity				
Agricultural productivity	SAGCOT Green Print 2012			Strategic Plan for the Development of the Agrarian Sector 2030
Carbon stored in forest and agricultural land including associated emissions reductions	SAGCOT Green Print 2012: Changwon Initiative			Strategic Plan for the Development of the Agrarian Sector 2030
Adoption of sustainable land management practices, certification and agroecological markets	Tanzania National Ecological Organic Agriculture Strategy 2023-2030 National Climate Change Response Strategy (2021-2026)			The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 Strategic Plan for the Development of the Agrarian Sector 2030
Healthy and diverse ecosystems.	SDG VNR 15.1.1 Bonn Challenge National Biodiversity Strategy and Action Plan 2015-2020 Strategic goal D, target 14 National Climate Change Response Strategy (2021-2026)			The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035 Bonn Challenge

Commented [GU37]: can we breakdown which ones are for Tanz. and which are for Moz. and the detailed indicators where possible. for all the indicators at the national level.

Thanks

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

Resilient communities to shocks, disasters and climate change	Tanzania Draft Development Vision 2050: A climate-resilience society that embraces climate-smart practices and targeted resilience building measures to adapt to the impacts of climate change	Strategic Plan for the Development of the Agrarian Sector 2030
Poverty rate	SDG VNR 1:2:1 Proportion of population living below the national poverty line AGCOT/SAGCOT: Lift 2 million out of poverty	Strategic Plan for the Development of the Agrarian Sector 2030; National Development Policy: SDG VNR Goal 1.1

DETAILED METHODOLOGY AND DATA COLLECTION OF LRI INDICATORS

SITE: LAND

INDICATOR: ABOVE GROUND BIODIVERSITY

SUB-INDICATOR: BIRD SPECIES RICHNESS

Bird species richness data was derived from point count surveys for all landscapes, following the methodology uses in Davis *et al.* (2024). Birds were identified and counted for 10 minutes following an initial 2-minute period. The number of unique bird species recorded was divided by the number of sites to get an average for each land cover type. Barue-Vanduzi, Ihemi and Sumbawanga data involved primary data collection. Kilombero bird species richness surveys was derived from Davis et al. (2024).

BARUE-VANDUZI

Bird species richness was sampled in 4 associations of Vanduzi district and 5 clubs of Samora Machel association in Barue district between 11th - 22nd August 2025. Bird surveys were conducted along the perimeter of the association/club patch at approximately 300m intervals.

KILOMBERO

Bird species richness data for the Kilombero landscape were obtained from Davis et al. (2024). In that study, bird surveys were conducted across four land uses: grassland (n = 47 plots), forest (n = 97 plots), smallholder sugarcane farms (n = 121 plots), and industrial sugarcane farms (n = 69 plots). Species richness was calculated as the number of bird species recorded per plot.

IHEMI

In Ihemi, 76-point count surveys were conducted in croplands across seven villages in Kilolo district, Iringa Region (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) from 13th - 28th October 2025.

SUMBAWANGA

In Sumbawanga, 73-point count surveys were conducted in croplands of eight villages distributed across five districts in the Sumbawanga cluster (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district - Mshani, Mleche; and Nsimbo district - Sitalike). Sampling was conducted 10th - 27th November 2025.

SUB-INDICATOR: MAMMAL SPECIES RICHNESS

KILOMBERO

Mammal species richness data for Kilombero landscape was obtained from the Agrisys Tanzania Project. Thirty-seven camera traps were deployed across cropland (n = 13), forest (n = 18) and grassland (n = 6) land use types between October 2019 - May 2020. The number of unique species identified per camera trap was recorded and averaged for each land use type.

SUB-INDICATOR: ARTHROPOD SPECIES RICHNESS

KILOMBERO

Arthropod species richness (pollinators and pests/predators) for the Kilombero landscape was obtained from the Agrisys Tanzania Project. Arthropods were sampled in 137 plots across cropland (n = 91 plots), forest (n = 23 plots), grassland (n = 12 plots) and woodland (n = 11 plots), using 20 × 20 m plots with a 5 × 5 m subplot and four 1 × 1 m sampling areas. Pollinators were assessed by selecting up to

four flowering plants (prioritising okra/pumpkin flowers where present) and conducting 15-minute timed observations per plant, recording flower visitors that contacted reproductive parts and collecting one specimen per morphospecies for later identification (pausing sampling during heavy rain and resuming 30 minutes after rainfall ceased). Pests/predators were assessed by timed manual searches of the four 1 × 1 m areas (20 minutes per area by two observers) collecting all morphospecies (predators, herbivores and pests). For both pollinator and pest/predator components, the number of unique species (morphospecies) recorded per plot was used as the richness metric and then averaged to generate a mean value for each land cover type.

INDICATOR: BELOW GROUND BIODIVERSITY

SUB-INDICATOR: EARTH WORM COUNTS

For earthworm counts, the dig and hand-sorting approach was adopted, which involved the excavation of a pit and hand sorting for a period of 10 minutes. Thresholds for each category were set using the typical range for East and Southern Africa from Phillips et al., 2019, which used data from over 9000 sites in 57 countries to predict global earthworm diversity, abundance and biomass using environmental models relating to climate, soil, vegetation and elevation.

BARUE-VANDUZI

Earthworm counts were performed during the Mozambique field campaign of August 2025. Sampling was not random due to the extremely dry soil conditions. Instead, counts were targeted in cropland where the soil was visibly dry (i.e. bare ground, n = 6 plots) and compared against soil that was covered in crop

residue or dense grass (i.e covered ground n = 10 plots)- where earthworms were expected to be. Earthworm counts were performed in 5 associations of Barue district: Chibuto (n= 4 plots); Macossa (n = 2 plots); Maputo (n = 3 plots); Matundo (n = 3 plots); Nhamadembe (n = 2 plots); Nhatsanze (n = 2 plots).

The excavated pits measured 50cm x 50cm to a depth of 25cm. The number of earthworms was converted to m² by dividing the sum of earthworms by 0.25.

KILOMBERO

Earthworms count data for 91 cropland plots, 25 forest plots, 13 grassland plots and 11 woodland plots were obtained from the Agrisys Project (Pfeifer et al., 2024) for the Kilombero landscape. Soil was excavated in a 20cm x 20cm area to a depth of 25cm at the northeast and southwest corner of a 10m x 10m plot. The number of earthworms was converted to m² by dividing the sum of earthworms (from the 2 pits) by 0.08.

IHEMI

In Ihemi, earthworm counts were performed in October 2025 on 5 farms in each of 7 villages of Kilolo district (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35. Three 50cm x 50cm pits to a depth of 25cm were excavated on each farm (SW corner, centre, and NE corner), with the number of worms per pit divided by 0.25 to convert to m².

SUMBAWANGA

Earthworm counts were performed in November 2025 in Sumbawanga landscape. Five farms were sampled in 8 villages distributed across 5 districts (Kalambo district

- Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district - Mshani, Mleche; and Nsimbo district - Sitalike), giving a total of 40 plots. Three 50cm x 50cm pits to a depth of 25cm were excavated on each farm (SW corner, centre, and NE corner), with the number of worms per pit divided by 0.25 to convert to m².

INDICATOR: CARBON STOCKS

SUB-INDICATOR: SOIL ORGANIC CARBON (SOC)

Soil organic carbon was obtained from secondary datasets for Barue-Vanduzi and Kilombero and primary data collection for Ihemi and Sumbawanga landscapes.

For Tanzania, thresholds were determined based on values for different land use types across Tanzania, using quartiles of data points from box and whisker plots (Mauya et al., 2022). Mauya et al., 2022 analysed soil organic carbon across different land use types of Mainland Tanzania using the NAFORMA dataset which took soil organic carbon measurements down to 30 cm at 1356 points in total (n=134 for cultivated land, n=59 for humid montane and lowland forest and n=725 for woodland). The excellent category for cropland was based on if the SOC was higher than 54.5 t C ha⁻¹ as this is the Changwon Initiative Tanzanian government target.

For Mozambique, the thresholds were set according to what was specified in the ISPM Soil and Plant Nutrition Lab Report.

BARUE-VANDUZI

Soil organic carbon (%) data for Barue-Vanduzi landscape was taken from ISPM Soil and Plant Nutrition Lab Report, provided by IUCN Mozambique. Sixty-one soil samples collected

from 18 farms (soybean, cabbage, chilli, pigeon pea, beans and maize) in the 7 Abril locality.

KILOMBERO

Soil organic carbon (t ha⁻¹) was obtained from the Agrisys project (Pfeifer et al., 2024) which collected samples in 13 grassland plots, 11 woodland plots, and 91 cropland (sugarcane, maize, intercrop, fallow land, pumpkin and okra) plots at sample depths of 0-20cm and 20-40cm.

To align cropland SOC values with the 0-30 cm depth used to define the poor-excellent thresholds, we derived a depth-weighted 0-30 cm estimate by combining measurements from 0-20 cm and 20-40 cm, weighting them by layer thickness (20 cm and 10 cm, respectively, with the 20-30 cm portion approximated from the 20-40 cm layer), and averaging across plots.

IHEMI

Soil organic carbon (SOC) data for forest was taken from Mwambala et al., 2019 which sampled SOC in the Udzungwa Scarp Nature Reserve across Ihemi and Kilombero Clusters. Samples were the average of the SOC from 0-30 cm from 12 sites (3 samples within each plot giving a sample size of 36).

Soil organic carbon (SOC) data for grassland in the Ihemi landscape were obtained from Laswai (2011), representing average SOC content from 0-30 cm soil depth. The number of plots sampled in this study was not specified.

Soil organic carbon (SOC) data for woodland were sourced from Mwakalukwa et al. (2024), who assessed carbon stocks in the 6,065 ha dry miombo woodland of the Gangalamtumba Village Land Forest Reserve (GVLFR) in Iringa

Rural District. In this study, SOC was measured to a depth of 30 cm from 35 sample plots established along transect lines approximately 2 km apart.

To estimate soil organic carbon on cropland in Ihemi, soil samples were collected on 5 farms in each of 7 villages of Kilolo district (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35. For each farm, three samples were taken from the 0-30cm layer and combined to provide a composite sample. Soil organic carbon (%) and soil bulk density was measured and used to estimate soil organic carbon in tons per hectare ($\text{SOCt/ha} = \text{SOC} (\%) \times \text{BD} (\text{g/cm}^3)$).

SUMBAWANGA

To estimate soil organic carbon on cropland in Sumbawanga, soil samples were collected from 5 farms in each of 8 villages distributed across 5 districts in November 2025 (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district – Mshani, Mleche; and Nsimbo district - Sitalike), giving a total of 40 plots. For each farm, three samples were taken from the 0-30cm layer and combined to provide a composite sample. Soil organic carbon (%) and soil bulk density was measured and used to estimate soil organic carbon in tons per hectare ($\text{SOCt/ha} = \text{SOC} (\%) \times \text{BD} (\text{g/cm}^3)$).

SUB-INDICATOR: ABOVE-GROUND CARBON

Above-ground biomass carbon was derived from secondary data sources and primary data collection for cropland above-ground biomass carbon in Ihemi and Sumbawanga. For primary data collection, above-ground carbon was estimated for each farm and converted to tonnes per hectare by calculating farm area

from farm-boundary shapefiles in QGIS. All trees on the farm were counted, but above-ground carbon was measured for a subsample of up to 10 trees per farm, where available. Total above-ground carbon for the farm was estimated by scaling the mean carbon stock of sampled trees by the total number of trees present.

First, the tree height was estimated using:

tree height = $\tan(\beta) \times \text{distance from the tree} + \text{eye height}$,

$$H = \frac{(\tan(\beta) \times D + E)}{100}$$

Where, H , is tree height (m), β is the angle to the top of the tree, D , is horizontal distance to the tree (cm) and E is the eye height (cm). All tree heights were estimated on level ground.

Second, tree diameter at breast height (DBH) was derived from stem circumference measured at 1.3 m above ground by dividing circumference by π . For multi-stemmed trees, an equivalent DBH was calculated by summing stem basal areas and converting the total basal area to a single diameter using the equation:

$$DBH_{eq} = \sqrt{\sum \left(\frac{Ci}{\pi}\right)^2}$$

Where, DBH_{eq} , is the equivalent diameter at breast height of multi-stemmed trees and Ci is the circumference of stem i .

Third, above-ground biomass (kg) was estimated using the allometric equation

$$1. \text{ } AGB_{est} = 0.0673 \times (\rho D^2 H)^{0.976}$$

Where, AGB_{est} , is above-ground biomass estimate, ρ , is wood density (g cm^3), D , is tree diameter (cm), and H , is tree height (m).

Fourth, carbon stock AGC_{est} was estimated using a constant carbon fraction of 0.55.

$$2. \ AGC_{est} = \frac{(AGB_{est} \times 0.55)}{1000}$$

For Tanzania, thresholds based on values for different land use types across Tanzania, using Mauya et al., (2019). Mauya et al., (2019) analysed above-ground carbon across different land use types of Mainland Tanzania using the NAFORMA dataset which measured trees within sampling plots and applied allometric models to calculate the estimated biomass at 1356 points in total which was then scaled to plot level. There were N=134 plots for cultivated land, N=59 plots for humid montane and lowland forest and N=725 plots for woodland.

KILOMBERO

Cropland above-ground carbon was taken from a systematic review of agroforestry practices in East Africa. The baseline value taken was the mean average for scattered trees which is common in Kilombero (Muthuri et al., 2023).

Woodland AGB was measured in Nyanganje and Kitonga Forests in four 1 ha sample plots where all trees >10 cm in diameter at breast height were measured (Shirima et al., 2011)

Grassland AGB for Kilombero was taken from Laswai (2011) who measured AGB in floodplain grassland who measured the AGB in a 1m² plot by cutting the herbaceous above ground vegetation at 2 cm above the soil surface, before weighing and sending for lab analysis. Ground litter was also included. The number of plots sampled is unclear from the data.

Forest AGB Ithemi and Kilombero was taken from Mwambala et al., 2023 which sampled AGB in the Udzungwa Scarp Nature Reserve across Ithemi and Kilombero Clusters. Within each plot, sub-plots of varying radii were

established. In the radius of 15 m, all trees with a diameter at the breast height greater than 20 cm were sampled, 10 m radius was used to sample trees with DBH ≥ 10 cm but ≤20 cm, and in the radius of 5 m, all trees with DBH ≥ 5 cm but ≤10 cm were sampled.

IHEMI

To estimate above ground carbon stocks of croplands in Ithemi, total carbon per farm was calculated on 5 farms in each of 7 villages of Kilolo district (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35.

Grassland AGB for Ithemi was taken from Laswai (2011) who measured AGB in upland grassland who measured the AGB in a 1m² plot by cutting the herbaceous above ground vegetation at 2 cm above the soil surface, before weighing and sending for lab analysis. Ground litter was also included. The number of plots sampled is unclear from the data.

Woodland AGB for Ithemi was taken from Mwakalukwa et al., 2024 who measured the carbon stocks of a 6065-ha dry miombo woodland Gangalamtumba Village Land Forest Reserve (GVLFR) in Iringa Rural district. Samples were done in 35 sample plots across the woodland, plots were along transect lines with approximately 2 km between plots. Trees and shrubs, deadwood, herbaceous vegetation and surface litter were all measured.

SUMBAWANGA

In Sumbawanga, above-ground carbon was estimated on a total of 40 farms in November 2025. Five farms were sampled in 8 villages distributed across 5 districts (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district - Mshani, Mleche; and Nsimbo district - Sitalike).

INDICATOR: SOIL FERTILITY

SUB-INDICATOR: ELECTRICAL CONDUCTIVITY

BARUE-VANDUZI

Soil electrical conductivity data was obtained from IUCN Mozambique. This dataset consisted of 61 soil samples collected from 18 farms (soybean, cabbage, chilli, pigeon pea, beans and maize farms in the 7 Abril locality).

Thresholds for excellent and good were set using ISPM Soil and Plant Nutrition Lab Report, and based on general salt tolerance of most crops (according to ISPM). The moderate and poor thresholds were determined based on salinity categories found in the [USDAs Soil Quality indicator factsheet](#).

SUB-INDICATOR: SOIL PH

Soil pH data was derived from secondary data sources for Barue-Vanduzi and Kilombero and primary data collection in Ihemi and Sumbawanga. The categories for scoring each baseline value were set using the optimal soil pH for *crop growth* - while this varies across crop types, we deliberately used a broad range of pH 5.5 - 7. The remaining subsequent categories were then determined as 0.5 units above or below this range.

BARUE-VANDUZI

Soil pH data was obtained from IUCN Mozambique. This dataset consisted of 61 soil samples collected from 18 farms (soybean, cabbage, chilli, pigeon pea, beans and maize farms in the 7 Abril locality). Additional soil pH measurements were obtained from 17 farms using an AgroCares Scanner by Solidaridad in November 2023.

KILOMBERO

Soil pH_(water 1:2.5 ratio) for the 0-20cm soil layer was obtained from the Agrisys project. The data was obtained from 91 farms (fallow land, intercrop, maize, okra and sugarcane), distributed across 7 villages and 1 industrial sugarcane plantation. The data was averaged across crop types to provide a baseline value for Kilombero.

IHEMI

In Ihemi, soil pH_(water 1:2.5 ratio) was estimated on 5 farms in each of 7 villages of Kilolo district in October 2025 (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35.

SUMBAWANGA

In Sumbawanga, soil pH_(water 1:2.5 ratio) was estimated on a total of 40 farms in November 2025. Five farms were sampled in 8 villages distributed across 5 districts (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district - Mshani, Mleche; and Nsimbo district - Sitalike).

SUB-INDICATOR: SOIL C:N

KILOMBERO

Soil C:N ratio was obtained from Ovenden et al., 2025. Samples were collected for the 0-20cm soil layer of 20 smallhold sugarcane across 3 villages between April - July 2022 and analysed at Newcastle University, UK to determine total carbon and total nitrogen using a LECO CHN628 instrument. The subsequent concentrations of total carbon and total nitrogen were used to estimate the C:N ratio of the soil.

SUB-INDICATOR: SOIL TOTAL N (KJEDABL)

Baseline values for soil total nitrogen were obtained from secondary data (Kilombero) and

primary data collection (Ihemi and Sumbawanga). Threshold values were determined from Sirappa et al. (2018).

KILOMBERO

Soil total nitrogen (%) for the 0-20cm soil layer was obtained from the Agrisys project. The data was obtained from 91 farms (fallow land, intercrop, maize, okra and sugarcane), distributed across 7 villages and 1 industrial sugarcane plantation. The data was averaged across crop types to provide a baseline value for Kilombero.

IHEMI

In Ihemi, soil total nitrogen was estimated on using samples from 5 farms in each of 7 villages of Kilolo district in October 2025 (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35.

SUMBAWANGA

In Sumbawanga, soil total nitrogen was estimated on a total of 40 farms in November 2025. Five farms were sampled in 8 villages distributed across 5 districts (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district - Mshani, Mleche; and Nsimbo district - Sitalike).

SUB-INDICATOR: SOIL PHOSPHORUS (BRAY I METHOD)

KILOMBERO

Soil phosphorus (Bray I method) was obtained from the Agrisys project (Pfeifer et al., 2024) which collected samples from 91 cropland (sugarcane, maize, intercrop, fallow land, pumpkin and okra) plots at sample depths of 0-20cm and 20-40cm. The data were averaged across crop type and depth to give an average

soil phosphorus value for cropland in the 0-40cm layer.

Threshold values were set from Heckman (1998).

SITE LEVEL: WATER

INDICATOR: SOIL-WATER AVAILABILITY

SUB-INDICATOR: PLANT AVAILABLE WATER CONTENT (%)

KILOMBERO

Intact soil cores were collected on 4 smallhold sugarcane farms in northern Kilombero in May 2022 with 3 repetitions giving a total of 12 plots. These samples were subsequently sent to Rothamsted Research, UK to determine the volumetric water content at varying levels of atmospheric pressure. The plant available water content (PAW) was calculated as the difference between the volumetric water content at 30 kPa (approximately field capacity) and 1480 kPa (permanent wilting point) and multiplied by 100 to give a percentage volume of soil as plant available water.

The threshold values were set from Datta *et al.*, (2017)

SUB INDICATOR: SOIL WATER HOLDING CAPACITY

The volumetric soil water holding capacity was estimated for cropland soils in Ihemi and Sumbawanga using the percolation method.

Procedure

1. **Weigh soil** – For each sample, weigh approximately **25 g** of the dry powdered soil and record the exact mass (*M_{dry}*, g). Place the weighed soil into its labeled filter-paper lined funnel. Record the sample ID and mass.

2. **Prepare water** –pour 50 mL of water into each of three labeled beakers (50 mL for each sample). Record $V_{\text{added}} = 50 \text{ mL}$ for each sample.
3. **Add water to soil** – Slowly add the water from each beaker to the corresponding funnel in **small increments**. *Avoid pouring so fast that water runs over the rim of the filter paper cone. Do not add water beyond the level of the filter paper cone.*
4. **Collect percolate** – As you add water it will seep through the soil and filter paper and collect in the graduated cylinder. When the entire 50 mL has been added, leave the setup undisturbed
5. **Allow drainage** – Wait until **no more water trickles from the stem of the funnel** (the video calls this "several minutes"). Record the time you finish pouring and the time you stop seeing drainage. Note: leaving for 5-20 minutes is common for this simple procedure; be consistent across samples.
6. **Record collected volume** – Read and record the volume of water collected in each graduated cylinder ($V_{\text{collected}}$, mL). Use the meniscus reading at eye level; if the collected water is below the lowest graduation, estimate to the best practical precision.
7. **Compute water retained** – For each sample:
 - Water retained (mL) = $V_{\text{added}} - V_{\text{collected}}$
 - Treat 1 mL water = 1 g water for these calculations.

1. Calculate WHC (volumetric) –

$$WHC (\%) = \frac{\text{water retained in soil (ml)}}{\text{volume of soil used (cm}^3\text{)}} \times 100$$

(Use M dry for the dry soil mass you recorded in step 1.)

Threshold values were set using:

<https://geologyworlds.wordpress.com/2017/06/13/water-holding-capacity/>

IHEMI

In Ihemi, volumetric soil WHC was estimated on 5 farms in each of 7 villages of Kilolo district in October 2025 (Idete, Lulindi, Mahenge, Masisiwe, Mtandika, Ruaha Mbuyuni and Wotalisoli) giving a total sample size of 35.

SUMBAWANGA

In Sumbawanga, volumetric soil WHC was estimated on a total of 40 farms in November 2025. Five farms were sampled in 8 villages distributed across 5 districts (Kalambo district - Chisenga, Kafukoka; Nkasi district- Chala B; Mpanda district - Kakese, Kabwaga; Sumbawanga district – Mshani, Mleche; and Nsimbo district - Sitalike).

SUB-INDICATOR: SOIL MOISTURE (%)

BARUE-VANDUZI

Soil moisture estimates were obtained for 17 farms using an AgroCares Scanner by Solidaridad in November 2023. The thresholds were setting using those provided in the Solidarid reports, as it is unclear whether the soil moisture percentage is by volume or mass of soil.

INDICATOR: SITE-LEVEL WATER BIOLOGICAL ACTIVITY

SUB-INDICATOR: MINISASS

MiniSASS and TARISSfupi both use the same sampling protocol:

- Hold a small net in the current while disturbing the upstream substrate (stones, vegetation, sand, etc.) using your feet or hands.
- Lift stones from the current and gently

pick off any insects using your fingers or forceps. Continue sampling for approximately 5 minutes, moving across the river to cover different habitats (biotopes).

- Rinse the net and empty its contents into a plastic tray. Identify organisms to group level using the Identification Guide; if needed, use a Dichotomous Key initially, then refer back to the Identification Guide for more detailed information. Mark the identified insects on the identification guide.
- Record site information. Sum the sensitivity scores for the identified taxa to calculate the average score, following the scoring sheet.

MiniSASS baseline values were obtained from secondary data (Kilombero, Ihemi) and primary data collection (Barue-Vanduzi).

Thresholds were based on the ecological condition categories defined in the miniSASS protocol, excluding the “natural (unmodified)” category since sampling occurred exclusively outside protected areas where no sites were expected to remain fully unmodified.

BARUE-VANDUZI

MiniSASS assessments were conducted at 10 sites across four associations in Vanduzi (7 Abril 1, 7 Abril 2, Campo 4, Nhamadembe) and three associations in Barue (Macossa, Matundo, Simucaí) in August 2025. Three associations were sampled both upstream and downstream, and four associations sampled once. Sites were initially selected from the list of sites previously sampled by ARA-Centro. Many of these sites however were not free flowing (natural) streams and so unsuitable for miniSASS assessment. For associations not previously sampled by ARA-Centro for water quality analysis, we found the nearest natural free flowing stream.

KILOMBERO

MiniSASS data was obtained for Kilombero from www.miniSASS.org. There were 11

miniSASS assessments recorded for Kilombero between July 2017 to August 2021, covering the Ikela river (n = 2); Msolwa river (n = 3), Sanje river (n = 3) and Mchombe river (n = 3). At sites were classified as rocky, and 3 were classified as sandy.

IHEMI

miniSASS data were obtained for Ihemi from www.miniSASS.org. In Ihemi, 23 assessments were recorded between January 2017 and December 2020. In total 3 rivers were sampled in Kilolo district, namely: Mtongolosi (n = 6); Ilombavana (n = 4); and Mtanga (n = 13). There were 10 sandy sites and 13 rocky sites sampled.

SUB-INDICATOR: TARISFUPI

TARISfupi baseline values consisted of primary data collection and secondary data collection - from an IUCN training in August 2025.

Thresholds were based on the ecological condition categories defined in the TARISfupi protocol, excluding the “natural (unmodified)” category since sampling occurred exclusively outside protected areas where no sites were expected to remain fully unmodified.

SUMBAWANGA

The biological activity indicator for the Sumbawanga landscape was comprised of secondary data collected in the Katuma and Msaginya river in August 2025. A further 54 assessments were carried out in November 2025 across Sumbawanga district, Kalambo district, Nkasi district, Mpanda district and Nsimbo district in the Sumbawanga landscape. In total, 40 TARISfupi assessments were performed in sandy river types and 20 were performed in rocky river types.

INDICATOR: SITE-LEVEL WATER CHEMISTRY

SUB-INDICATORS: PH, ELECTRICAL CONDUCTIVITY, TOTAL DISSOLVED SOLIDS (PPM)

This indicator was comprised of secondary data (Kilombero) and primary data (Barue-Vanduzi, Ihem, Sumbawanga).

For water pH, thresholds were set using the Scottish Environmental Protection Agency's Environmental Quality Standards and Standards for Discharges to Surface Waters, revised in 2020. For water electrical conductivity ($\mu\text{S}/\text{cm}$) and total dissolved solids (ppm) thresholds were set based on the potential plant injury and necessary management for use as irrigation water (Kadhem, 2013).

BARUE-VANDUZI

Initially, we intended to use record chemical properties in small streams as a site-level indicator and larger rivers as a landscape level indicator, however differentiating between the two was not feasible because of the apparent lack of 'large' rivers in the vicinity of the associations where we were conducting fieldwork. Consequently, assessing water quality: chemical properties at the site level was achieved using still bodies of water, which were typically man-made wells dug on or around the farm - except for one natural pool of water in Nhatsanze. Water pH, electrical conductivity and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 11 sites across 4 associations in Barue (Chibuto $n = 4$; Maputo $n = 2$; Matundo $n = 2$; Nhatsanze $n = 3$) during the field campaign to Mozambique in August 2025. In Vanduzi such wells were not present as there is an extensive irrigation system.

KILOMBERO

This data has been anonymised due to a confidentiality agreement. Water electrical conductivity, pH, and total dissolved solids (ppm) were measured at 18 sampling sites covering three rivers of Strahle order 2 or 3) on three separate dates in November and December 2024, with three replicate measurements per site, resulting in a total of 156 observations.

IHEMI

Water pH, electrical conductivity and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 11 sites in rivers of Strahler order 2 or 3 across 5 villages (Idete, $n = 5$; Lulindi, $n = 1$; Mahenge, $n = 1$; Masisiwe, $n = 3$; Wotalisoli, $n = 1$). Per site, three repeat measurements were taken and then all were averaged to get a single value for site level water quality: chemical properties.

SUMBAWANGA

Water pH, electrical conductivity and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 28 sites in rivers of Strahler order 2 or 3 in November 2025. The sites were distributed across 9 villages (Kalambo district: Chisenga, Kafukoka; Mpanda district: Kabwaga, Kakese; Nsimbo district: Sitalike; and Sumbawanga district: Ilemba, Kalambanzite, Kilembo, Mleche). Per site, three repeat measurements were taken and then all were averaged to get a single value for site level water quality: chemical properties.

LANDSCAPE LEVEL: WATER

INDICATOR: LANDSCAPE WATER AVAILABILITY

SUB-INDICATOR: RIVER DISCHARGE

Commented [GU38]: chemical properties

River discharge was recorded to indicate water availability in each landscape using the **GEOGLOWS v2 retrospective simulation dataset**, which comprises approximately 7 million river segments and 125 computational watersheds globally. Each river segment is assigned a unique nine-digit identification number. To select the most appropriate segment for analysis, streams were first mapped according to their **Strahler order**, a hierarchical classification of stream size. Strahler order 1 represents the smallest headwater streams with no tributaries, and when two streams of the same order merge, the downstream segment increases by one order. We selected the river segment with the **highest Strahler order** whose upstream waterways were contained within the boundaries of the cluster, as this segment best represents the combined flow of the upstream network.

Average river discharge was calculated for March and September 2024, corresponding to the long rainy and long dry seasons, respectively. The selected segments for each cluster were:

- **Sumbawanga:** Katuma River (ID 110579596) and Leiche River (ID 110479908)
- **Ihemi:** Great Ruaha River (ID 110540244) and Lukosi River (ID 110540252)
- **Kilombero:** Great Ruaha River (ID 110540262) and Kilombero River (ID 110500912)
- **Barue-Vanduzi:** Dicuti River (ID 20622959) and Nhazonia River (ID 120752743)

Thresholds for scoring river discharge in March and September 2024 were set **arbitrarily** by comparing the average daily discharge for each segment in 2024 to the corresponding 30-

year monthly averages. Excellent 25% +/-; good = 50% +/-; moderate = 75% +/-; poor = >75% +/-

INDICATOR: LANDSCAPE WATER ACCESS

For this indicator the sub-indicators relate to the phrasing of the survey question. The thresholds for all was set arbitrarily, split from 0 - 100%, given that target for SDG indicator 6.1.1. 'Proportion of the population using safely managed drinking water services' is 100%.

SUB-INDICATOR: % OF PEOPLE WHO HAVE ACCESS TO X WATER SOURCE FOR CONSUMPTION

BARUE-VANDUZI:

The percentage of people who have access to mains or well water sources for consumption was taken from the IPSOS baseline report (2023) for Barue-Vanduzi. We consider covered wells or mains the most clean, safe and sustainable water source and use this as the indicator for access to water.

SUB-INDICATOR: MAIN SOURCE OF WATER FOR HOME CONSUMPTION

KILOMBERO AND IHEMI

The percentage of people who have access to covered wells, springs or tap water for consumption was taken from the IPSOS baseline report (2023) for Ihemi. We consider covered wells or springs or tap the most clean, safe and sustainable water source and use this as the indicator for access to water.

SUB-INDICATOR: % OF PEOPLE REPORTING A PROBLEM WITH ACCESSING CLEAN WATER FOR CONSUMPTION

SUMBAWANGA

The sub-indicator data is ‘percentage of people reporting a problem with accessing clean water for consumption’ that was taken from the IPSOS baseline report (2022) for Sumbawanga.

INDICATOR: LANDSCAPE WATER QUALITY: CHEMICAL PROPERTIES

SUB-INDICATORS: PH, EC AND PPM

This indicator was comprised of secondary data (Kilombero) and primary data (Barue-Vanduzi, Ihem, Sumbawanga).

For water pH, thresholds were set using the Scottish Environmental Protection Agency’s Environmental Quality Standards and Standards for Discharges to Surface Waters, revised in 2020. For water electrical conductivity ($\mu\text{S}/\text{cm}$) and total dissolved solids (ppm) thresholds were set based on the potential plant injury and necessary management for use as irrigation water (Kadhem, 2013)

BARUE-VANDUZI

Water pH, electrical conductivity ($\mu\text{S cm}^{-1}$), and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 12 sites across the Barue–Vanduzi landscape, encompassing three associations in Vanduzi (Campo 4, $n = 2$; 7 Abril 2, $n = 2$; Nhamadembe, $n = 3$) and four associations in Barue (Chibuto, $n = 1$; Simucaí, $n = 1$; Macossa, $n = 2$; Matundo, $n = 1$). Measurements were taken in free-flowing surface waters (rivers, streams, and irrigation channels) to characterise landscape-level chemical water quality, as few large rivers were present near the associations where SUSTAIN-PRO are active.

KILOMBERO

This data has been anonymised due to a confidentiality agreement. Water electrical conductivity, pH, and total dissolved solids (ppm) were measured at 13 sampling sites covering two rivers of Strahler order 4 or above on three separate dates in November and December 2024, with three replicate measurements per site, resulting in a total of 114 observations.

IHEMI

Water pH, electrical conductivity and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 13 sites in rivers of Strahler order 4 or above across 5 villages in Kilolo district (Lulindi, $n = 3$; Mahenge, $n = 2$; Mtandika, $n = 3$; Ruaha Mbuyuni, $n = 3$; Wotalisoli, $n = 2$). Per site, three repeat measurements were taken and then all were averaged to get a single value for landscape level water quality: chemical properties.

SUMBAWANGA

Water pH, electrical conductivity and total dissolved solids (ppm) were measured using a Hanna HI 98129 multiparameter instrument at 6 sites in rivers of Strahler order 4 or above in November 2025. The sites were distributed across 4 villages (Mpanda district: Kabwaga, Kakese; Nsimbo district: Sitalike; and Sumbawanga district: Mshani). Per site, three repeat measurements were taken and then all were averaged to get a single value for landscape level water quality: chemical properties.

LANDSCAPE: ADAPTIVE CAPACITY

INDICATOR: AGRICULTURAL, AGROFORESTRY AND TREE PLANTATION PRODUCTIVITY

CROP YIELD

Where possible, the thresholds were based on the SUSTAIN Africa target yields for Tanzania and Mozambique. A crop would be scored as poor if it dropped below the current baseline yield values and would be scored as excellent if it yields above the SUSTAIN Africa target. A SUSTAIN Africa target was not available for maize and beans in Sumbawanga and so the excellent threshold was based on the yield being above the highest regional average across Tanzania.

BARUE-VANDUZI

The yield data for maize, soybean pigeon pea, green bean, and cabbage were collected by IUCN SUSTAIN PRO in June 2025 and used in this report.

KILOMBERO

SUSTAIN Africa provided baseline data for rice, sunflower, soybean, and sugarcane (contract farmers, KSC estate irrigated and KSC estate rainfed)

IHEMI

SUSTAIN Africa provided baseline data for Ihemi for rice, sunflower and soybean.

SUMBAWANGA

For Sumbawanga, the [Open Data for Africa data portal](#) was used which sources data by region from the Annual Agricultural Sample Survey 2022/2023. The average crop yield for maize, rice, sunflower and beans for Rukwa and Katavi was used.

INDICATOR: ADOPTION OF SUSTAINABLE LAND MANAGEMENT PRACTICES

SUB-INDICATOR: ADOPTION OF SUSTAINABLE LAND MANAGEMENT PRACTICES

For this indicator, we used secondary data previously collected by IUCN-SUSTAIN PRO. There were slight differences between landscapes of the types of practices included. The threshold values set for this indicator were arbitrary, split from 0 - 100%.

BARUE-VANDUZI

For Barue-Vanduzi, data was taken from the IPSOS Mozambique 2023 report, which surveyed 180 people in the landscape. The sustainable land management practices included as sub-indicators for Barue-Vanduzi were manure application, crop rotation, practices to reduce soil erosion; compost application.

KILOMBERO, IHEMI AND SUMBAWANGA

For Tanzania, the following practices were chosen to assess this indicator: adoption of practices to enhance soil fertility, apply manure, crop rotation, practices to enhance soil erosion, add compost. Practices to enhance soil fertility were related to adding or working with the soil to make crops grow better and included adding organic and inorganic fertiliser, burning fields, crop rotation, intercropping, planting legumes and minimising tillage. Practices to reduce soil erosion include digging terraces, planting grasses and cover crops, planting trees, using sand filled sacks and digging contour lines. We used the raw data from the IPSOS baseline report (2022) as provided by IUCN to calculate the proportion of respondents doing each practice. N=117 for Ihemi, n=250 for Kilombero and N=176 for Sumbawanga.

Advancing the Landscape Resilience Index under the SUSTAIN Initiative

For Sumabwanga, we also included the practice of Agroforestry as this data was available through the SUSTAIN Eco Agroforestry Assessment that was conducted in August 2025 for 468 farmers across 18 villages.

INDICATOR: HOUSEHOLD POVERTY

SUB-INDICATOR: PPI

KILOMBERO, IHEMI, SUMBAWANGA

PPI data for Tanzania was taken from SUSTAIN Africa baseline data for Kilombero and Ihemi and the IPSOS Eco baseline data for Sumbawanga as the percentage of likelihood of households living below poverty line \$1.90 per person per day.

The threshold values were based on the PPI 2022-2023 poverty benchmark percentiles from each region for the bottom 20th, 40th, 60th, 80th percentile for the region.

SUB-INDICATOR: LACK OF MONEY FOR FOOD

BARUE-VANDUZI

Data for this sub-indicator was collected in 2025 IUCN Mozambique. Two hundred and seventeen participants were asked:

“During the past 12 months, was there a time when you or others in your household were hungry but did not eat because there was not enough money or other resources for food?”

INDICATOR: LAND COVER PERCENTAGE

SUB-INDICATOR: TREE COVER LOSS (%)

Shapefiles of landscape were converted to a KML file in QGIS, and uploaded to the Global

Forest Watch. Tree cover loss was then selected in the Global Forest Watch dashboard, which provides annual estimates for percentage tree cover loss, using the year 2000 as a baseline.

Note: tree cover loss models improved ~2015. As both of the excellent thresholds were for years prior to this the actual percent tree cover loss for these years may be higher - and so the threshold for excellent may be too low.

KILOMBERO, IHEMI, SUMBAWANGA

The categories were obtained using the annual average national % tree cover loss for all of Tanzania. the lowest was 0.25% which was used for excellent. The highest was 0.87% which was used for poor. The remaining categories were obtained as halfway between the two.

BARUE-VANDUZI

The categories were obtained using the annual average national % tree cover loss for all of Mozambique. the lowest was 0.21% which was used for excellent. The highest was 1.24% which was used for poor. The remaining categories were obtained as halfway between the two.

INDICATOR: LANDSCAPE BIODIVERSITY

SUB-INDICATOR: BIODIVERSITY HABITAT INDEX (PROPORTION OF SPECIES)

ALL LANDSCAPES

The Biodiversity Habitat Index (BHI) provides an estimate of the species diversity likely to persist within a given spatial unit (at 1km resolution)—such as a country, biome, ecosystem type, or the entire planet—based on

Commented [GU39]: Calvin indicated that a more recent survey/collection of data will be done by the end of the project. so we (iucn) should update these numbers

the extent, condition, and connectivity of its natural ecosystems. Indicator results can be reported in two ways: (1) as the **effective proportion of habitat** remaining, which accounts for habitat condition, functional connectivity, and spatial variation in species composition (beta diversity); or (2) as the **proportion of species expected to persist** over the long term, calculated using a species-area relationship applied to the effective habitat remaining. We deemed the latter more interpretable so have used this as our indicator of landscape biodiversity. The average BHI score for 2020 was calculated for each landscape using QGIS.

We set the thresholds arbitrarily, based on the present range of values in the data (0.60 - 0.99), whereby 0.99 indicates 99% of species are expected to persist.

INDICATOR: PERCEPTION OF SHOCKS AND STRESSES

ALL LANDSCAPES

For both countries, data for this indicator was taken from the IPSOS Baseline report data which looked at the percentage of respondents who consider drought or floods/storms a big problem.

The threshold values set for this indicator were arbitrary, split from 0 - 100%.

INDICATOR: EXTREME WEATHER

SUB-INDICATOR: FLOODING

ALL LANDSCAPES

Data for Flooding was taken from the [FAO EVE Global Flood Monitoring System](#) because this data goes down to the district level and is updated every 2 weeks. The average

percentage of the land area affected by flooding per month in 2025 and the average percentage of the population exposed to flooding were used as the indicators. Spatial coordinates are not provided for the data set, so it was not possible to take the percentage area for the whole year.

Barue-Vanduzi: Barué and Vanduzi districts

Kilombero: Kilombero and Kilosa districts

Ihemi: Kilolo and Iringa districts

Sumbawanga: Sumbawanga Urban, Sumbawanga, Mpanda Urban, Nkasi, Mlele, Mpanda districts

The threshold values set for this indicator were arbitrary, split from 0 - >25% affected by flooding.

SUB-INDICATOR: COMBINED DROUGHT INDEX

KILOMBERO, IHEMI AND SUMBAWANGA

For each landscape in Tanzania, the **Combined Drought Index (CDI)** was used as the drought sub-indicator, downloaded from the [Humanitarian Data Exchange](#). Developed by the East Africa Drought Watch, the CDI identifies areas currently affected by agricultural drought and areas at risk. It has a spatial resolution of 10 km, with each cell classified as: **Watch** (precipitation deficit observed), **Warning** (precipitation deficit with soil moisture anomaly), **Alert** (precipitation and soil moisture anomalies accompanied by vegetation anomalies), **Partial recovery** (meteorological conditions normalized, vegetation not fully recovered), or **Full recovery** (both meteorological and vegetation conditions normalized).

Each cell integrates anomalies in precipitation, soil moisture, and satellite-derived vegetation condition, and is scored 1-10, with **scores of 7-10 indicating drought conditions**. To calculate the proportion of each landscape affected by drought in 2024,

cells with CDI values 7-10 were selected in QGIS, monthly rasters were merged, and the resulting raster was clipped using the landscape shapefiles to determine the drought-affected area.

SUB-INDICATOR: RISK OF DROUGHT IMPACT FOR AGRICULTURE

BARUE-VANDUZI

The East Africa Drought Watch does not cover Mozambique; therefore, for the Barue-Vanduzi landscape, we used the Risk of Drought Impact for Agriculture (RDri-Agri) index, developed by the [Global Drought Observatory](#). The RDri-Agri index provides a global drought risk assessment for each dekad (10-day period) of the year at a spatial resolution of approximately 110 km.

The RDri-Agri index categorizes drought risk as low, medium, or high and is calculated by integrating drought hazard, exposure, and vulnerability. Exposure refers to the total population, livelihoods, and assets in an area susceptible to drought, while vulnerability reflects the likelihood that these exposed elements will experience adverse effects. Drought hazard is quantified by combining rainfall anomalies (Standardized Precipitation Index, SPI), vegetation photosynthetic activity anomalies (fAPAR anomaly - fraction of Absorbed Photosynthetically Active Radiation), and soil moisture anomalies.

To calculate the area in Barue-Vanduzi impacted by drought, we selected the cells with categorised as high, and merged the raster for each dekad. The resulting raster was then clipped using the Barue-Vanduzi shapefile to quantify the drought-affected area.

The threshold values set for this indicator were arbitrary, split from 0 - >25% affected by drought.

Barue-Vanduzi landscape: Vanduzi district

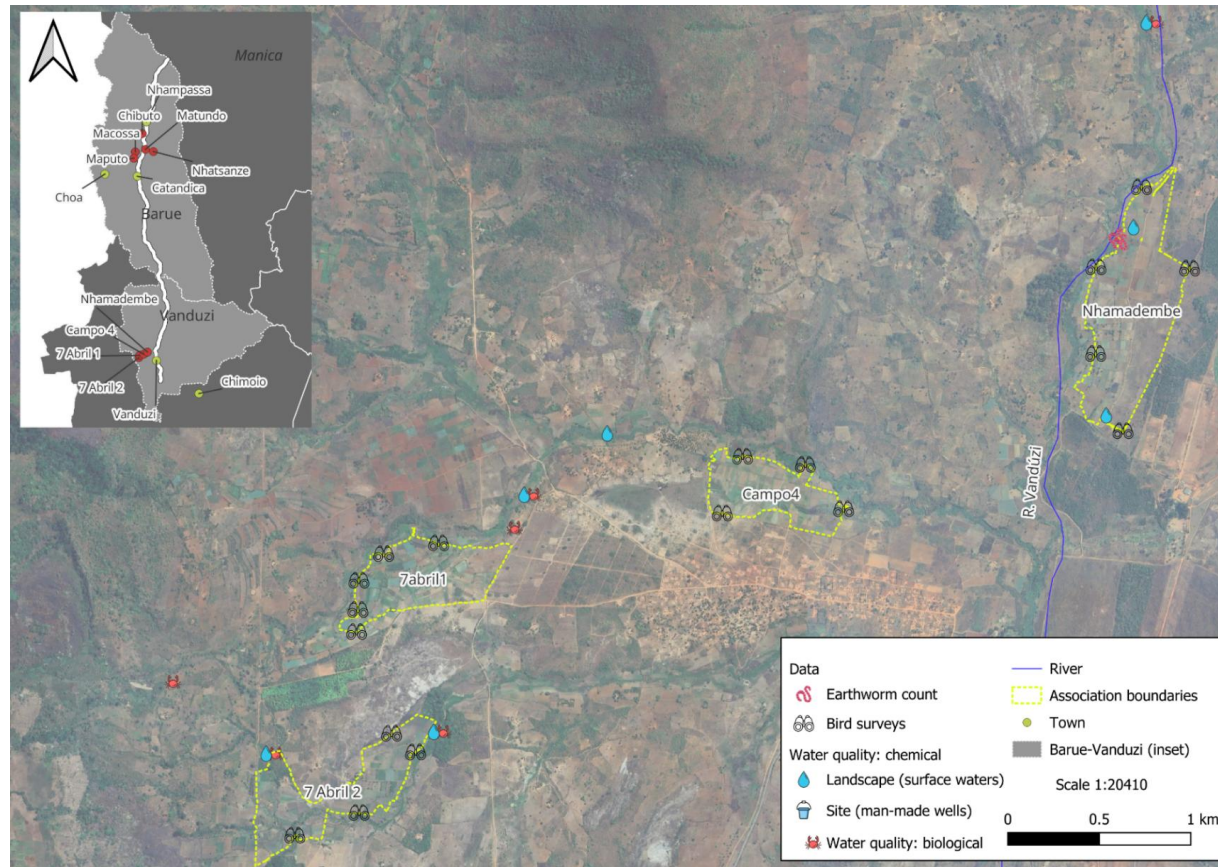


Figure 3. Primary data collection sites in Vanduzi district, Barue-Vanduzi landscape, Mozambique

Barue-Vanduzi landscape: Barue district

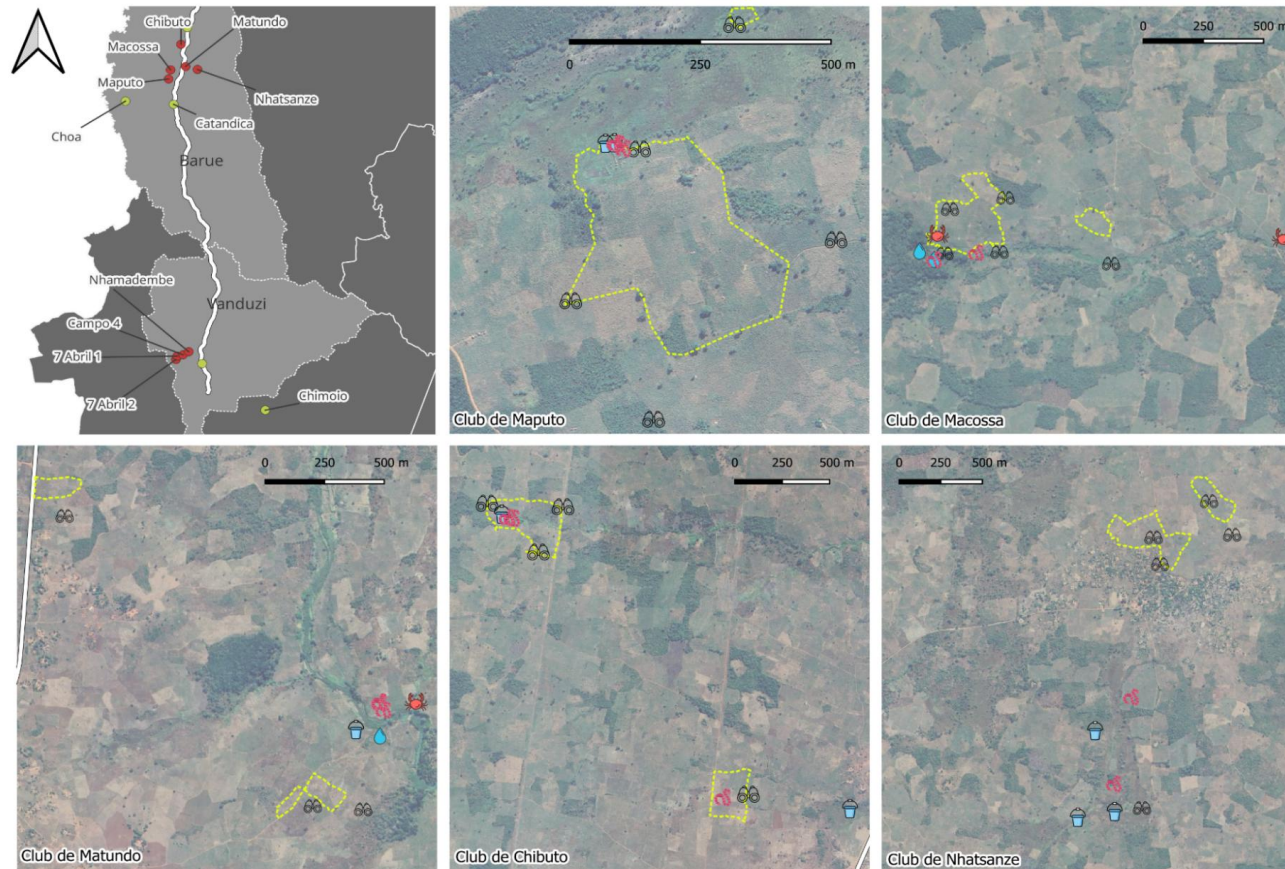


Figure 4. Primary data collection sites in Barue district, Barue-Vanduzi landscape, Mozambique

Ihemi landscape

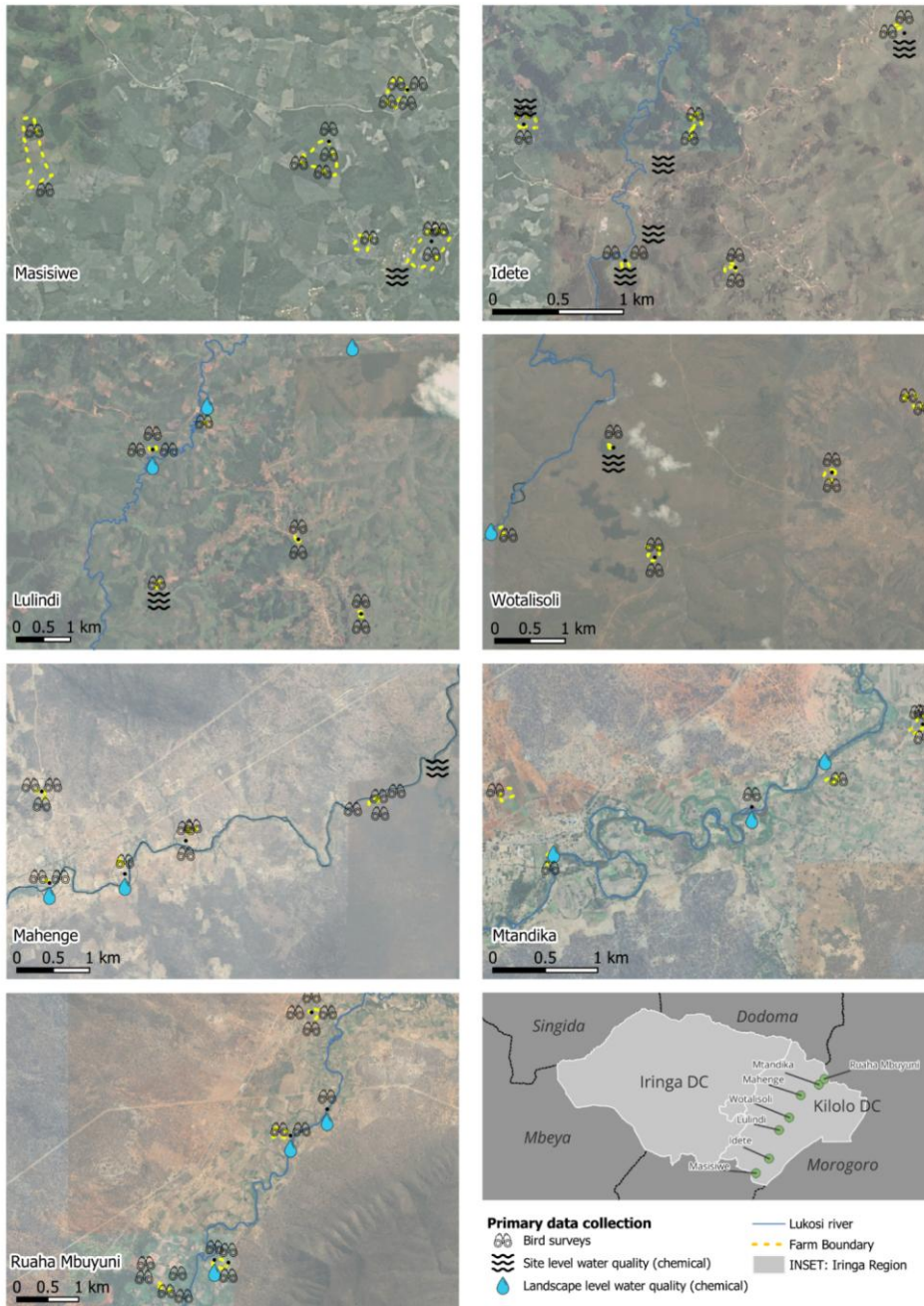


Figure 5. Primary data collection sites in Ihemi landscape, Tanzania

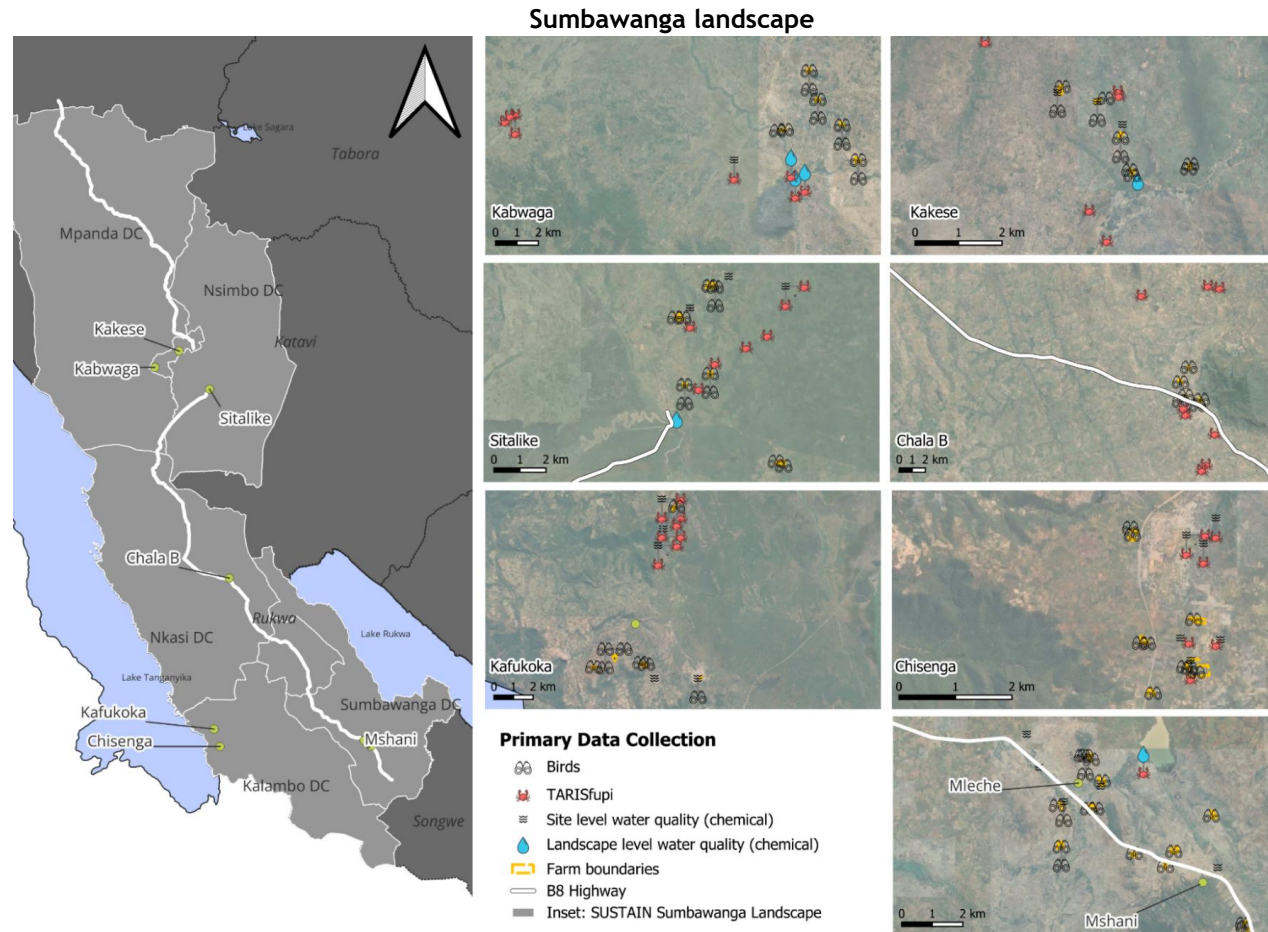


Figure 6. Primary data collection sites in Vanduzi district, Sumbawanga landscape, Tanzania

BASELINE ASSESSMENT OF LANDSCAPE RESILIENCE

BARUÉ-VANDUZI



Commented [GU40]: 1. Generally for all the baselines of the landscapes, i would recommend adding a paragraph on the national level indicators. Yes, we do not monitor the national level but we should include an update if it exists. in the tables of the policy analysis section you had a target and baseline set for some of these policies, are we able to access any kind of reporting on their progress? if that is possible i would like to include it to have a full picture for each of the landscapes. perhaps this is something that Evodius can inquire about or search for in Swahili.

2. Plus, adding a note for all indicators that still relied on the 2023 IPSOS studies to see which ones can and should be updated once we finalise the MEL result indicator data collection.

BASELINE ASSESSMENT SUMMARY

Land

Soil fertility was rated as *good*, but above-ground biodiversity and below-ground biodiversity were rated as *moderate*. Carbon stocks are rated as *poor*.

Water

Site level water quality: chemical properties was rated as *excellent*, soil-water availability is rated as *moderate*, and small stream biological activity is *poor*.

Landscape level water quality: chemical properties was rated as *excellent*, while landscape water access and landscape water availability is *moderate*.

Adaptive Capacity

Perception of shocks and stresses and extreme weather is rated as *excellent*, whereas adoption of SLM practices, landscape biodiversity, extreme weather and household poverty are rated as *good*. Agricultural, agroforestry, and tree plantation productivity is rated as *moderate* and land cover percentage is rated as *poor*.

Table 7. Indicator score and subsequent health category for Barué-Vanduzi. Landscape Resilience categories are Poor: <1.50; Moderate: 1.50 - 2.49; Good: 2.50 - 3.49; Excellent: =>3.50

Pillar	Indicator	Average Score	Health Category
Site			
Land	Above-ground biodiversity	2.00	Moderate
	Below-ground biodiversity	1.50	Moderate
	Carbon stocks	1.00	Poor
	Soil fertility	3.00	Good
	Soil-water availability	2.00	Moderate
Water	Water quality: Small stream biological activity	1.00	Poor
	Water quality: Small stream chemical properties	3.67	Excellent
Landscape			
Water	Landscape water access	2.00	Moderate
	Landscape water availability	2.00	Moderate
	Water quality: Large River chemical properties	3.67	Excellent
Adaptive capacity	Adoption of Sustainable Land Management (SLM) practices	2.50	Good
	Agricultural, agroforestry and tree plantation productivity	2.00	Moderate
	Extreme weather	2.50	Good
	Household poverty	3.00	Good
	Landcover percentage	1.00	Poor
	Landscape biodiversity	3.00	Good
	Perception of shocks and stresses	3.50	Excellent





INDICATOR HEALTH

SITE: LAND

Above-ground biodiversity

The average number of bird species recorded at each site was 4. With a maximum number of bird species recorded at any one plot being 9, this sub-indicator was rated as moderate.

Note: in total 48 species were recorded during the fieldwork, 16 species were unique to Barue, 17 were unique to Vanduzi and 15 were recorded in both.

Below-ground biodiversity

Soils with a crop residue layer supported markedly higher earthworm densities (approximately 20 individuals per m²) compared to bare soils (about 2 individuals per m²). Based on these observations, bare ground was rated as *poor* and covered soil as *good*, resulting in an overall indicator rating of moderate.

Carbon stocks

Cropland soil organic carbon averaged 0.86% resulting in a poor carbon stocks score for the Barue-Vanduzi landscape.

Soil fertility

Soil fertility was good overall: mean electrical conductivity (68.99 µS/cm) was scored in the moderate range, while mean pH (5.88) scored excellent.

SITE: WATER

Soil-water availability

Soil moisture in sampled croplands averaged 16.34%, giving soil-water availability a moderate rating.

Water quality: chemical properties

Well-water chemistry indicated high suitability for irrigation overall (mean pH 6.08, EC 187.64 $\mu\text{S}/\text{cm}$, TDS 95.18 ppm), producing an excellent site-level chemical water quality score.

Water quality: biological activity

MiniSASS scores were poor in both sandy- and rocky-bed channels, indicating limited biological condition despite generally favourable chemistry.

LANDSCAPE: WATER

Landscape water access

Only 37% of respondents reported access to mains or well water for home consumption, placing landscape water access in the moderate category.

Landscape water availability

Relative to 30-year seasonal averages, March 2024 flows were below average in both representative rivers, while September 2024 conditions improved (Dicui moderate, Nhazonia excellent). Overall water availability was moderate.

Water quality: chemical properties

Surface-water chemistry at landscape scale was favourable (mean pH 6.58, EC 100.25 $\mu\text{S}/\text{cm}$, TDS 49.33 ppm), resulting in an excellent rating.

LANDSCAPE: ADAPTIVE CAPACITY

Extreme weather

Flood exposure was low in both districts in 2025 (monthly average 0.02% of land area affected in Barue and 0.02% in Vanduzi), scoring excellent.



By contrast, drought affected 100% of land area in 2024 in both districts (scoring poor). Combined, the extreme weather indicator was good.

Agricultural, agroforestry and tree plantation productivity

Reported yields varied widely across key crops (maize 2.37 t/ha, soybean 1.68 t/ha, pigeon pea 2.07 t/ha, green bean 1.00 t/ha, cabbage 48.3 t/ha), giving an overall moderate productivity score.

Pigeon pea scored excellent and soybean good, while maize, green bean and cabbage scored poor, pulling the overall rating to moderate.

Adoption of Sustainable Land Management (SLM) practices

SLM uptake was relatively strong: manure use (52%) and crop rotation (69%) scored good, while erosion-control practices (48%) and compost use (41%) were moderate, producing an overall good score.

Household poverty

In the previous 12 months, 43 people surveyed (19.81%) said that they had experience hunger due to insufficient money to buy food.

Perception of shocks and stresses

Drought was reported as a major problem for 41% of respondents and scored as good, whereas flooding was reported as a major problem by only 21% of respondents and scored excellent. Overall, perception of shocks and stresses indicator was scored good.

Landcover percentage

Tree cover loss in 2024 was 1.6% relative to 2000 baseline cover, substantially above Mozambique's recent annual range used for scoring, and therefore rated poor.

Landscape biodiversity

The Biodiversity Habitat Index was 0.85, suggesting around 85% of species are expected to persist long term; this scored good.

SYNTHESIS

The Barue-Vanduzi landscape shows encouraging signs of ecological functionality and community-level adaptation, but also clear weaknesses that, which if left unaddressed, may erode landscape resilience over time.

Good landscape biodiversity persistence, measured using the Biodiversity Habitat Index (BHI), suggests that overall, the Barue-Vanduzi landscape still retains habitat structure capable of supporting biodiversity despite cropland being the predominant land use. However, within croplands field-based bird species richness surveys indicated relatively low above-ground biodiversity at the site level with four or fewer species recorded at more than half of cropland sites.

The high annual tree cover loss rate (1.6%) used to indicate land cover change is particularly concerning, as it directly threatens long-term biodiversity stability, ecosystem service provision, and local microclimates. This risk is further exacerbated by the absence of any legally protected areas within the Barue-Vanduzi landscape.

Although above-ground carbon stocks were not included in the carbon indicator for this landscape due to data limitations, the observed tree cover loss is likely to have substantially reduced landscape-level carbon stocks. This compounds the already poor carbon stock rating driven by low soil organic carbon (SOC) levels in cropland soils. Low SOC is also likely to have contributed to the moderate below-ground biodiversity observed, as reflected in earthworm abundance (Schon *et al.*, 2020). The stark contrast between earthworm densities under covered versus bare soils demonstrates that soil biological health responds strongly to land management practices (Viaene *et al.*, 2025). Bare soils on community demonstration plots were visibly drier during sampling, and the lower earthworm abundance was unsurprising given both low SOC and moderate soil-water availability.

Although soil-water availability data were limited to community demonstration plots, where soils were largely bare or only loosely covered, it is likely that the measured soil moisture levels were also influenced by management practices, since bare or loosely covered soils were visibly drier than covered soils during the earthworm counts. Seasonal variability in rainfall is also likely to have played a role- soil moisture data were collected in November 2023, which typically marks the onset of the rainy season in Mozambique. However, landscape water availability data derived from the GEOGLOWS retrospective river discharge dataset indicate that river discharge in both Barue and Vanduzi was considerably lower during this period, suggesting a delayed onset of the rainy season. The moderate rating for landscape water availability in 2024 further indicates a drier-than-average year, with reduced river flows across the landscape.

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Site-level water quality indicators presented contrasting results, with poor scores for biological activity but excellent average scores for chemical properties. This divergence suggests that chemical pollution or agricultural runoff is not a major issue in the landscape. This interpretation is supported by moderate soil electrical conductivity and excellent soil pH values, which point to limited chemical inputs in farming systems. The poor performance of the biological water quality indicator may instead reflect localized habitat degradation, as sampling was conducted only at sites where people have access to water, where physical disturbance of riverbanks and substrates is likely to have already occurred. Such degradation can negatively affect macroinvertebrate communities independently of chemical pollution.

Water quality was assessed primarily in relation to its suitability for irrigation; however, field observations and the 2023 IPSOS survey indicate that streams and rivers are also widely used for domestic consumption.

The good rating for adoption of sustainable land management practices indicates promising current land management trajectories and likely reflects the success of SUSTAIN ~~PR~~^Q~~ro~~ community demonstration plots, where association leaders establish learning sites for other members. Continued monitoring of agricultural, agroforestry, and tree plantation productivity will be important in the coming years to assess the longer-term impacts of increased adoption of these practices.

Commented [GU41]: I wanted to include a comment before by the indicator on what tree plantation refers to/tracks? is that in relation to afforestation efforts or perhaps restoration works within SUSTAIN? or wood volume/production, found it a bit unclear.

Finally, the extreme weather indicator produced mixed results. While drought emerged as a significant issue, the finding that 100% of the landscape was affected is likely influenced by the coarse spatial resolution (100km) of the underlying data. Nevertheless, landscape water availability data from GEOGLOWS confirm that 2024 was drier than the 30-year average, with reduced river flows. The 2023 IPSOS survey data on perception of drought and flooding as a major problem further support the importance of drought relative to flooding, although survey results were pooled across Barue and Vanduzi. Given the extensive irrigation infrastructure in Vanduzi, the impacts of drought are likely more localized and pronounced in Barue.



RECOMMENDATIONS

While the baseline assessment of landscape resilience presents a mixed picture, it also highlights clear opportunities for intervention that would simultaneously improve multiple site- and landscape-level indicators while contributing to national targets.

Reversing tree cover loss

Land cover change, indicated by tree cover loss in the Barue-Vanduzi landscape is disproportionately higher than the national average and nature-based solutions aimed at halting and reversing deforestation, such as agroforestry, would therefore have significant co-benefits. Firstly, at the site-level, promoting agroforestry would increase carbon stocks of croplands and likely enhance habitat quality for birds (Davis *et al.*, 2024) and invertebrates (Campera *et al.*, 2024).

Second at the landscape-level, an increase in farmers practicing agroforestry would also contribute to reversing the tree cover loss across Barue-Vanduzi and maintain or improve the habitat connectivity which could be tracked using the landscape biodiversity indicator (biodiversity habitat index). The effectiveness of agroforestry promotion efforts can also be monitored through the adoption of sustainable land management (SLM) practices, providing a measure of uptake over time.

Third, at the national level, promoting agroforestry would contribute to both the Strategic Plan for the Development of the Agrarian Sector 2030, which seeks to reduce greenhouse gas emissions from agriculture and The Strategy and Action Plan for the Conservation of Biological Diversity in Mozambique 2015-2035, which aims to increase the percentage of agricultural area land benefiting from sustainable management practice and climate-smart agriculture.

Improving soil carbon

Carbon stocks could further be improved through interventions that increase soil organic carbon in cropland soils. The SUSTAIN ~~PRO~~Pro initiative has already introduced relevant practices through training at community demonstration plots, including the application of manure and retention of organic residues. However, continued follow-up is essential to ensure these practices are consistently implemented. Increasing SOC would contribute directly to national carbon storage targets and is also likely to generate multiple site-level benefits. Higher SOC can reduce the impacts of drought by improving soil water-holding capacity (Feifel *et al.*, 2024; Ramirez *et al.*, 2023), enhance crop yields through improved nutrient availability (Lal, 2004; Oldfield *et al.*, 2019), and support higher earthworm abundance and overall soil biological activity (Van Vliet *et al.*, 2007; Viaene *et al.*, 2025), all of which could be tracked with the LRI. These benefits would also contribute to the Strategic Plan for the Development of the Agrarian Sector 2030 at the national level.

Addressing water access

Although water quality was assessed primarily in relation to irrigation suitability, the widespread use of surface water for drinking and domestic purposes indicates a need for further action. Installing safe, managed water access points and establishing buffer zones along rivers and streams to reduce contamination would support progress towards the SDG indicator 6.1.1, which aims to increase the proportion of population using safely managed water services. Field observations also highlighted stark differences in irrigation infrastructure between Vanduzi and Barue. Expanding irrigation infrastructure in Barue would improve soil-water availability at the site-level, which would likely facilitate improved agricultural activity at the landscape level. The social benefits of such interventions are measurable using the landscape-level water access indicator and at the national level, improved irrigation infrastructure would also contribute towards Mozambique's progress on

SDG indicator 6.4, which aims to increase water-use efficiency across all sectors and The Strategic Plan for the Development of the Agrarian Sector 2030 target of doubling agricultural productivity.

Interventions in water sources management would also reduce risks to human safety, particularly where rivers are used for bathing and washing despite the presence of crocodiles, which have been reported in the Fudze river. Investment in water infrastructure would therefore deliver benefits for livelihoods, health, and safety while strengthening overall landscape resilience.

KILOMBERO



BASELINE ASSESSMENT SUMMARY

Land	Water	Adaptive capacity
Above-ground biodiversity, below-ground biodiversity and carbon stocks are rated as <i>good</i> . Soil fertility was rated as <i>moderate</i> .	Site level water chemistry and soil-water availability are rated as <i>excellent</i> , but small stream biological activity is <i>poor</i> . Landscape level chemistry was rated as <i>excellent</i> , landscape water availability was rated as <i>good</i> , and landscape water access is <i>moderate</i> .	Extreme weather, household poverty, landscape biodiversity and perception of shocks and stresses are rated as <i>good</i> . However, agricultural, agroforestry, and tree plantation productivity, adoption of SLM practices and landcover percentage is rated as <i>moderate</i> .

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Table 8. Indicator score and subsequent health category for Kilombero. Landscape Resilience categories are Poor: <1.50; Moderate: 1.50 - 2.49; Good: 2.50 - 3.49; Excellent: =>3.50

Pillar	Indicator	Average Score	Health Category
Site			
Land	Above-ground biodiversity	2.64	Good
	Below-ground biodiversity	3.00	Good
	Carbon stocks	2.50	Good
	Soil fertility	2.00	Moderate
Water	Soil-water availability	4.00	Excellent
	Water quality: Small stream biological activity	3.50	Good
	Water quality: Small stream chemical properties	4.00	Excellent
Landscape			
Water	Landscape water access	2.00	Moderate
	Landscape water availability	2.75	Good
	Water quality: Large River chemical properties	4.00	Excellent
Adaptive Capacity	Adoption of Sustainable Land Management (SLM) practices	2.00	Moderate
	Agricultural, agroforestry and tree plantation productivity	2.00	Moderate
	Extreme weather	3.50	Excellent
	Household poverty	3.00	Good
	Landcover percentage	2.00	Moderate
	Landscape biodiversity	3.00	Good
	Perception of shocks and stresses	2.50	Good

INDICATOR HEALTH

SITE: LAND

Above-ground biodiversity

Bird species richness in sugarcane plantations (11 species), smallholder sugarcane farms (12 species), and grasslands (12 species) were categorised as good, whereas forest (10 species) was categorised as moderate.

Arthropoda species richness was categorised as moderate in cropland (30 species), forest (32 species) and grassland (33 species). In Woodland it was categorised as good (28 species). The functional group of each invertebrate species was also recorded (but not included in the scoring). Across all land uses, invertebrate communities fulfil a wide range of ecological roles—including biocontrol,

decomposition, nutrient recycling, pollination, predation, and disease/vector dynamics—with slight variation in functional groups such as ectoparasites in cropland and woodland and kleptoparasites unique to grassland.

Mammal species richness was categorised as moderated in cropland (3 species) and grassland (4 species), whereas forest (10 species) was categorised as good.

Overall, above ground biodiversity was categorised as moderate for Kilombero.



Below-ground biodiversity

Earthworm abundance was scored as excellent (38 individuals per m²) in grassland, good (29 individuals per m²) in cropland and moderate (19 individuals per m²) in forest. The overall below-ground biodiversity indicator was categorised as good.

Carbon stocks

Above-ground carbon in cropland scored good (8.2 t C ha⁻¹), whilst forest (323.71 t C ha⁻¹) and grassland (17.98 t C ha⁻¹) were scored as excellent. In contrast, woodland (24.2 t C ha⁻¹) was scored poor.

For soil organic carbon each land use was scored as moderate- cropland 36.99 t C ha⁻¹; Forest 32.33 t C ha⁻¹; grassland 38.60 t C ha⁻¹; woodland 29.29 t C ha⁻¹

Overall, this indicator was scored as good.

Soil fertility

Soil fertility was assessed in croplands only. The average C:N ratio (15:1) was classed as good; the average soil pH (5.95) was classed as excellent; the average soil total N (0.08%) and

average soil phosphorus (11 ppm) was classed as poor. Overall soil fertility was scored as moderate.

SITE: WATER

Soil-water availability

Plant-available water content at field capacity was 30.8% of soil volume and scored excellent.

Site level water quality: biological activity

Using the miniSASS protocol, sandy river types were rated as good, whereas rocky river types were rated as excellent.

Site level water quality: chemical properties

Water chemistry in rivers of Strahler order 2 or 3 in Kilombero was categories as excellent, with average values of EC 159.40 µS/cm, pH 8.20 and total dissolved solids 79.20 ppm.

LANDSCAPE: WATER

Landscape water access

Landscape water access was moderate with 36% of respondents reporting access to improved sources such as wells, springs, or taps.

Landscape water availability

The Kilombero river average daily discharge was rated as poor in March 2024 with more than double the 30-year average discharge but excellent in September.

The average daily discharge for the Ruaha river in March 2024 was 69% higher than its 30 years monthly average and so was rated moderate but was only 12% higher than average in September so rated excellent.

Overall, this indicator received a rating of good

Landscape water quality: chemical properties

Water chemistry in rivers of Strahler order 4+ Kilombero was categories as excellent overall, with average values of EC 247.28 $\mu\text{S}/\text{cm}$, pH 8.53 and total dissolved solids 127.61 ppm.

LANDSCAPE: ADAPTIVE CAPACITY

Extreme weather

The average percentage of the land area affected by flooding in Kilombero district was 5.46 % and classed as good, the average percentage land area affected by flooding in Kilosa district was 0.33% and classed as excellent.

The percentage land area affected by drought at any stage during 2024 was 1.39% for Kilombero district and scored as excellent, whereas 14.33% of Kilosa district was affected by drought in 2024, so this was scored as good.

Overall this indicator was scored as excellent.

Agricultural, agroforestry and tree plantation productivity

Each crop yield sub-indicator was scored as moderate (rice- 3.29 tons ha^{-1} ; sunflower- 0.97 tons ha^{-1} ; soybean- 0.61 tons ha^{-1} ; smallhold sugarcane- 47 tons ha^{-1} ; industrial rainfed sugarcane- 55 tons ha^{-1} ; industrial irrigated sugarcane- 85 tons ha^{-1}).

Adoption of Sustainable Land Management (SLM) practices

Adoption of SLM practices was scored moderate overall. Specifically adoption of practices to increase soil fertility (52%) scored as good whilst crop rotation (30%), practices to reduce soil erosion (27%) and adding compost (45%) scored as moderate. Adoption of the application of manure (15%) was poor.

Household poverty

Household poverty was scored as good with 20% of households in Kilombero likely to be living below poverty line of \$1.90 per person per day.

Perception of shocks and stresses

Drought was reported as a major problem for 61% of respondents and scored as moderate, whereas flooding was reported as a major problem by 44% of respondents and scored good. Overall, perception of shocks and stresses indicator was scored good.

Landcover percentage

In 2024, the Kilombero landscape recorded a tree cover loss of 0.62% and was therefore rated as moderate.

Landscape biodiversity

The Biodiversity Habitat Index was 0.85, scoring good for expected long-term species persistence.

SYNTHESIS



The Kilombero landscape demonstrates generally good ecological condition and moderate-to-good resilience across site- and landscape-level indicators, with particularly strong performance in water availability and quality, carbon storage, and biodiversity persistence. However, several indicators point to emerging pressures related to land management, soil fertility constraints, and uneven adoption of sustainable practices that may limit long-term resilience if not addressed.

Biodiversity

Above-ground biodiversity in Kilombero was categorised as moderate overall, reflecting variation among taxa and land uses. Bird species richness was consistently good in sugarcane plantations, smallholder sugarcane, and grassland, but only moderate in forests. However, several bird species were found exclusively within forest habitats, highlighting the importance of forest habitat for supporting disturbance-sensitive species or those which are forest specialist. Arthropod species richness was moderate across most land uses and good in woodland, indicating relatively intact invertebrate communities that support a broad range of ecological functions, including nutrient cycling, pollination, predation, and biological control (Pfeifer *et al.*, 2024). Mammal species richness was highest in forested areas, where it scored as good, while cropland and grassland supported fewer species, reflecting the importance of forest habitats for maintaining larger vertebrate biodiversity (Brockhoff *et al.*, 2017; Pfeifer *et al.*, 2017).

Below-ground biodiversity was stronger, with earthworm abundance rated as good overall and particularly high in grasslands. This suggests that soil biological processes remain functional across much of the landscape, providing a solid foundation for nutrient cycling and soil structure, especially outside intensively managed cropland. The ecological value of the landscape recorded at the site-level is further evidenced by the good rating of landscape biodiversity indicator at the landscape-scale.

Carbon stocks and soil condition

Carbon stocks in Kilombero were rated as good, underpinned by excellent above-ground carbon stocks in forest and grassland and moderate soil organic carbon across all land cover types. In contrast, above-ground carbon stocks in woodland were rated as poor, indicating comparatively low woody biomass in these habitats compared to other woodland habitats in Tanzania. This poor rating may be a result of less protection offered to woodlands than that of forests, which could explain the moderately rated land cover percentage, which suggests that tree cover loss is high relative to the rest of Tanzania. The importance here is that deforestation not only threatens to undermine the above-ground biodiversity and carbon stocks measured at the site-level but also the landscape biodiversity persistence.

Soil organic carbon levels were moderate across all land uses, including cropland, forest, grassland, and woodland, suggesting a relatively uniform but only intermediate capacity for below-ground carbon storage throughout the landscape. Together, these results highlight the critical role of forest and grassland in maintaining carbon stocks, while also indicating potential for improvement, particularly through land management practices that enhance biomass accumulation and soil organic carbon.

However, soil fertility in croplands was only moderate, constrained by poor soil total nitrogen and plant available phosphorus levels - despite favourable soil pH and C:N ratios. This imbalance suggests that while soils are chemically suitable for nutrient cycling, current nutrient limitations may restrict crop productivity and responsiveness to management inputs.

Water availability and quality

Water-related indicators were a key strength of the Kilombero landscape. Site-level soil-water availability on smallholder sugarcane farms was excellent, indicating strong potential for crop growth under current conditions - this can be expected since the Kilombero Valley is a historic flood plain. However, this indicator only demonstrates that the soil has the capacity to hold water, not necessarily that it does year-round. Drought, when measured by the percentage of land area affected, was scored as excellent and good for Kilombero and Kilosa districts, respectively. However, across the whole Kilombero cluster, drought was perceived as a major problem affecting 61% of respondents, indicating a problem with water access for farmers, as opposed to availability. This issue was also prevalent for water access for home consumption, with only 36% of people accessing improved water sources, resulting in a moderate rating and highlighting a social vulnerability despite strong environmental water indicators.

Climate resilience and livelihoods

Adoption of sustainable land management practices was also moderate overall. While practices aimed at enhancing soil fertility showed relatively good uptake, adoption of manure application was low, and crop rotation, erosion control, and composting were only moderately adopted. This uneven uptake suggests opportunities for strengthening farmer support and extension.

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RECOMMENDATIONS



Although the baseline assessment indicates that landscape resilience in the Kilombero cluster is generally **strong**, targeted interventions at the site and landscape scales would significantly enhance long-term sustainability and contribute directly to Tanzania's national policy and development commitments. Two priority intervention areas emerge: improving access to water for multiple uses and sustainable intensification of agriculture.

Improving access to water for multiple uses

At the site level, expanding small-scale irrigation infrastructure would improve crop productivity and reduce sensitivity to rainfall variability and drought. These actions contribute to SDG VNR 2.4.1 by increasing the area under irrigation and support national goals for climate-resilient agricultural growth. At the landscape level, improving access to safe drinking water through protected wells, irrigation schemes, and managed water access points would address a key vulnerability identified in the baseline assessment. These interventions directly contribute to national progress under SDG VNR 1.4.1 (population with access to drinking water and sanitation) and support Tanzania's Nationally Determined Contribution (2021) commitment to achieve universal access to clean and safe water in rural areas by 2030. Strengthening water monitoring and basin-level management would further support national objectives under SDG VNR 6.3.2 and the National Climate Change Response Strategy (2021-2026), which prioritises water resilience across all basins.

Sustainable intensification of agriculture

At the site level, addressing soil fertility constraints through improved nutrient management, organic amendments, crop rotations, and composting would enhance yields while building resilience to climate stress. These interventions support national agricultural productivity objectives under the SAGCOT Green Print and contribute to emissions reduction and carbon storage targets by improving

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soil organic carbon, in line with the Changwon Initiative. At the landscape level, scaling up adoption of sustainable land management and agroecological practices strengthens ecosystem function, stabilises production, and reduces vulnerability to shocks. Increased uptake directly contributes to national commitments under the Tanzania National Ecological Organic Agriculture Strategy (2023-2030) and the National Climate Change Response Strategy (2021-2026), which targets widespread adoption of climate-smart land management.

Collectively, these site- and landscape-level interventions strengthen community resilience, improve livelihoods, and support national ambitions for climate adaptation, poverty reduction, and sustainable development as articulated in Tanzania's Draft Development Vision 2050.

Supporting ongoing initiatives

In addition to identifying priority areas for intervention, the LRI can be used to monitor the co-benefits of restoration activities already underway in the landscape. For example, the Nyerere-Udzungwa Wildlife Corridor (NUWC), launched in 2021, is a long-term restoration initiative that aims to reconnect the Udzungwa Mountains National Park with Julius Nyerere National Park, maintaining connectivity between Tanzania's southern and western elephant populations. The LRI could track improvements in landscape carbon stocks and biodiversity while also capturing social outcomes such as changes in human-wildlife interaction incidence and household poverty levels associated with participating villages.

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BASELINE ASSESSMENT SUMMARY

Land	Water	Adaptive Capacity
Soil fertility is rated as Good, but above-ground biodiversity and below-ground biodiversity was rated as Moderate, and carbon stocks are rated as Poor.	Site level water quality: chemical properties was rated as excellent, whereas soil-water availability and small stream biological activity is moderate. Landscape level water quality: chemical properties was rated as <i>excellent</i>, while landscape water availability is rated as <i>good</i> and landscape water access is <i>poor</i>.	The extreme weather indicator is rated as <i>excellent</i>, the adoption of SLM practices, household poverty and landscape biodiversity are rated as <i>good</i>, landcover percentage is rated as <i>moderate</i>, whereas agricultural, agroforestry, and tree plantation productivity and the perception of shocks and stresses are rated as <i>poor</i>.

Table 9. Indicator score and subsequent health category for Ihemi. Landscape Resilience categories are Poor: <1.50; Moderate: 1.50 - 2.49; Good: 2.50 - 3.49; Excellent: =>3.50

Pillar	Indicator	Average Score	Health Category
Site			
Land	Above-ground biodiversity	2.00	Moderate
	Below-ground biodiversity	1.00	Poor
	Carbon stocks	2.29	Moderate
	Soil fertility	2.50	Good
Water	Soil-water availability	3.00	Good
	Water quality: Small stream biological activity	3.00	Good
	Water quality: Small stream chemical properties	4.00	Excellent
Landscape			
Water	Landscape water access	1.00	Poor
	Landscape water availability	2.75	Good
	Water quality: Large River chemical properties	4.00	Excellent
Adaptive Capacity	Adoption of Sustainable Land Management (SLM) practices	3.20	Good
	Agricultural, agroforestry and tree plantation productivity	2.00	Moderate
	Extreme weather	3.75	Excellent
	Household poverty	3.00	Good
	Landcover percentage	2.00	Moderate
	Landscape biodiversity	3.00	Good
	Perception of shocks and stresses	1.00	Poor

INDICATOR HEALTH

SITE: LAND

Above-ground biodiversity

The average number of bird species recorded at each site was 6.39. With a maximum number of bird species recorded at any one plot being 19, this sub-indicator was rated as moderate.

Below-ground biodiversity

Average earthworm abundance across 35 cropland plots was 1 per m², resulting in a poor below-ground biodiversity score.

Carbon stock

Above-ground carbon was categorised as excellent in forest (323.71 t C ha⁻¹) good in grassland (7.51 t C ha⁻¹), moderate in woodland (39.18 t C ha⁻¹) and poor in cropland (2.75 t C ha⁻¹).

Soil organic carbon was scored as moderate in each of the land use types: cropland- 36.58 t C ha⁻¹; woodland- 29.46 t C ha⁻¹; forest- 63.4 t C ha⁻¹; and grassland- 47.5 t C ha⁻¹.

Overall, this indicator was scored as moderate.

Soil fertility

Soil fertility was assessed in croplands only. The average soil total nitrogen (0.04%) was classed as poor, whereas the average soil pH (6.72) was classed as excellent. Overall soil fertility was scored as good.

SITE: WATER

Soil-water availability

Volumetric soil water holding capacity at field capacity in cropland soils was 0.70 cm³/cm³ and classed as good.

Site level water quality: biological activity

Biological water quality in both sandy and rocky river types were scored as good, according to the miniSASS protocol.

Site level water quality_chemical properties:

Water chemistry in rivers of Strahler order 2 or 3 in Ihemi was categorised as excellent, with average values of EC 90.46 µS/cm, pH 7.07 and total dissolved solids 52.62 ppm.

LANDSCAPE: WATER

Landscape water access

Landscape water access was categorised as poor with 36% of respondents reporting access to improved sources such as wells, springs, or taps.

Landscape water availability

In March 2024, the average daily discharge of the Lukosi river and Ruaha river were 98% and 63% higher than their respective 30-year average for March, and were rated poor and moderate, respectively. In September, both stretches of river were close to their 30-year average and rated as excellent.

Overall, this indicator received a rating of good.

Landscape water quality: chemical properties

The landscape-level indicator for water chemical properties was assessed in different rivers of Strahler order 4 or above in Ihemi. Average electrical conductivity (103.23 µS/cm), pH (7.46) and total dissolved solids (40.46 ppm) were all classed as excellent, so the overall score for this indicator was excellent.

LANDSCAPE: ADAPTIVE CAPACITY

Extreme weather

The average percentage of the land area affected by flooding in any one month in 2025 was 0.02% for Kilolo and 0.54% for Iringa and both sub-indicators scored excellent.

The percentage land area affected by drought at any stage during 2024 was 4.78% for Iringa and scored as excellent, whereas 5.10% of Kilolo was affected by drought in 2024, so this was scored good.

Overall this indicator was scored as excellent.

Agricultural, agroforestry and tree plantation productivity.

Each crop yield sub-indicator was scored as moderate (rice- 3.29 tons ha⁻¹; sunflower- 0.97 tons ha⁻¹; soybean- 0.61 tons ha⁻¹)

Adoption of Sustainable Land Management (SLM) practices.

Adoption of practices to enhance soil fertility (80%) and crop rotation (75%) scored excellent, whilst practices to reduce soil erosion (67%) and application of manure (67%) scored good. Adding compost (41%) scored as moderate. Overall, this indicator scored as good.

Household poverty

Household poverty was scored as good with 15% of households in Ihemi likely to be living

below poverty line of \$1.90 per person per day.

Perception of shocks and stresses.

Drought was reported as a major problem for 59% of respondents and scored as moderate, whereas flooding was reported as a major problem by only 14% of respondents and scored excellent. Overall, perception of shocks and stresses indicator was scored good.

Landcover percentage

In 2024, the Ihemi landscape recorded a tree cover loss of 0.79% and was therefore rated as moderate.

Landscape biodiversity

The Biodiversity Habitat Index was 0.82, scoring good for expected long-term species persistence



SYNTHESIS

The Ihemi landscape exhibits a generally functional and resilient socio-ecological system, underpinned by strong water quality, low exposure to extreme weather events, and relatively high uptake of sustainable land management practices. However, this overall resilience masks important weaknesses in soil health, carbon storage, and equitable access to water, which may limit long-term sustainability if not addressed.

Biodiversity and ecosystem functioning

Below-ground biodiversity in Ihemi was poor, with low earthworm abundance recorded across cropland plots, indicating limited soil biological activity in cultivated areas. This suggests that soil processes such as mineralisation of organic matter and nutrient cycling may be constrained under current agricultural practices (Badalucco, 2017; Li et al., 2018; Byers et al., 2024) and may account for the moderate rating of soil organic carbon and extremely low concentrations of soil total nitrogen in croplands.

At the landscape scale, the biodiversity habitat index indicates good long-term biodiversity persistence, suggesting that Ihemi retains sufficient habitat connectivity and quality to support species over time. However, site-level above-ground biodiversity tells a more nuanced story. Bird species richness in croplands was moderate, with average species counts substantially lower than the maximum recorded at individual sites.



The land cover change indicator suggests moderate tree cover loss has occurred across the Ihemi landscape and was evidenced by a widespread absence of trees within or around farm boundaries fields. Not only does tree cover loss impact a landscape's delivery of supporting ecosystem services, such as above-ground biodiversity, it also impacts its delivery of regulating ecosystem services, particularly carbon sequestration. In Ihemi, relative carbon sequestration was strongly shaped by land cover, with excellent carbon storage in forests, moderate in woodlands and poor in croplands. These patterns point to a landscape where carbon storage is concentrated in remaining natural and semi-natural habitats, presenting an opportunity for sustainable land management practices as a key priority area for SUSTAIN. Encouragingly, adoption of sustainable land management practices is good overall in Ihemi. High uptake of soil fertility enhancement, crop rotation, and widespread erosion control and manure application, indicates strong farmer engagement and awareness with SLM practices.

Water availability, access, and quality

Water quality across Ihemi is a clear strength. Both site- and landscape-level water chemistry indicators scored excellent, and biological water quality in small streams was rated good, indicating relatively intact aquatic ecosystems and limited chemical pollution.

Both site- and landscape-level water availability were rated good. At the site-level, this reflects favourable water holding capacity of the soil and at the landscape-level reflects river flows that were above long-term averages in March 2024 and returned to near-average conditions by September, despite considerable seasonal and interannual variability. However, these favourable environmental conditions contrast sharply with limited access to water for people and production. Only 17% of households have access to safe drinking water, and just over half of farmers have access to irrigation. This disconnect helps explain why drought is perceived as a major problem by nearly 60% of respondents, despite low measured exposure to drought across the landscape. In Ihemi, water scarcity appears to be less about physical availability and more about infrastructure, access, and management, creating social vulnerability even in a relatively water-secure landscape.



RECOMMENDATIONS

The baseline assessment highlights that while Ihemi shows strong water quality, climate resilience, and adoption of sustainable land management practices, key opportunities exist to strengthen water access, soil health, and carbon sequestration. Targeted interventions at the site and landscape levels can also advance national targets and policy commitments.

Expand access to safe and reliable water

At the site and landscape level, improving access to safe drinking water is critical, as only 17% of households currently use improved sources. Investments in protected wells, piped schemes, and managed community water points would improve human health and resilience. These actions directly support SDG VNR 1.4.1 (population with access to drinking water and sanitation) and Tanzania's Nationally Determined Contribution (2021) commitment to achieve 100% access to clean and safe water in rural areas by 2030.

At the landscape level, expanding irrigation infrastructure would also enhance agricultural productivity, support crops against seasonal rainfall variability, and contribute to SDG VNR 2.4.1 (area under irrigation), supporting national objectives to increase irrigated agricultural land and build climate-resilient food systems.

Improve soil fertility and below-ground biodiversity

At the site level, low earthworm abundance and pending soil fertility results indicate the need for targeted interventions to improve soil health. Practices such as organic amendments, composting, manure application, and diversified crop rotations would enhance soil fertility, boost crop productivity, and increase below-ground biodiversity. These interventions support national targets under the Changwon Initiative (increase soil organic carbon in cropland to 54.5 t ha⁻¹ by 2030) and contribute to emissions reductions in line with the SAGCOT Green Print (2012).

At the landscape level, widespread adoption of sustainable land management and agroecological practices would enhance ecosystem functioning, stabilise yields, and reduce vulnerability to climate shocks. This aligns with the Tanzania National Ecological Organic Agriculture Strategy (2023-2030), which aims to increase the number of farmers practicing Ecological Organic Agriculture from 1 million to 3 million by 2030, and with the National Climate Change Response Strategy (2021-2026) target of at least 50% of communities adopting climate-smart land management practices.

Enhance carbon stocks and forest management

At the site level, targeted planting of trees, agroforestry integration, and restoration of degraded cropland areas can improve above-ground and soil carbon stocks, particularly in woodlands and croplands where carbon storage is currently low. At the landscape level, protecting forests and woodlands through community-based forest management and restoration would safeguard ecosystem services, biodiversity, and hydrological regulation. These interventions support national and international commitments including SDG VNR 15.1.1 (area of natural forest cover), the Bonn Challenge (restore 5.2 million hectares by 2030), the Changwon Initiative (11,011,950 ha of forest restoration by 2030), and targets under the National Climate Change Response Strategy (2021-2026) to restore at least 3 million ha of degraded forest land by 2026.

Strengthen climate and livelihood resilience

Integrated interventions that combine improved water access, soil fertility enhancement, and ecosystem restoration will strengthen community resilience to drought and flooding, which remain important concerns despite the overall excellent rating for extreme weather. These measures support Tanzania's Draft Development Vision 2050, which aims to create a climate-resilient society, and contribute to SDG 1.2.1 and SAGCOT objectives to reduce rural poverty and build adaptive livelihoods.

SUMBAWANGA



BASELINE ASSESSMENT SUMMARY

Land

Soil fertility scored good, while above-ground biodiversity and carbon stocks were moderate. Below-ground biodiversity was scored as poor

Water

Site level water quality: chemical properties was rated as excellent and soil-water availability good and biological water poor.

Landscape level water quality: chemical properties, and water availability were good, but access to water was moderate.

Adaptive Capacity

Extreme weather and perception of shocks and stresses was rated as excellent, landscape biodiversity was good and adoption of SLM practices, agricultural, agroforestry and tree plantation productivity, household poverty and landcover change was moderate.

Table 10. Indicator score and subsequent health category for Sumbawanga. Landscape Resilience categories are Poor: <1.50; Moderate: 1.50 - 2.49; Good: 2.50 - 3.49; Excellent: =>3.50

Pillar	Indicator	Average Score	Health Category
Site			
Land	Above-ground biodiversity	2.00	Moderate
	Below-ground biodiversity	1.00	Poor
	Carbon stocks	2.00	Moderate
	Soil fertility	2.50	Good
Water	Soil-water availability	3.00	Good
	Water quality: Small stream biological activity	1.50	Moderate
	Water quality: Small stream chemical properties	4.00	Excellent
Landscape			
Water	Landscape water access	2.00	Moderate
	Landscape water availability	3.25	Good
	Water quality: Large River chemical properties	3.33	Good
Adaptive Capacity	Adoption of Sustainable Land Management (SLM) practices	2.33	Moderate
	Agricultural, agroforestry and tree plantation productivity	2.00	Moderate
	Extreme weather	3.57	Excellent
	Household poverty	2.00	Moderate
	Landcover percentage	2.00	Moderate
	Landscape biodiversity	3.00	Good
	Perception of shocks and stresses	3.50	Excellent

INDICATOR HEALTH

SITE: LAND

Above-ground biodiversity

The average number of bird species recorded at each site was 5. With a maximum number of bird species recorded at any one plot being 10, this sub-indicator was rated as moderate.

Note: in total 115 species were recorded during the point count surveys.

Below-ground biodiversity

Average earthworm abundance across 35 cropland plots was 6 per m², resulting in a poor below-ground biodiversity score

Carbon stocks

This indicator was assessed through above-ground carbon and soil organic carbon on croplands only.

Above-ground carbon on cropland scored poor (2.50 t C ha⁻¹), whereas for soil organic carbon (41.05 t C ha⁻¹) scored good. Overall this indicator was scored as moderate.

Soil fertility:

Soil fertility was assessed in croplands only. The average soil total nitrogen (0.02%) was classed as poor, whereas the average soil pH (6.34) was classed as excellent. Overall soil fertility was scored as good.

SITE: WATER

Soil-water availability:

Volumetric soil water holding capacity at field capacity in cropland soils was 0.70 cm³/cm³ and classed as good.

Site level water quality: biological activity.

Using the TARISfupi protocol, biological water quality was moderate in sandy river types and rocky river types were scored as poor. Overall, this indicator was scored moderate.

Site-level water chemistry

Water chemistry in rivers of Strahler order 2 or 3 in Sumbawanga was categorised as excellent, with average values of EC 210.25 µS/cm, pH 6.91 and total dissolved solids 105.40 ppm.

LANDSCAPE: WATER

Landscape water access

Landscape water access was categorised as moderate with 70% of respondents reporting problems with access to clean water for consumption in Sumbawanga.

Landscape water availability

In March 2024, the average daily discharge of the Katuma river and Leiche river were 21.82% and 3.84% higher than their respective 30-year average for March, and were both rated as excellent

In September, the Katuma river was 64% higher than the 30 year average and rated as moderate, whereas the Leiche river was 33.97% higher than the 30 year average and rated as good.

Overall, this indicator received a rating of good.

Landscape level water chemistry

The landscape-level water chemistry of rivers of Strahler order 4+ was categorised as good overall.

Average electrical conductivity (304.22 µS/cm) and total dissolved solids (153.67 ppm) were all classed as good, pH (7.65) was rated as excellent.

LANDSCAPE: ADAPTIVE CAPACITY

Extreme weather

The average percentage of the land area exposed to flooding in any one month scored as excellent for Kalambo (0.61%), Mlele (1.035%), Mpanda (0.40%), Nkasi (0.42%) and Sumbawanga urban (0.06%) and rated as good for Mpanda Urban (8.38%) and Sumbawanga rural (6.95%).

The percentage of area affected by drought was rated as excellent for Nsimbo (3.54%), Sumbawanga Urban (0%), Sumbawanga rural (0%), Kalambo (0%) and Mpanda rural (0%) and rated as moderate for Nkasi (17.71%) and Mpanda rural (22.22%).

Overall, this indicator was scored excellent.

Agricultural, agroforestry and tree plantation productivity

Each crop yield sub-indicator was scored as moderate (rice- 2.75 tons ha⁻¹; sunflower- 0.6 tons ha⁻¹; maize- 1.55 tons ha⁻¹; and beans- 0.65 tons ha⁻¹).

Adoption of Sustainable Land Management (SLM) practices.

Adoption of practices to enhance soil fertility (63%), crop rotation (61%), practices to reduce soil erosion (68%) and adding manure (54%) scored as good. Adoption of the application of compost (20%) and agroforestry (23%) was poor. Overall, this indicator scored as moderate.

Household poverty

Household poverty was scored as moderate with 31% of households in Sumbawanga likely to be living below poverty line of \$1.90 per person per day.

Perception of shocks and stresses

Drought was reported as a major problem for 45% of respondents and scored as good, whereas flooding was reported as a major problem by only 3% of respondents and scored excellent. Overall, perception of shocks and stresses indicator was scored good.

Landcover percentage

In 2024, the Sumbawanga landscape recorded a tree cover loss of 0.56% and was therefore rated as moderate.

Landscape biodiversity

The Biodiversity Habitat Index was 0.85, scoring good for expected long-term species persistence



The Sumbawanga landscape shows a broadly functional socio-ecological system with comparatively strong adaptive capacity to climate extremes and generally good water availability and quality. However, several indicators point to constraints that may limit long-term resilience and livelihood outcomes, particularly poor soil biological health, uneven aquatic ecosystem condition, moderate land-use pressure on tree cover, and continued poverty and water access challenges.

Biodiversity and ecosystem functioning

Similar to the other SUSTAIN landscapes, the biodiversity habitat index indicates Sumbawanga has the capacity to support biodiversity across the landscape overall, but the site-level biodiversity metrics, which were targeted in croplands highlight the influence of local-level pressures.

The above-ground biodiversity indicator for Sumbawanga, bird species richness, was considered moderate with an average of 5 different species identified during point count surveys. In contrast, a previous survey conducted by Seki *et al.* (2025) found high bird species richness in Sumbawanga landscape. Here, several land cover types were assessed but it is not clear what constituted *high* bird species richness, and the data was not available at the time of drafting the report.

Low bird species richness may in part be due to disturbance caused by tree cover loss or a preexisting lack of trees in and around cropland habitats, which was evidenced by the moderate above-ground biomass carbon. Bird species richness is also likely reflective of the limited below-ground biodiversity as a food source, although the type of bird species (e.g. granivore, insectivore, vermivore) was not established. The low earthworm abundance points to constrained soil biological functioning in cropland areas, likely reflecting limited organic matter inputs, soil disturbance, and declining soil structure.

Poor below-ground biodiversity is an important warning sign because it often precedes or accompanies reduced nutrient cycling efficiency (Badalucco, 2017; Li *et al.*, 2018; Byers *et al.*, 2024); however, soil organic carbon was high in croplands. Consequently, low earthworm abundance may reflect temporary variation in soil conditions, such as seasonal effects or cultivation-related disturbance, particularly where slash-and-burn agriculture is common (Lalthanzara & Zodinpui, 2021).

Water availability, access, and aquatic ecosystem condition

Sumbawanga scored similar to Ihemi and Kilombero across indicators in the water pillar. Site- and Landscape-level water availability were rated as good, with favourable soil water holding capacity on croplands and a river discharge generally close to or above long-term averages in 2024, though with notable seasonal variation. Chemical water quality was also strong: excellent in small streams and good-to-excellent in larger rivers, indicating that water chemistry remains suitable for irrigation purposes.

However, biological water quality tells a different story. Moderate biological condition in sandy rivers and poor condition in rocky rivers despite positive water chemistry suggest that riverbanks degradation may be occurring, potentially due to sedimentation or impacts of physical disturbance. Riverbank degradation from cattle was evident in the landscape, since pastoralists frequently access different points along the river.

Despite relatively good water availability and quality, access remains a key constraint: 70% of households reported problems accessing clean water. This points to an important vulnerability in infrastructure and service provision, meaning that water stress in Sumbawanga is likely experienced less as a shortage of water in the landscape and more as a shortage of safe, reliable, and convenient access.

Climate resilience and livelihoods

Sumbawanga scored highly for extreme weather resilience overall, with low actual exposure and perceived impacts of flooding and drought-affected areas across several districts. This suggests that acute climate hazards are not currently the dominant driver of vulnerability across the landscape.

Adoption of sustainable land management practices was moderate overall and while many farmers report using practices to enhance soil fertility, crop rotation, erosion control, and manure application and while sustainable soil management practices are common, soil total nitrogen was markedly low. This may be a symptom of nutrient leaching, since total nitrogen indicates the nitrogen-supplying capacity of the soil.

Finally, livelihood outcomes remain a concern, with a substantial share of households living below the poverty line, reinforcing that resilience and ecosystem condition are closely tied to development challenges.



RECOMMENDATIONS

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Thank you.



The baseline results indicate key priorities for interventions in Sumbawanga, namely: improving water access, supporting livelihoods to reduce household poverty, increasing tree cover in croplands and reducing riverbank degradation.

Improve access to safe and reliable drinking water

Although water availability and chemistry are generally good, access is a major constraint, with 70% of households reporting difficulties. A major priority is therefore infrastructure and service provision, rather than increasing water supply. Interventions should include protected/covered water points, small, piped schemes where feasible, and community-managed systems for operation and maintenance. These actions directly support SDG VNR 1.4.1 (access to drinking water and sanitation) and Tanzania's Nationally Determined Contribution (2021) commitment to expand access to clean and safe water in rural areas by 2030.

Increase tree cover within croplands to strengthen biodiversity and above-ground carbon

Increasing on-farm trees through agroforestry, boundary planting, and farmer-managed natural regeneration would raise above-ground carbon, improve habitat structure for birds (supporting

above-ground biodiversity), and reduce erosion (supporting aquatic ecosystem condition). Since agroforestry adoption is currently low, targeted extension and demonstration plots should prioritise practical, low-cost tree integration models. These actions contribute to SDG VNR 15.1.1 (natural forest cover), the Bonn Challenge (restore 5.2 million hectares by 2030), restoration commitments under the National Climate Change Response Strategy (2021-2026), and forest restoration targets under the Changwon Initiative.

Rebuild soil biological functioning while addressing nitrogen constraints

Overall, adoption of sustainable land management (SLM) practices in Sumbawanga is encouraging yet soil total nitrogen remains extremely low. The relatively high uptake of several practices demonstrates strong community engagement and awareness, suggesting that nature-based solutions (NbS) to improve soil nitrogen through strengthened SLM could be a high-impact opportunity for SUSTAIN. Strengthening these practices would contribute to national priorities on sustainable agriculture and resilience (SDG VNR 2.4.1) and aligns with the National Climate Change Response Strategy (2021-2026) objective to scale up climate-smart land management at community level.

Composting (using plant material) remains limited in the landscape, partly due to seasonal shortages of available biomass and partly due to limited understanding of its benefits, as noted in the SNV report. At the same time, crop residues left after harvest represent a substantial nutrient resource but are often burned—a common post-harvest practice in the landscape. While ash can contribute some nutrients, burning results in significant losses of key elements, particularly nitrogen and sulphur, and releases avoidable emissions. Shifting away from residue burning would therefore support both productivity and climate objectives, including national commitments on emissions reduction and carbon management articulated in the SAGCOT Green Print.

Promoting residue retention and incorporation, rather than burning, would allow nutrients to be released more gradually, improve soil structure, and increase nitrogen retention in the soil. These measures could be complemented by targeted nitrogen replenishment through legume rotations, green manures, and improved manure and compost management. Together, these actions would strengthen soil fertility and farm productivity (supporting SDG VNR 2.4.1), help build and maintain soil organic matter in line with the Changwon Initiative soil organic carbon ambitions and contribute to the national agenda on climate-smart, low-emissions agriculture under the National Climate Change Response Strategy (2021-2026).

Link sustainable intensification to livelihoods and poverty reduction

With moderate crop yields and significant levels of household poverty (31% below the \$1.90/day line), strengthening resilience must also improve livelihoods. Integrating water access improvements, soil biological recovery and nitrogen management, and tree-based production systems provides a pathway to raise productivity while reducing vulnerability. This supports poverty reduction objectives under SDG VNR 1.2.1 and aligns with broader national ambitions for climate-resilient development under Tanzania's Draft Development Vision 2050.

Supporting ongoing initiatives to restore riverbanks

Rivers in Tanzania are legally protected by a 60-metre buffer zone, yet in practice riverbanks are frequently used—particularly by pastoralists—leading to trampling, vegetation loss, bank erosion, and declining aquatic habitat quality. Building on existing community-led initiatives to restore riverbank corridors offers a practical pathway to reduce these pressures while maintaining access. For example, in Kabwaga, communities have designated controlled water-access points along riverbanks and introduced a payment mechanism for pastoralists, with revenues reinvested into buffer restoration; notably, beehives have been installed to deter cattle from entering protected areas, helping to protect vegetation and stabilise banks. Scaling and supporting similar approaches would enable site- and catchment-level actions focused on restoring riparian buffers, stabilising banks at access points, limiting disturbance in sensitive reaches, and reducing sediment delivery through improved ground cover and upslope erosion control. These interventions align with national objectives on water resilience and catchment management under the National Climate Change Response Strategy (2021–2026) and support progress towards SDG VNR 6.3.2 on water quality monitoring and management. In this context, the LRI could track improvements in site-level biological water quality (e.g., macroinvertebrate indicators) alongside implementation indicators such as kilometres of buffer restored, number of controlled access points established, and observed reductions in bank disturbance.



LIMITATIONS TO THE LRI INDICATORS

LIMITATIONS OF THE INDICATORS THEMSELVES

In developing the LRI, indicators were designed to be adaptable across diverse landscapes and data contexts. Sub-indicators were selected to balance data availability and collection feasibility with ecological representativeness. As a result, the specific sub-indicators applied under each indicator may differ between landscapes.

For example, in Ihemi and Sumbawanga, the soil fertility indicator comprises only two sub-indicators (soil pH and soil total nitrogen) limited to cropland soils as defined in the sampling design. In contrast, in Kilombero, the same indicator includes a wider suite of sub-indicators (e.g. soil pH, soil C:N ratio, available phosphorus, and soil total nitrogen), providing a more comprehensive assessment. This variation reflects differences in landscape characteristics, sampling objectives, and logistical feasibility, rather than inconsistency in the underlying conceptual framework.

Where possible, sub-indicators were selected to reflect ecological processes and functional attributes of landscape resilience. For instance, bird species richness serves as a proxy for above-ground biodiversity and ecosystem functioning, as birds respond sensitively to habitat disturbance and trophic changes. Similarly, indices such as TARISSfupi/MiniSASS were used to assess water quality through the diversity and composition of aquatic macroinvertebrates, which respond predictably to changes in hydrological and chemical conditions.

It is important to recognise that the LRI is designed to provide a broad, integrated picture of landscape resilience rather than a detailed diagnostic tool for any single ecosystem component. When applying LRI results to inform decision-making, users should consider the representativeness and scope of each sub-indicator. For example, while bird species richness can indicate general trends in

biodiversity, targeted biodiversity management decisions should draw on additional taxonomic or habitat-specific data to ensure ecological accuracy and effectiveness.

Indicator scoring

Where possible, scoring thresholds were derived from scientific literature and/or locally relevant policy standards to ensure consistency and transparency. However, such reference information was not available for several indicators and sub-indicators, specifically *above-ground biodiversity*, *extreme weather*, *adoption of sustainable land management practices*, *water holding capacity*, and *household poverty*.

In these cases, expert judgement was applied to define provisional thresholds, based on the consultants' professional experience and understanding of local conditions. These thresholds should therefore be considered *interim* and *adaptive*. As new scientific evidence or policy benchmarks become available, we recommend revising the scoring system to reflect these updates. Alternatively, thresholds could be refined using IUCN SUSTAIN's internal targets or those developed by their landscape partners, to better align with ongoing programme objectives and monitoring frameworks.

Data limitations

The baseline assessment draws on multiple data sources, including primary field data, secondary datasets, and ongoing monitoring programmes. While the indicators were designed to be adaptable across different landscapes and contexts, some require specialised expertise, equipment, and financial resources that may limit the feasibility of consistent long-term monitoring. For example, soil fertility measurements are costly and technically demanding, requiring laboratory analysis and trained personnel.

These factors may constrain future data collection efforts unless dedicated resources are secured or simplified proxy measures are identified.

Ecological indicators also operate across different spatial and temporal scales, reflecting the timeframes at which natural processes and management outcomes become observable. Most LRI indicators are expected to show meaningful trends only over multi-year periods. Therefore, we recommend collecting data prior to NbS implementation and then at three-year intervals to capture medium-term change.

However, some indicators, such as total soil and above-ground carbon stocks, are likely to respond more slowly and may require longer monitoring horizons. In contrast, annual indicators (e.g., crop yields or vegetation cover) can provide valuable insights into short-term variability and early signs of change, though these depend on available resources. Ultimately, to attribute improvements to management interventions, the magnitude of change due to NbS implementation must exceed the natural inter-annual variation.

Seasonality also exerts a strong influence on soil and water-related indicators within the LRI. Because primary data collection was conducted during the dry season, the results provide only a partial snapshot of land health conditions. We therefore recommend repeating data collection during the wet season to obtain a more balanced and representative assessment of land and ecosystem dynamics.

PATHWAYS FOR UPTAKE

This report first outlined the key challenges SUSTAIN-Africa has faced in operationalising and monitoring landscape resilience to prioritise NbS and ILM interventions. It then showed how the LRI addresses these challenges by applying a multi-scale, multi-pillar framework and integrating data from multiple sources, including spatially explicit studies, global datasets, and primary data collection. Using this approach, we conducted a baseline assessment of landscape resilience

to identify priority NbS and ILM interventions aligned with national policies and commitments.

In this section, we discuss how the LRI can bring multiple stakeholders together to collaborate on strengthening landscape resilience through three key entry points: (i) establishing an evidence base for investment, (ii) ongoing monitoring, and (iii) institutional alignment (Table 11). Kilombero is used as a theoretical case study to illustrate these pathways.

Table 11. Tabulation of how LRI can be used to support decision-making in practice by stakeholders at different scales

Entry Point	Scale	Primary Users	How the LRI Is Used in Practice
Establishing Evidence for Investment	Site	Farmers, cooperatives, community groups, NGOs	Use LRI baselines to identify priority fields, communities, or sites where NbS and SLM interventions can deliver the most immediate resilience and livelihood benefits.
	Landscape	District authorities, NGOs, corridor platforms (e.g. SAGCOT, Beira Agricultural Growth Corridor)	Aggregate site-level data to identify hotspots of land and water stress or opportunity, and to prioritise interventions at strategic locations within the landscape.
	National	National ministries, coordinating bodies, national programmes	Use aggregated baselines to inform strategic investment priorities, programme design, and resource allocation aligned with land, water, and climate objectives.
Monitoring, Reporting, and Adaptive Management	Site	Farmers, community groups, local implementers	Use monitoring results to assess whether interventions are delivering expected outcomes and to adapt practices over time.
	Landscape	District governments, basin authorities, and multi-stakeholder platforms	Coordinate interventions across value chains and ecosystems, align implementers around shared landscape objectives, and manage trade-offs.
	National	Ministries, national monitoring and reporting units	Track progress towards national land, water, climate, and development targets and inform adaptive policy responses.
Institutional Integration and Policy Alignment	Site & landscape	Public and private actors, NGOs, basin organisations	Embed LRI evidence into routine planning, coordination, and co-investment decisions for NbS and ILM interventions.
	National	Government institutions, policy and planning bodies	Integrate LRI indicators into policy frameworks, planning processes, and reporting systems to support institutional uptake and long-term scaling.

THEORETICAL CASE STUDY: IMPROVING WATER ACCESS FOR MULTIPLE USES IN THE KILOMBERO VALLEY

Historically, investment in system-level landscape interventions has been constrained not by a lack of need, but by the absence of decision-relevant evidence that links local conditions to system-wide risks, co-benefits, and policy or value-chain priorities. The LRI addresses this gap by integrating land, water, and adaptive capacity indicators into a single, multi-level decision-support framework. By translating fragmented sectoral data into coherent, investment-relevant insights, the LRI enables more coordinated prioritisation, reduces investment risk, and supports the scaling of integrated landscape approaches.

The LRI baseline assessment identified water access and water variability as critical constraints on landscape resilience in the Kilombero Valley. Vulnerabilities were evident across drinking water access, irrigation reliability, and basin-level water management, with direct implications for food security, livelihoods, and agricultural value chains. By consolidating findings across Land, Water, and Adaptive Capacity, the LRI enabled the prioritisation of improving access to water for multiple uses as a high-impact intervention area where public policy objectives and private sector interests converge.

PUBLIC AND PRIVATE INTEREST IN IMPROVING WATER ACCESS FOR MULTIPLE USES

Improving access to water for multiple uses in the Kilombero Valley involves a diverse set of public authorities and actors and private actors operating at different spatial and institutional levels (Table 12). A key mechanism linking actors across public and private institutions is the Water Use Associations (WUAs).

In this context, WUAs act as an operational interface between public authorities, private actors, and local communities. They can coordinate water access and support the

maintenance and restoration of water sources and riverine areas. They could also take the lead in monitoring water quality since previous SUSTAIN-led initiatives have provided the training and equipment required to conduct water sampling.

By integrating LRI indicators into WUA monitoring and decision-making processes, a shared evidence base can be established to support coordination, adaptive management, and informed decision-making across stakeholders, helping to balance competing water demands and strengthen long-term landscape resilience.

Public sector institutions play a central role through responsibilities for service delivery, resource governance, and policy implementation. At district level, water authorities identify underserved communities, expand access to safe drinking water, and support small-scale irrigation infrastructure.

At landscape levels, institutions oversee water allocation, quality monitoring, and the management of trade-offs between domestic use, agricultural demand, and ecosystem requirements. LRI baselines and monitoring outputs support prioritisation, planning, and reporting across these levels, including alignment with SDGs and Tanzania's Nationally Determined Contribution.

Private sector organisations also have a direct interest in water availability and reliability in the Kilombero landscape. Agricultural producers such as Kilombero Sugar Company depend on consistent irrigation supply to sustain productivity and manage climate risk, while downstream buyers face increasing expectations related to water stewardship and supply-chain resilience. At the site level, farmer cooperatives (such as AMCOS) rely on improved access to irrigation to enhance yields and reduce vulnerability to rainfall variability.

The LRI enables these actors to engage with shared water risks at landscape scale, supporting collective approaches such as Water User Associations (WUAs), efficiency

Commented [GU47]: I do not find that the case study to be clear and i am not sure what it communicates exactly? - what i gathered is that you suggest for the case of Kilombero to institutionalise the LRI within WUAs and their inner workings with support from the private sector players that exist in the landscape and consume water resources for their businesses? as in a collaborative manner or each of them using LRI separately?
anyway we can discuss this in the meeting.

improvements, and complementary Nature-based Solutions.

Table 12. Key Stakeholders, Interests, and Use of LRI Evidence

Stakeholder	Primary Level of Operation	Interest in Water Access	How the LRI Supports Their Decisions
District Water Authorities: Kilombero District	Site / Landscape	Drinking water access, small-scale irrigation	Identify underserved communities and prioritise infrastructure investments
Rufiji Basin Water Office	Landscape	Water allocation, quality, and long-term availability	Use aggregated indicators to inform basin planning and manage competing water uses
National Ministries (Water, Agriculture, Environment)	National	Policy delivery, SDG and NDC reporting	Use LRI baselines and monitoring data to inform programmes and track national targets
Water User Associations (WUAs)	Site/Landscape	Water allocation, water monitoring, irrigation and water management	Use LRI to monitor water availability and water quality, coordination of access, guide restoration activity their impact
Private actor: Kilombero Sugar Company (KSC)	Landscape	Reliable irrigation water, supply stability	Identify water-related risks and support investments that strengthen long-term productivity
Private actor: Downstream Buyers (e.g. Nestlé)	Landscape	Water stewardship, ESG commitments	Use landscape-level evidence to support responsible sourcing and shared water initiatives
AMCOS and Farmer Cooperatives	Site / Landscape	Irrigation access, climate resilience	Target on-farm water management and advocate for support
NGOs (e.g. Reforest Africa, conservation partners)	Site / Landscape	Catchment protection, NbS delivery	Align restoration and NbS interventions with priority water and ecosystem areas
Local communities	Site	Safe drinking water, domestic use for livelihoods, climate resilience	Use LRI-informed evidence (via WUAs) to understand availability and quality, supporting equitable access and participation in local water management decisions.

MONITORING, LEARNING, AND ADAPTIVE MANAGEMENT

Beyond initial prioritisation, the LRI provides a framework for tracking outcomes over time. Monitoring indicators related to water access, irrigation coverage, land condition, and adaptive capacity allow stakeholders to assess whether interventions are delivering the intended benefits and to adjust approaches as conditions change.

At the site level, farmers and local implementers can use monitoring results to adapt practices, such as irrigation scheduling or soil and water conservation measures. At the landscape level, district authorities and multi-stakeholder platforms can use aggregated data to coordinate interventions across communities and ecosystems, reducing duplication and managing trade-offs between uses. For both public and private actors, consistent monitoring strengthens accountability and provides credible evidence of progress.

SUPPORTING INSTITUTIONAL UPTAKE AND SCALING

The Kilombero case demonstrates how LRI-enabled evidence can be embedded within existing planning, governance, and investment processes. By aligning monitoring outputs with district planning cycles, basin-level water management, and national reporting frameworks, local interventions become visible beyond individual projects. This strengthens the case for sustained public and private investment and creates conditions for multi-stakeholder engagement aligned with value-chain resilience and landscape resilience objectives.

Importantly, the LRI does not function as a financing mechanism in itself. Instead, it strengthens the foundations for investment by improving the quality, coherence, and usability of evidence. In landscapes such as Kilombero that provide resources and homes for stakeholders from small holders to corporate organisations, this shared evidence base is essential for scaling Integrated Landscape Management and ensuring that improvements in water access deliver lasting resilience outcomes.

RISKS, ENABLERS AND MITIGATION MEASURES FOR SCALING—

Scaling the LRI as an investment-enabling tool presents both opportunities and challenges. Key risks include data continuity, institutional capacity constraints, and coordination across actors. However, these risks can be mitigated through phased implementation, integration with existing monitoring systems, and partnerships with research institutions and basin authorities.

Enablers include strong policy alignment, growing private sector interest in water stewardship, and emerging blended finance models for nature-based solutions. As demonstrated in similar African contexts (WWF, 2025), combining public capital, private investment, and shared measurement frameworks can unlock significantly larger flows of finance for nature. The LRI provides the evidence backbone required to support such models at scale.

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CONCLUSION

The Landscape Resilience Index (LRI) provides a unified framework that integrates indicators across the three pillars: Land, Water, and Adaptive Capacity. By combining site-level data with landscape scale data, the LRI bridges fragmented monitoring systems to create a coherent picture of landscape resilience and enables future interconnections between indicators to be identified. By linking local interventions, and their anticipated impacts to national policy commitments and global targets the LRI ensures that improvements in landscape resilience are not only measurable but also directly connected to different targets across regions and nations. In doing so, the LRI transforms technical data into actionable insights that can be used to guide investment, policy integration, and long-term planning.

NEXT STEPS

In this report we have demonstrated how the LRI can be applied to inform NbS and ILM intervention decisions across different contexts. However, research activities and environmental initiatives are widespread in much of the landscapes meaning that new data will constantly be available so to remain up-to-date the LRI must take advantage of this data. Furthermore, the threshold values in many cases could only be set arbitrarily at this stage, so it is important that they are reviewed and revised, if necessary, by local stakeholders in each of the landscapes.

We recommend two key priorities for SUSTAIN further advance the LRI:

1. a participatory workshop with local stakeholders for threshold evaluation; clarify roles and responsibilities for applying LRI in practice and feedback prior to embedding in existing planning, monitoring and investment processes.
2. and the appointment of a data co-ordinator whose responsibility will be to track the ongoing data collection activities and incorporate it into the dataset.

For the latter, the workshop should also serve as an opportunity for practitioners to discuss and agree a *best practice* guide for data collection so the dataset can be seamlessly integrated. Together, these steps would strengthen the LRI as a shared decision support tool, enabling to support co-ordinated action, attract investment and drive sustained improvement in landscape resilience.

Commented [GU49]: is this planned to take place from your side? i assume yes as calvin mentioned a validation workshop is yet to take place. will this be done from Evodius's side?

ANNEXES

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ORIGINAL LHMI INDICATOR LIST

Table S. 1. Original LRI indicator list

Scale	Pillar	Indicator
Site	Soil	Bird species richness per site
		Invertebrate species richness per site (pollinators and ground dwelling)
		Earthworm count (individuals and type)
		Total Carbon stock (above and below ground)
		Soil C:N ratio
		Soil pH
		Rate of soil loss
Site	Water	Infiltration rate of water
		Depth to water table
		Tanzania River Scoring System (sensitivity score using macroinvertebrate assemblages)
Landscape	Water	Tanzania River Scoring System (sensitivity score using macroinvertebrate assemblages)
		Access to water per different use types
Landscape	Resilience	Agricultural, agroforestry and tree plantation productivity
		Adoption of Sustainable Land Management (SLM) practices

		Household poverty
		Natural ecosystem connectivity
		Threatened species: % of species that are critically endangered
		Perception of shocks and stresses
		Extreme weather

Table S. 2. Scoring table with sensitivity scores for macroinvertebrates included in Tanzania River Scoring System (TARISSfupi) and Mini Stream Assessment Scoring System (miniSASS) assessments

Groups	Description	Sensitivity score	
		TARISSfupi	miniSASS
1	Flat worms	3	3
2	Worms	2	2
3	Leeches	2	2
4	Crabs/Shrimps	6	6
5	Stoneflies	15	17
6	Minnow Mayflies	5	5
7	Other Mayflies	11	11
8	Brushlegged Mayflies (Oligoneuridae)	17	-
9	Oval Mayflies (Prosopistomatidae)	17	-
10	Damselflies	4	4
11	Dragonflies	6	6
12	Bugs/Beetles	5	5
13	Caddisflies	9	9
14	True Flies	2	2
15	Snails	4	4
Total score			
Number of groups			
Average score			

MOZAMBIQUE STAKEHOLDER WORKSHOP

A stakeholder workshop was held in Chimoio city, Mozambique, on Tuesday, August 26th, hosted by IUCN-Mozambique Project Manager Mohammed Esmael. The workshop brought together representatives from various local organizations (table A7) to discuss the integration of the Land Health Monitoring Index (LHMI) into regional activities.

Objectives and Prioritization

The workshop was designed to achieve the following core objectives:

- **LHMI Introduction:** Provide stakeholders with an overview of the LHMI and introduce SUSTAIN's holistic concept of land health.
- **Data Exchange and Capacity:** Establish how each organization could contribute to the LHMI through previously collected data and by leveraging existing capacity for future data collection.
- **National Policy Integration:** Solicit feedback on proposed national-level indicators and discuss pathways for the LHMI to inform national policies.
- **Indicator Review:** Receive expert feedback on the comprehensiveness and representativeness of the sub-indicators included in the LHMI methodology.

Due to a delayed start caused by the DPAP Director's prior engagement, focus was placed on the data exchange and policy alignment sessions (table A6) to immediately establish how the LHMI could integrate with organizational activities.

Key Outcomes and Next Steps

The session on data exchange was productive, and we established various types of data that the local organisations have the capacity to collect (A5). Discussions around national policy integration were more challenging, as participants expressed a need for formal written authorization from their institution before commenting on national policies.

During this session it was agreed that we would follow up with Mohammed Esmael who will officially contact the participants' respective organizations to formally request access to collected data and solicit comments regarding national policy alignment. This ensures that the next steps respect organizational protocols and secure the necessary institutional buy-in for data access and collaboration.

Table S. 3. Local organisation capacity to collect data relevant for the LRI in Mozambique

Organization	Data Collection Focus
Land pillar	
IIAM	Studies pedology (soil science) and soil fertility, and measures carbon stocks.
ISPM	Performs soil fertility analysis.
Campo 4	Performs soil fertility analysis.

ADEM	Collects data on soil fertility and soil management.
Munhrare Association	Surveys forest plant species richness.
Unizambeze	Performs biodiversity analysis and studies, and maps soil and watercourses.
Water pillar	
ARA-CENTRO, IP.	Available data includes water quality monitoring (chemical, physical, and biological). Data that could be collected includes water flow measurement (m ³ /s), analysis of macroinvertebrates (aquatic species), and identification of resources for field irrigation.
ISPM	Conducts water salinity studies.
Munhrare Association	Performs water analysis.
ADEM	Performs water analysis.
Adaptive capacity pillar	
IIAM	Collects data related to climate change, including data analysis for the LHMI project (covering agroforestry systems, multiplication of nitrogen-rich crops, and analysis of soil microorganisms).
ADEM	Focuses on climate change adaptation by introducing plant species capable of adapting to climate changes to serve as windbreaks and help retain moisture.

Table S. 4. Workshop Agenda

Session	Time	Tasks
Introductions	08:45 - 09:00	Introduction to project team and attendees
LHMI Presentation	09:00 - 09:30	Overview: -SUSTAIN-Africa -LHMI
Breakfast	09:30 - 10:00	-
Session 1: data exchange	10:00 - 10:40	Discussion points: •Do any of your company's activities feed into the LHMI indicators? •Do you know of any other available data that is relevant for the LHMI?
Session 1: feedback	10:40 - 11:20	Group feedback on discussion points
Comfort break	11:20 - 11:30	

Session 2: LHMI integration into national policy	11:30 - 12:15	Discussion points: •Which national commitments are relevant to the LHMI? •On a practical level, how could the LHMI feed into national policies?
Session 2: feedback	12:15 - 13:00	Group feedback on discussion points
Lunch	13:00 - 14:00	-
Session 3: Evaluating the LHMI sub-indicators	14:00 - 15:00	Discussion points: •What other indicators should be considered? •What other factors should be considered? E.g. seasonality, land use type
Session 3: feedback	15:00 - 15:45	Group feedback on discussion points
Closing remarks	15:45 - 16:00	

Table S. 5. Mozambique Workshop attendees

Organisation	Representative
IIAM - Mozambique Institute of Agricultural Research (IIAM)	Alcidio Vilanculos
ISPM - Instituto Superior Politecnico de Manica	Helder Juliao
ARA-CENTRO - Administração Regional de Águas do Centro	Delton
SDAE Vanduzi - Serviço Distrital de Actividades Economicas de Vanduzi	Salva Nhalungo

SDAE Barue Serviço Distrital de Actividades Económicas de Barue	Cristovao Nhamalezi
UCAMA - União Provincial de Camponeses de Manica	Zacarias
DPAP - Direção Provincial de Agricultura e Pescas	Muzaja e Director
ADEM - Agencia de Desenvolvimento Economico da Provincia de Manica	Manuel Queiroz
IUCN - International Union for the Conservation of Nature	Mahomed Esmael